

Use-Case Specification — Dormify Dormitory Management System

Dormitory Management System

Field	Details
Document type	Use-Case Specification
Version	1.0 — PA3
Date	25 July 2026
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Reviewed by	Đào Duy Anh
Edited by	Đào Duy Anh

Specification Snapshot

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Review item	Summary
Product	Dormify
Scope	Functional behavior for the PA3 Dormitory Management System
Use-case count	72 use cases
Functional groups	11 functional groups plus traceability appendices
Selected PA3 focus	Maintenance Management
Related use-case model	use-case-model.md and use-case-model.pdf
Main readers	Development team, reviewers, testers, and PA3 evaluators
Modeling status	Intended system behavior; implementation status is not claimed by this document

PA3 Structure Checklist

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PA3 requirement	How this document addresses it
Clear use-case ordering	Use cases are grouped by workflow and functional area, with a functional group index before the detailed specifications.

PA3 requirement	How this document addresses it
Use-case model alignment	The detailed specifications are linked to the separate Mermaid use-case model and PDF export.
Use-case name and ID	Every detailed use case starts with Use-case ID and Use-case name .
Actor(s)	Every use case lists the initiating actor and supporting actor when applicable.
Description and preconditions	Each specification explains the user goal, trigger, and conditions required before execution.
Basic flow	The normal successful path is written as numbered, step-by-step behavior.
Alternative flows	Recoverable alternatives and exception paths are listed under a dedicated field.
Postconditions and special requirements	Outcome state, constraints, business rules, validation, and security notes are captured per use case.
Prototype screenshots	Screenshots from PA3/assets are included under every use case so PA3 prototype evidence appears beside the relevant specification.

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Functional Group Index

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Section	Functional group	Use-case count
2	Authentication and Personal Profile	10
3	System Administration	4
4	Room and Student Management	11
5	Contract Management	6
6	Checkout and Deposit Refund	4
7	Finance and Meter Management	11
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1. Introduction

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1.1 Purpose

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This document specifies the functional behavior of the Dormify Dormitory Management System. It expands the approved use-case model into complete use-case descriptions that can be used for design, implementation, testing, and PA3 evaluation.

1.2 Scope and Modeling Decisions

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- The document contains **72 use cases** across 11 functional groups.
- The related use-case model is maintained in `use-case-model.md`, with a PDF export in `use-case-model.pdf`.
- System Admin and Dormitory Manager remain separate actors.
- Floor Manager responsibilities are represented under Dormitory Manager in the diagrams and specifications, while the implementation may retain a separate role.
- FR08 (Back up and restore data) is outside the current scope.
- Payment methods are alternative flows of UC-FIN-09, not separate use cases.
- Automatic notifications are treated as postconditions of their originating use cases.
- The specification describes the intended system behavior and does not claim implementation status.
- Each use case includes a **Prototype Screens:** field with one or more embedded screenshots from `PA3/assets` beside the relevant behavior.

1.3 Use-Case Template

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Each use-case specification follows the same structure so design, implementation, testing, and review work can trace behavior consistently.

Template field	Purpose
Use-case ID and name	Provides the stable identifier and clear title required by PA3.
Actor(s) and supporting actors	Identifies who starts or supports the behavior.
Description and trigger	Explains the goal and the event that starts the use case.
Preconditions	Lists what must be true before the use case begins.
Basic flow (main success scenario)	Defines the normal successful path step by step.
Alternative flows	Defines recoverable alternatives, exceptions, and error paths.
Postconditions	States what is true after the use case finishes.
Special requirements	Captures constraints that affect design, security, data, or validation.
Prototype screens	Includes one or more PA3 asset screenshots covering the related basic-flow and alternative-flow screens.
Relationships	Records include, extend, or related-use-case links.
Traceability	Links the use case to functional requirements or feature IDs.

1.4 Actors

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Actor	Responsibility
Applicant / Guest	Submits a preliminary residence profile and authenticates before becoming a resident.
Student	Uses residence, room, contract, invoice, maintenance, feedback, conduct, and communication functions.
System Admin	Controls accounts, roles, permissions, logs, and administration monitoring.
Dormitory Manager	Performs dormitory operations across rooms, students, contracts, finance, residence, maintenance, feedback, and conduct.
Maintenance Staff	Receives and processes assigned repair work.
Google OAuth Provider	Provides external Google authentication.

Actor	Responsibility
Email / SMS Service	Delivers confirmation and password-recovery messages.
Payment Gateway	Processes electronic invoice payments.
Scheduled Trigger	Starts time-based debt reminder and overdue processes.

2. Authentication and Personal Profile

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Account registration, authentication, profile self-service, and student residence information.

ID	Use case	Primary actor
UC-AUTH-01	Submit Preliminary Residence Profile	Applicant / Guest
UC-AUTH-02	Log In	Applicant / Guest, Student
UC-AUTH-03	Log Out	Student
UC-AUTH-04	Reset Forgotten Password	Applicant / Guest, Student
UC-PRO-01	View Personal Profile	Student
UC-PRO-02	Update Contact Information	Student
UC-PRO-03	Change Password	Student
UC-STAY-01	View Current Residence Information	Student
UC-STAY-02	View Roommates	Student
UC-STAY-03	View Residence History	Student

UC-AUTH-01 — Submit Preliminary Residence Profile

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Use-case ID: UC-AUTH-01

Use-case name: Submit Preliminary Residence Profile

Actor(s): Applicant / Guest

Supporting Actor(s): Email / SMS Service

Description: A prospective resident creates a preliminary profile so the dormitory can contact and evaluate the application.

Trigger: The applicant selects the registration or residence-application function.

Preconditions:

1. The applicant is not authenticated.
2. No active account already uses the submitted email or citizen identification number.

Basic Flow (Main Success Scenario):

1. The applicant enters preliminary personal data, contact information, accommodation preferences, and a password.
2. The system validates required fields and checks uniqueness.
3. The system creates an applicant/student account in a pending or active state according to the configured admission policy.
4. The system sends a confirmation or contact notification through email or SMS.
5. The system displays a successful submission message and directs the applicant to log in or await further contact.

Alternative Flows:

- A1. Duplicate email or identification number: the system rejects the submission and identifies the conflicting field.
- A2. Invalid or incomplete information: the system highlights validation errors and keeps the entered data.
- A3. Notification service unavailable: the profile is saved and the system records the delivery failure for retry or manual follow-up.

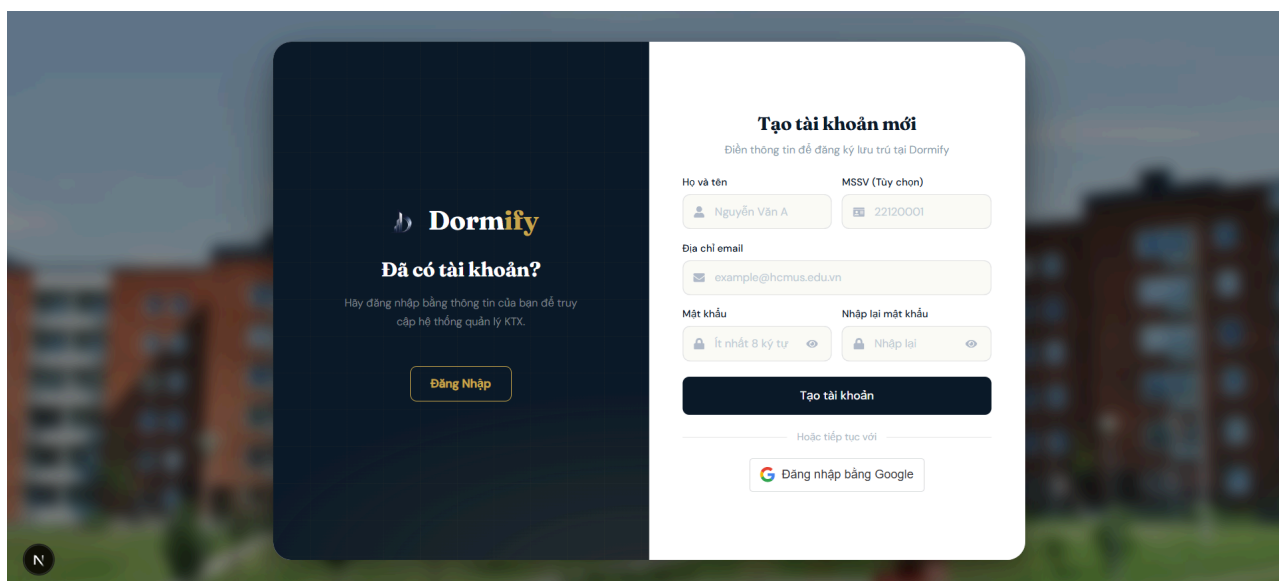
Postconditions:

- A preliminary residence profile and user account are stored.
- A confirmation/contact notification is queued or sent.

Special Requirements:

- Passwords must be stored as secure hashes.
- Sensitive identity data must be protected in transit and at rest.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-AUTH-01.

Relationships:

- None.

Traceability: FR01

UC-AUTH-02 — Log In

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-AUTH-02

Use-case name: Log In

Actor(s): Applicant / Guest, Student

Supporting Actor(s): Google OAuth Provider

Description: A registered user authenticates using email, citizen identification number, or Google authentication.

Trigger: The user submits the login form or selects Google sign-in.

Preconditions:

1. The user account exists.
2. The account is not locked, deactivated, or deleted.

Basic Flow (Main Success Scenario):

1. The user enters an email or citizen identification number and password.
2. The system validates the credentials and account status.
3. The system creates an authenticated session or access token.
4. The system identifies the user role and redirects the user to the appropriate workspace.

Alternative Flows:

- A1. Google login: the Google OAuth Provider authenticates the user; the system links or creates the local account according to policy.
- A2. Invalid credentials: the system denies access without revealing which field is incorrect.
- A3. Locked/deactivated account: the system denies access and displays the permitted reason or support guidance.
- A4. OAuth failure: the user returns to the login page with a non-sensitive error message.

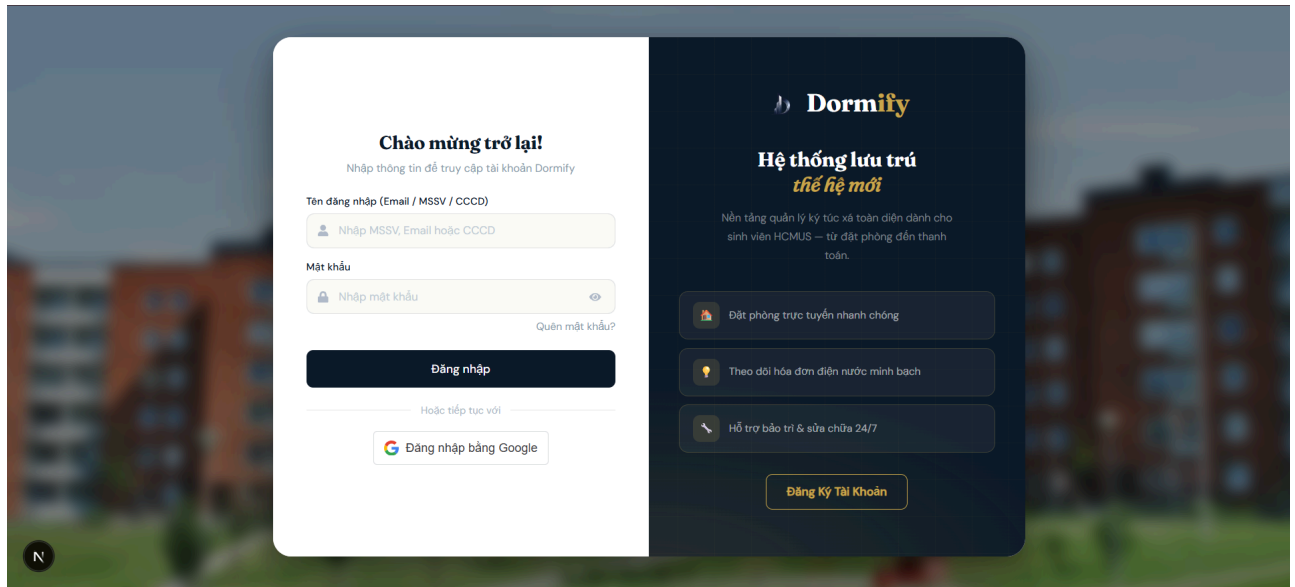
Postconditions:

- A valid authenticated session exists.
- The login event may be written to the audit log.

Special Requirements:

- Authentication attempts should be rate-limited.
- Session data must not expose passwords or sensitive credentials.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-AUTH-02.

Relationships:

- None.

Traceability: FR02

UC-AUTH-03 — Log Out

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-AUTH-03

Use-case name: Log Out

Actor(s): Student

Supporting Actor(s): None

Description: An authenticated user ends the current session.

Trigger: The user selects Log out.

Preconditions:

1. The user is authenticated.

Basic Flow (Main Success Scenario):

1. The system invalidates or removes the local session credentials.
2. The system clears protected user context from the client.
3. The system redirects the user to the login page.

Alternative Flows:

- A1. Server-side logout is unavailable: the client still clears local credentials and prevents access to protected pages.
- A2. Session already expired: the system keeps the user logged out and redirects to the login page without showing an error.

- A3. Unsaved local form state exists: the system warns the user before clearing the session and protected client context.

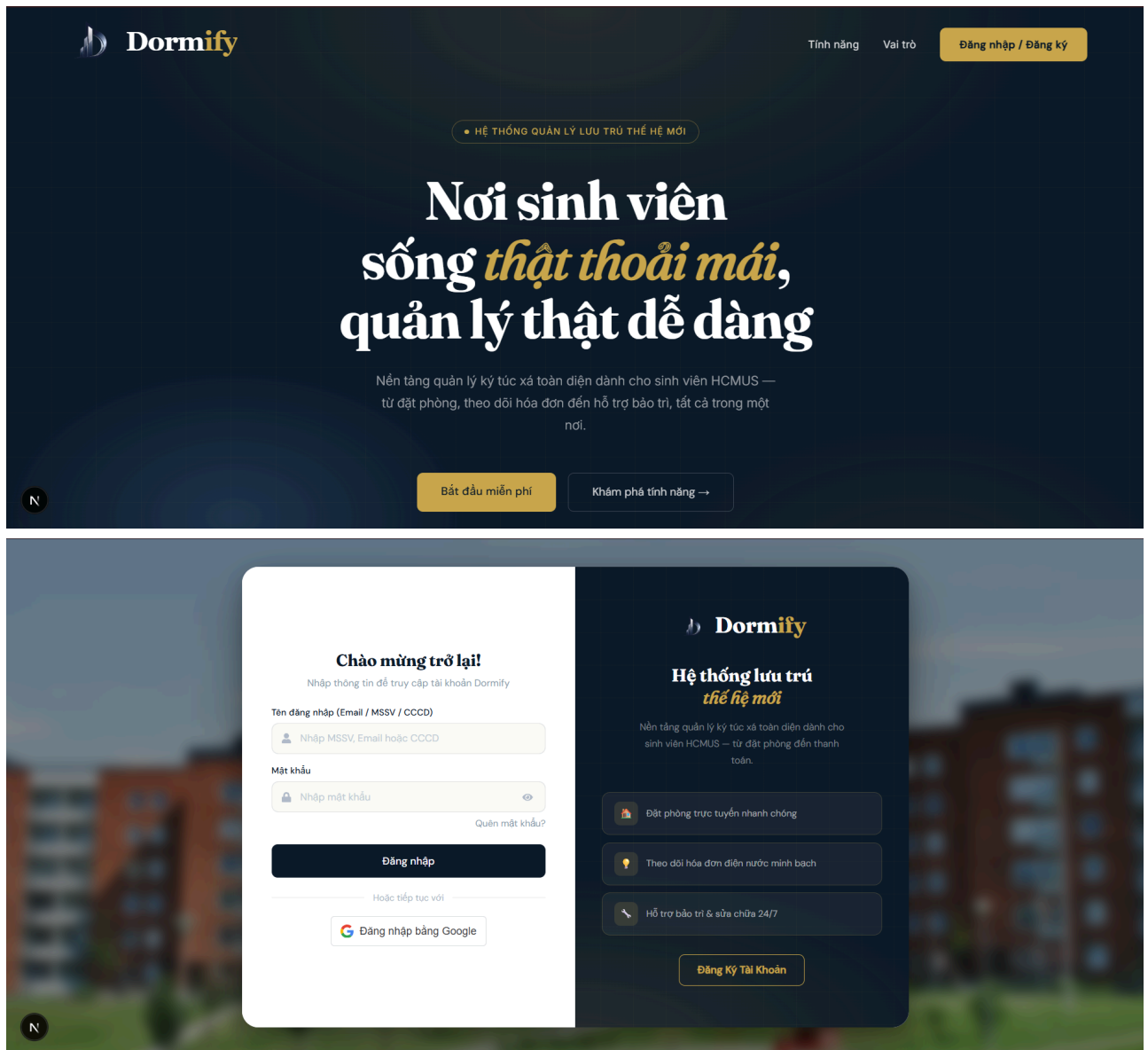
Postconditions:

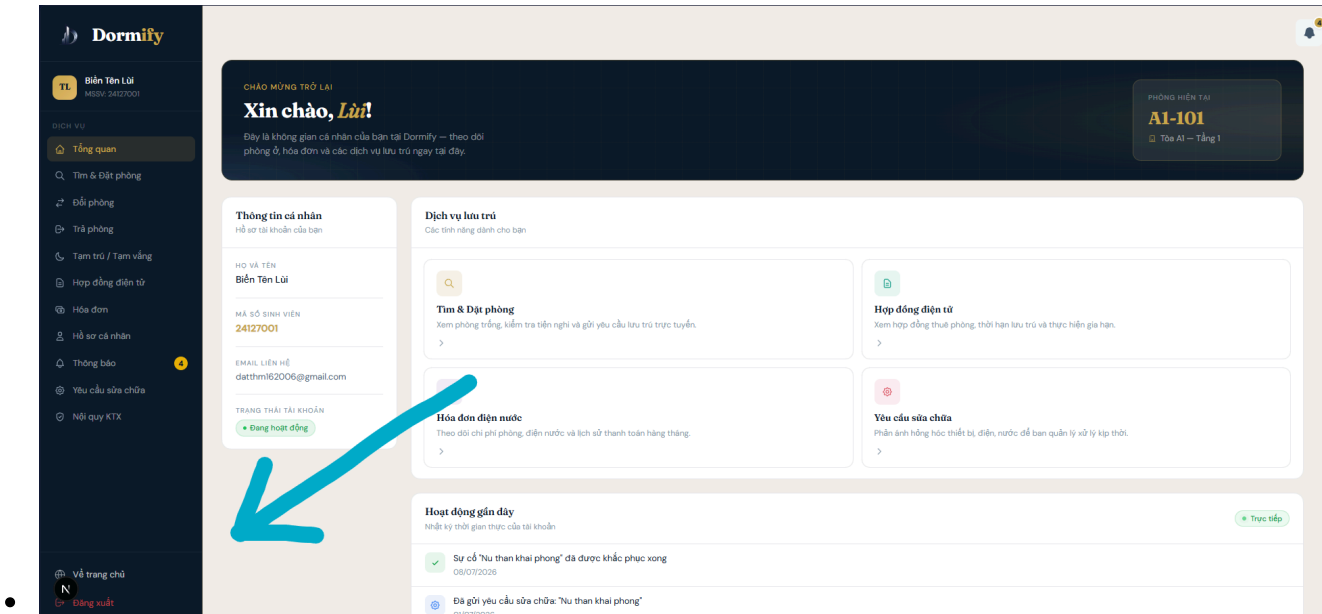
- The current device no longer has an active user session.

Special Requirements:

- Protected pages must require re-authentication after logout.

Prototype Screens:





- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-AUTH-03.

Relationships:

- None.

Traceability: FR02

UC-AUTH-04 — Reset Forgotten Password

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-AUTH-04

Use-case name: Reset Forgotten Password

Actor(s): Applicant / Guest, Student

Supporting Actor(s): Email / SMS Service

Description: A user who cannot remember the password verifies account ownership and sets a new password.

Trigger: The user selects Forgot password.

Preconditions:

1. An account exists for the submitted email or phone number.

Basic Flow (Main Success Scenario):

1. The user submits the registered email or phone number.
2. The system generates a time-limited OTP or reset token.
3. The Email / SMS Service delivers the verification code or reset link.
4. The user submits the OTP/token and a new password.
5. The system validates the request, updates the password hash, and invalidates the OTP/token.
6. The system confirms the reset and directs the user to log in.

Alternative Flows:

- A1. Unknown account: the system returns a generic response to prevent account enumeration.
- A2. Invalid or expired OTP/token: the reset is rejected and the user may request a new code.

- A3. Weak password: the system requests a password that satisfies the configured policy.

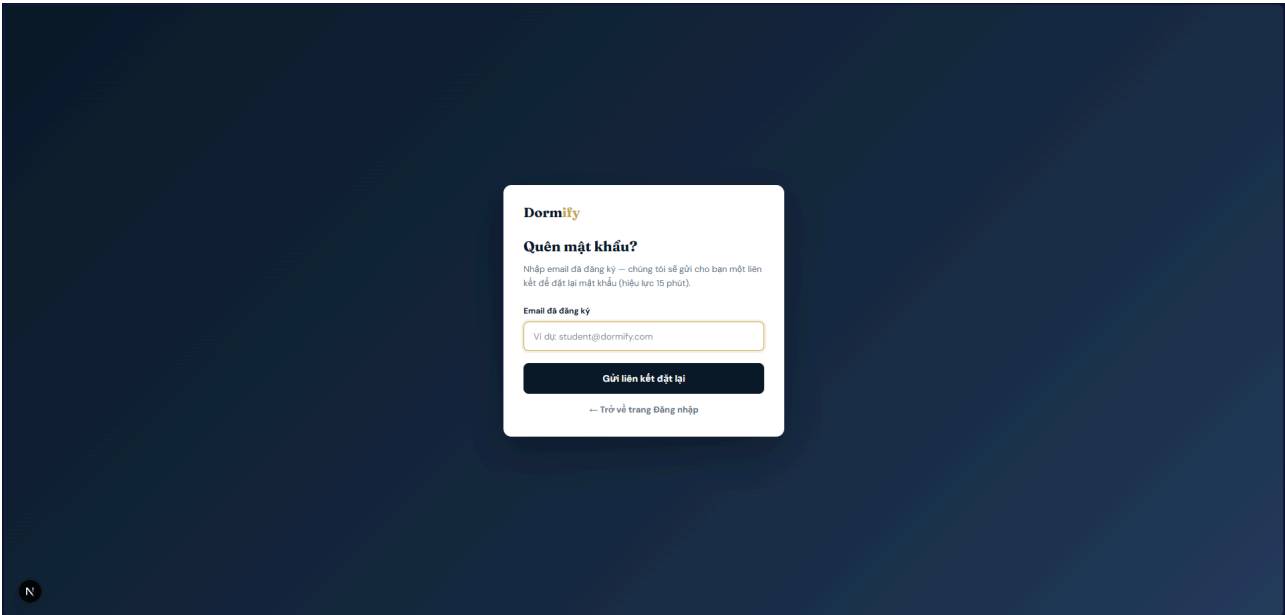
Postconditions:

- The account password is changed.
- The OTP/token cannot be reused.

Special Requirements:

- OTP/token lifetime and retry limits must be enforced.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-AUTH-04.

Relationships:

- None.

Traceability: FR02

UC-PRO-01 — View Personal Profile

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-PRO-01

Use-case name: View Personal Profile

Actor(s): Student

Supporting Actor(s): None

Description: A student views personal and identity information stored by the dormitory.

Trigger: The student opens the Personal Profile page.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system retrieves the student profile.
2. The system displays identity data, contact data, permanent address, student identifier, and account information.
3. The system visually distinguishes editable contact fields from manager-controlled identity fields.

Alternative Flows:

- A1. Profile data is incomplete: the system shows empty or pending fields without failing the page.
- A2. Profile service unavailable: the system shows a retry message and does not expose stale sensitive data beyond the active session policy.
- A3. Unauthorized profile access attempt: the system denies access and records the attempt for audit review.

Postconditions:

- No data is changed.

Special Requirements:

- Only the profile owner and authorized management roles may view the complete record.

Prototype Screens:

-  UC-PRO-01 PA3 prototype screen - studentprofile
- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-PRO-01.

Relationships:

- None.

Traceability: FR03, ST01

UC-PRO-02 — Update Contact Information

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-PRO-02

Use-case name: Update Contact Information

Actor(s): Student

Supporting Actor(s): None

Description: A student updates permitted contact details without modifying protected identity information.

Trigger: The student selects Edit contact information.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system displays editable fields such as phone number, email where permitted, current address, and avatar.
2. The student enters changes and submits.

3. The system validates the values and uniqueness constraints.
4. The system saves only fields allowed for student self-service.
5. The system confirms the update.

Alternative Flows:

- A1. Invalid phone/email format: the system rejects the affected field.
- A2. Protected identity field included in the request: the system ignores or rejects the unauthorized change.
- A3. Duplicate contact identifier: the system requests a different value.

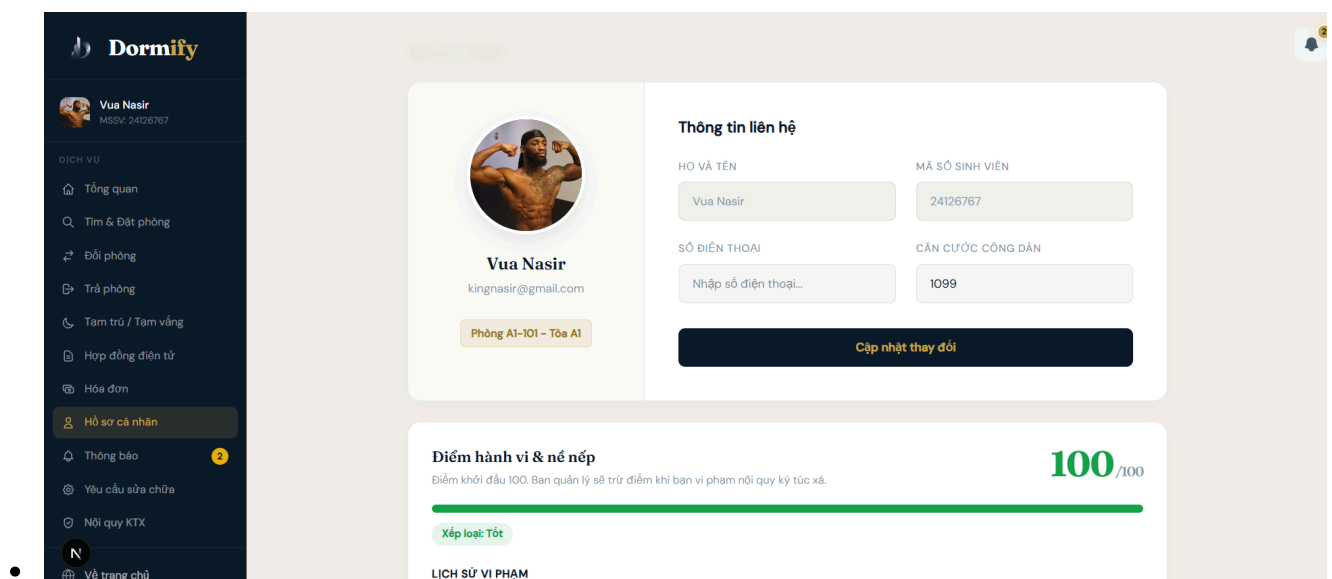
Postconditions:

- Permitted contact information is updated.

Special Requirements:

- Full name, date of birth, citizen identification number, and other identity fields require an authorized manager.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-PRO-02.

Relationships:

- None.

Traceability: FR03, ST02

UC-PRO-03 — Change Password

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-PRO-03

Use-case name: Change Password

Actor(s): Student

Supporting Actor(s): None

Description: An authenticated student changes the account password by providing the current password.

Trigger: The student selects Change password.

Preconditions:

1. The student is authenticated.
2. The student knows the current password.

Basic Flow (Main Success Scenario):

1. The student enters the current password, new password, and confirmation.
2. The system verifies the current password.
3. The system validates password strength and matching confirmation.
4. The system stores the new password hash.
5. The system confirms the change and may require re-login on other devices.

Alternative Flows:

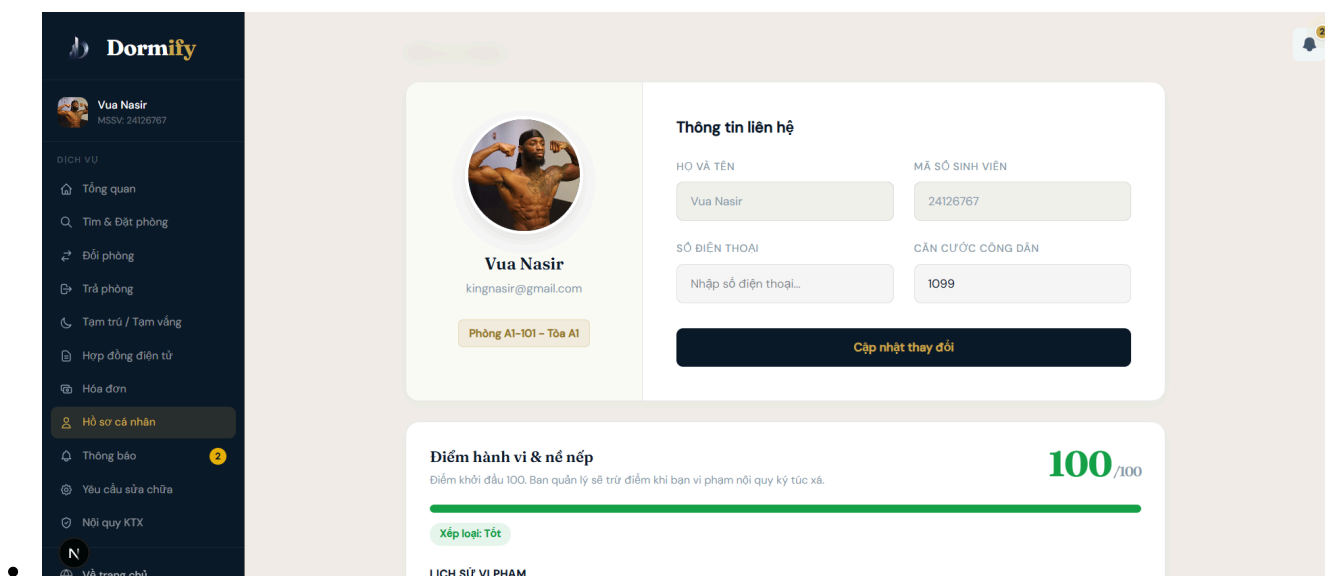
- A1. Current password incorrect: no change is made.
- A2. New password and confirmation differ: the system requests correction.
- A3. New password violates policy: the system explains the unmet requirements.

Postconditions:

- The new password becomes valid and the old password becomes invalid.

Special Requirements:

- Password values must never be logged or returned.

Prototype Screens:

- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-PRO-03.

Relationships:

- None.

Traceability: FR02, ST03

UC-STAY-01 — View Current Residence Information

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-STAY-01

Use-case name: View Current Residence Information

Actor(s): Student

Supporting Actor(s): None

Description: A resident views the currently assigned building, floor, room, bed, and stay status.

Trigger: The student opens Current Residence.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system finds the student's active room assignment or contract.
2. The system displays building, floor, room, bed, room type, contract period, and residence status.

Alternative Flows:

- A1. No active assignment: the system displays a message and a link to search or apply for a room.
- A2. Assignment is expired or inactive: the system shows the latest known residence record with an inactive status label.
- A3. Residence details are partially unavailable: the system displays available fields and provides a contact option for dormitory support.

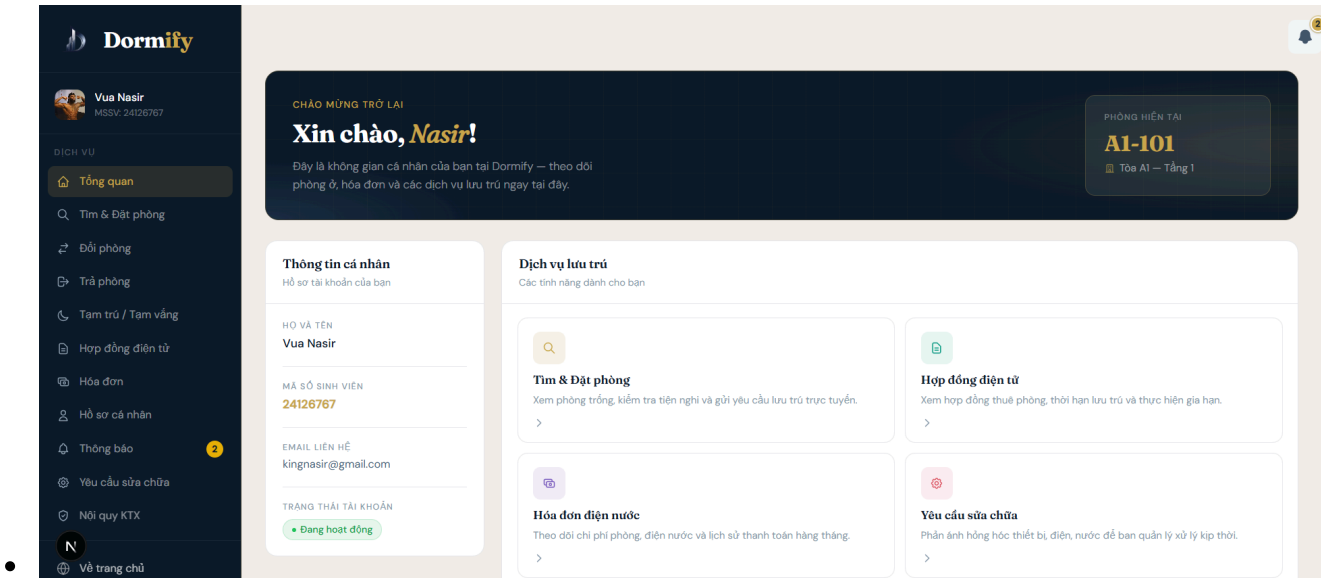
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-STAY-01.

Relationships:

- None.

Traceability: FR03, ST04

UC-STAY-02 — View Roommates

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-STAY-02

Use-case name: View Roommates

Actor(s): Student

Supporting Actor(s): None

Description: A resident views basic information about students currently sharing the assigned room.

Trigger: The student selects Roommates.

Preconditions:

1. The student is authenticated.
2. The student has an active room assignment.

Basic Flow (Main Success Scenario):

1. The system retrieves active occupants of the same room.
2. The system displays each roommate’s permitted public information, such as name and student identifier.

Alternative Flows:

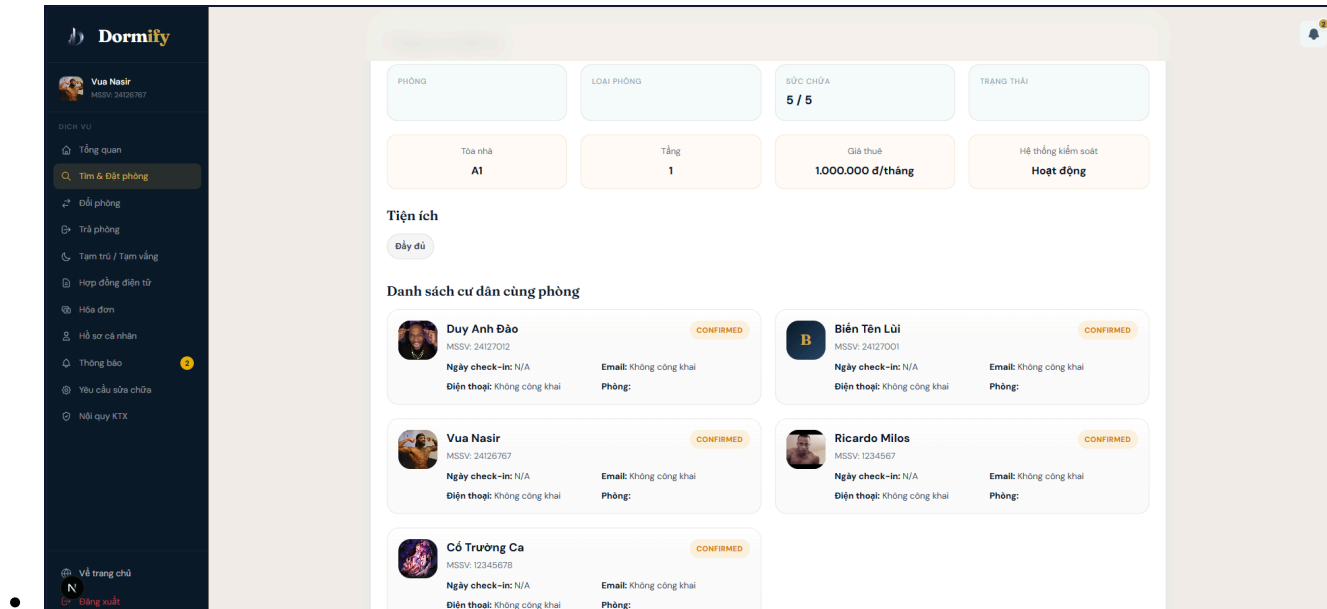
- A1. The student is the only occupant: the system displays an appropriate empty state.
- A2. A roommate has privacy-restricted contact data: the system omits those fields.

Postconditions:

Special Requirements:

- Sensitive personal information must not be exposed to roommates.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-STAY-02.

Relationships:

- None.

Traceability: FR03, ST05

UC-STAY-03 — View Residence History

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-STAY-03

Use-case name: View Residence History

Actor(s): Student

Supporting Actor(s): None

Description: A student reviews previous room assignments, transfers, contracts, and checkout records.

Trigger: The student opens Residence History.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system retrieves historical room assignments and relevant dates.
2. The system orders records from newest to oldest.
3. The student may open an entry to view room, contract, transfer, and checkout details.

Alternative Flows:

- A1. No previous residence records: the system displays an empty state.
- A2. Filters return no history: the system displays an empty filtered state and offers a clear-filter action.
- A3. Selected historical detail is archived or unavailable: the system keeps the list visible and explains that the detail cannot be opened.

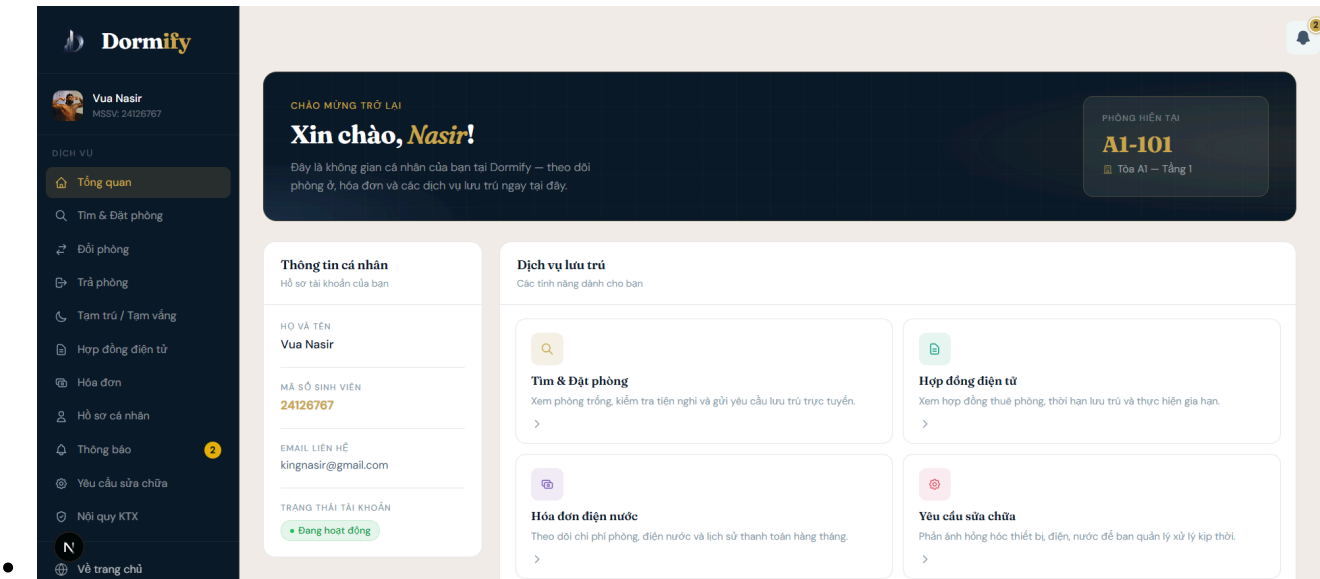
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-STAY-03.

Relationships:

- None.

Traceability: FR03, ST06

3. System Administration

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

System account control, authorization, auditability, and administrative monitoring.

ID	Use case	Primary actor
UC-ADM-01	Manage User Accounts	System Admin
UC-ADM-02	Manage Roles and Permissions	System Admin
UC-ADM-03	View and Filter Audit Logs	System Admin

ID	Use case	Primary actor
UC-ADM-04	View Administration Dashboard	System Admin

UC-ADM-01 — Manage User Accounts

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-ADM-01

Use-case name: Manage User Accounts

Actor(s): System Admin

Supporting Actor(s): None

Description: The System Admin views accounts and performs lifecycle actions such as lock, unlock, deactivate, or delete.

Trigger: The admin opens User Account Management.

Preconditions:

1. The actor is authenticated as System Admin.

Basic Flow (Main Success Scenario):

1. The system displays searchable and filterable user accounts.
2. The admin selects an account and an allowed action.
3. For a restrictive action, the admin enters a reason where required.
4. The system validates permissions and account dependencies.
5. The system updates the account status or removes the account according to retention policy.
6. The system records the action in the audit log.

Alternative Flows:

- A1. Lock account: access is denied until unlocked.
- A2. Unlock account: active access is restored.
- A3. Deactivate account: login is disabled while records are retained.
- A4. Delete account with active obligations: the system blocks deletion and recommends deactivation.
- A5. Admin attempts to remove the last System Admin or their own essential access: the system rejects the action.

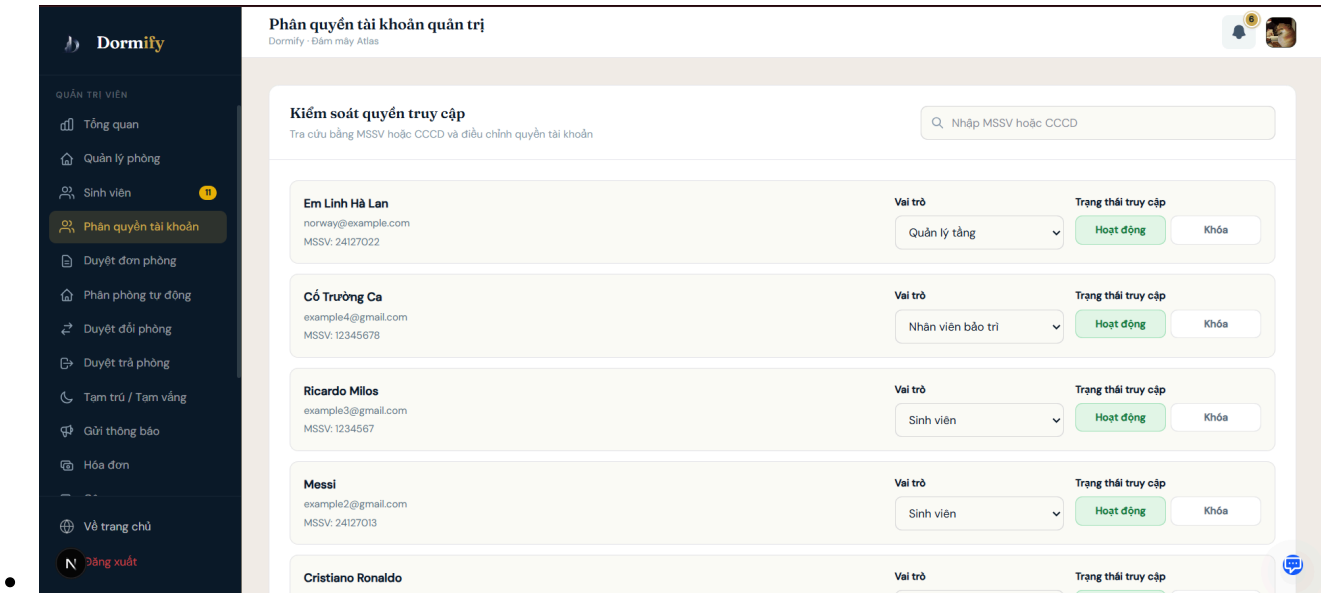
Postconditions:

- The account reflects the selected lifecycle status.
- An audit record exists.

Special Requirements:

- Referential integrity with contracts, invoices, and history must be preserved.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ADM-01.

Relationships:

- None.

Traceability: FR04

UC-ADM-02 — Manage Roles and Permissions

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-ADM-02

Use-case name: Manage Roles and Permissions

Actor(s): System Admin

Supporting Actor(s): None

Description: The System Admin defines roles, assigns permissions, assigns roles to users, and revokes access.

Trigger: The admin opens Roles and Permissions.

Preconditions:

1. The actor is authenticated as System Admin.

Basic Flow (Main Success Scenario):

1. The system displays roles and their permission sets.
2. The admin creates or edits a role and selects allowed functions.
3. The admin assigns one or more roles to a user.
4. The system validates that critical permissions remain assigned to at least one administrator.
5. The system saves the configuration and applies it to subsequent authorization checks.
6. The system records the change in the audit log.

Alternative Flows:

- A1. Duplicate role name: the system requests another name.
- A2. Invalid permission combination: the system blocks the change and explains the conflict.

- A3. Revoke access: the system removes the selected role/permission and terminates access where required.

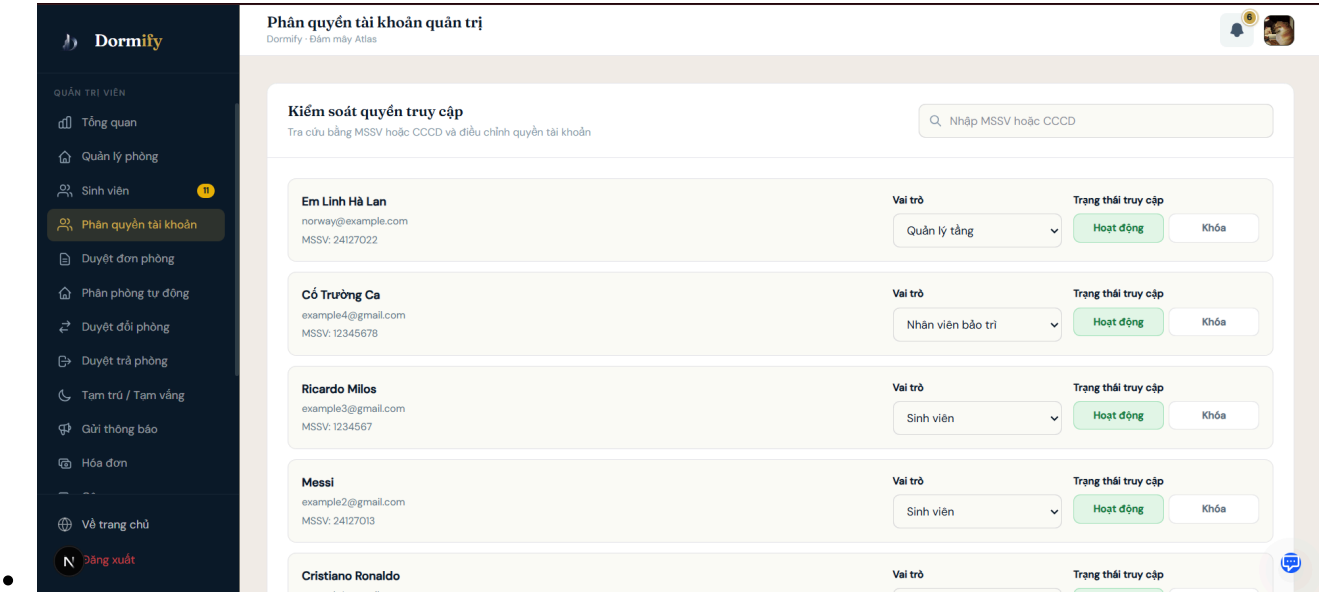
Postconditions:

- Role and permission assignments are updated.

Special Requirements:

- Authorization must follow least-privilege principles.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ADM-02.

Relationships:

- None.

Traceability: FR05

UC-ADM-03 — View and Filter Audit Logs

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-ADM-03

Use-case name: View and Filter Audit Logs

Actor(s): System Admin

Supporting Actor(s): None

Description: The System Admin reviews recorded system actions for accountability and troubleshooting.

Trigger: The admin opens Audit Logs.

Preconditions:

1. The actor is authenticated as System Admin.
2. Audit records exist or the system can return an empty result.

Basic Flow (Main Success Scenario):

- 1. The system displays audit records ordered by time.
- 2. The admin filters by user, role, date range, action, function, result, or affected object.
- 3. The admin opens a record to view permitted before/after metadata and technical context.
- 4. The system preserves the log as read-only.

Alternative Flows:

- A1. No records match: the system displays an empty result.
- A2. Sensitive values are present in the original request: the system redacts passwords, tokens, and protected fields.

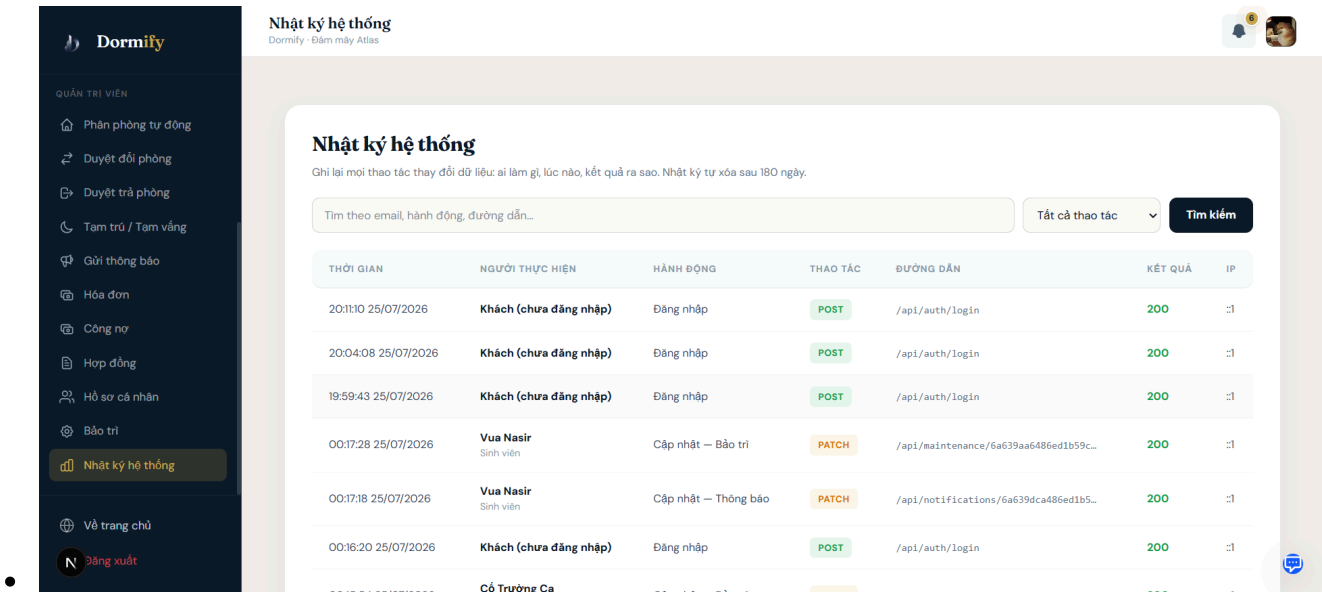
Postconditions:

- No operational data is changed.

Special Requirements:

- Audit logs should be tamper-resistant and retained according to project policy.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ADM-03.

Relationships:

- None.

Traceability: FR06

UC-ADM-04 — View Administration Dashboard

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-ADM-04

Use-case name: View Administration Dashboard

Actor(s): System Admin

Supporting Actor(s): None

Description: The System Admin views system-wide statistics, warnings, and operational summaries.

Trigger: The admin enters the administration workspace.

Preconditions:

1. The actor is authenticated as System Admin.

Basic Flow (Main Success Scenario):

1. The system aggregates account, occupancy, finance, request, maintenance, and alert data.
2. The system displays key indicators and recent warnings.
3. The admin selects a metric to navigate to the relevant management page.

Alternative Flows:

- A1. A data source is temporarily unavailable: the dashboard marks the affected widget and displays the remaining metrics.
- A2. No data for a period: the widget displays zero or an empty state.

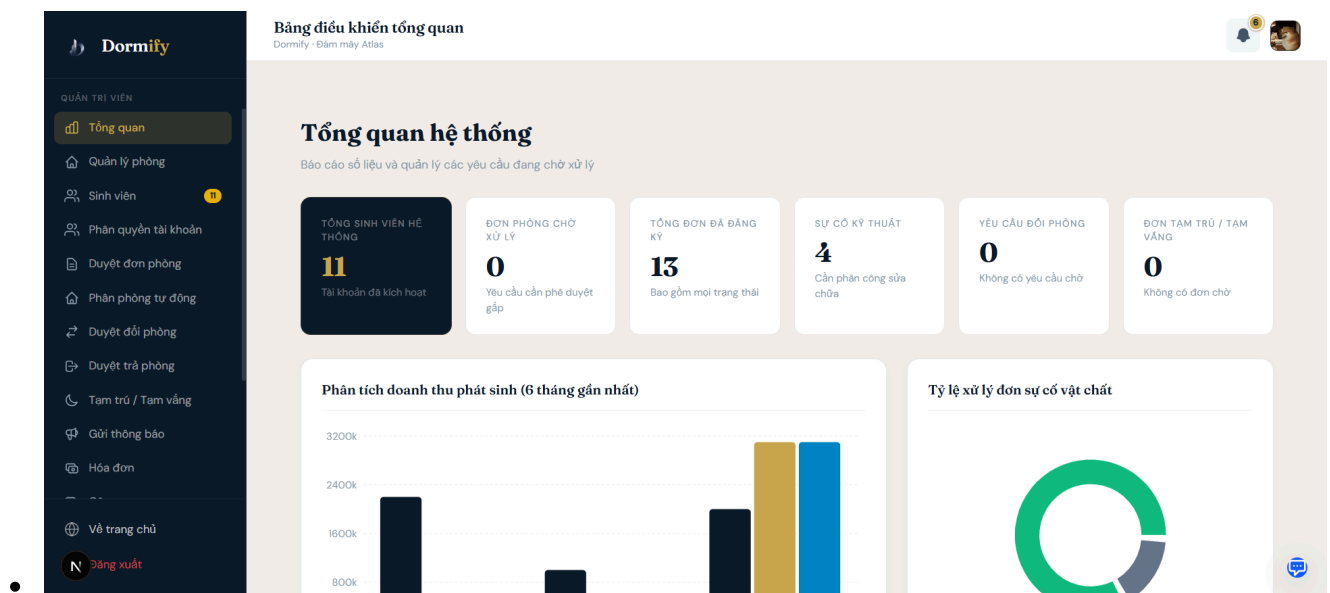
Postconditions:

- No data is changed.

Special Requirements:

- Dashboard values should identify their update time and reporting period.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ADM-04.

Relationships:

- None.

Traceability: FR07

4. Room and Student Management

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Room inventory, student records, applications, automatic allocation, and transfers.

ID	Use case	Primary actor
UC-ROOM-01	Manage Rooms	Dormitory Manager
UC-STU-01	Manage Student Records	Dormitory Manager
UC-ROOM-02	Search Available Rooms	Student
UC-ROOM-03	View Room Details	Student
UC-ROOM-04	Submit Room Application	Student
UC-ROOM-05	Track Room Application Status	Student
UC-ROOM-06	Review Room Application	Dormitory Manager
UC-ROOM-07	Run Automatic Room Allocation	Dormitory Manager
UC-ROOM-08	Submit Room Transfer Request	Student
UC-ROOM-09	Track Room Transfer Status	Student
UC-ROOM-10	Review Room Transfer Request	Dormitory Manager

UC-ROOM-01 — Manage Rooms

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-ROOM-01

Use-case name: Manage Rooms

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager maintains the dormitory room catalog and room operating status.

Trigger: The manager opens Room Management.

Preconditions:

- 1. The actor is authenticated as Dormitory Manager.

Basic Flow (Main Success Scenario):

- 1. The system displays rooms with building, floor, type, capacity, occupancy, price, facilities, and status.
- 2. The manager adds a new room or selects an existing room to edit.
- 3. The system validates uniqueness, capacity, price, and required data.
- 4. The manager may mark a room as available, full, unavailable, or under maintenance.
- 5. The system saves the room and records the action.

Alternative Flows:

- A1. Delete room with active occupants/contracts: the system rejects deletion.
- A2. Reduce capacity below current occupancy: the system rejects the change.
- A3. Mark occupied room under maintenance: the system requires a relocation plan or confirmation according to policy.

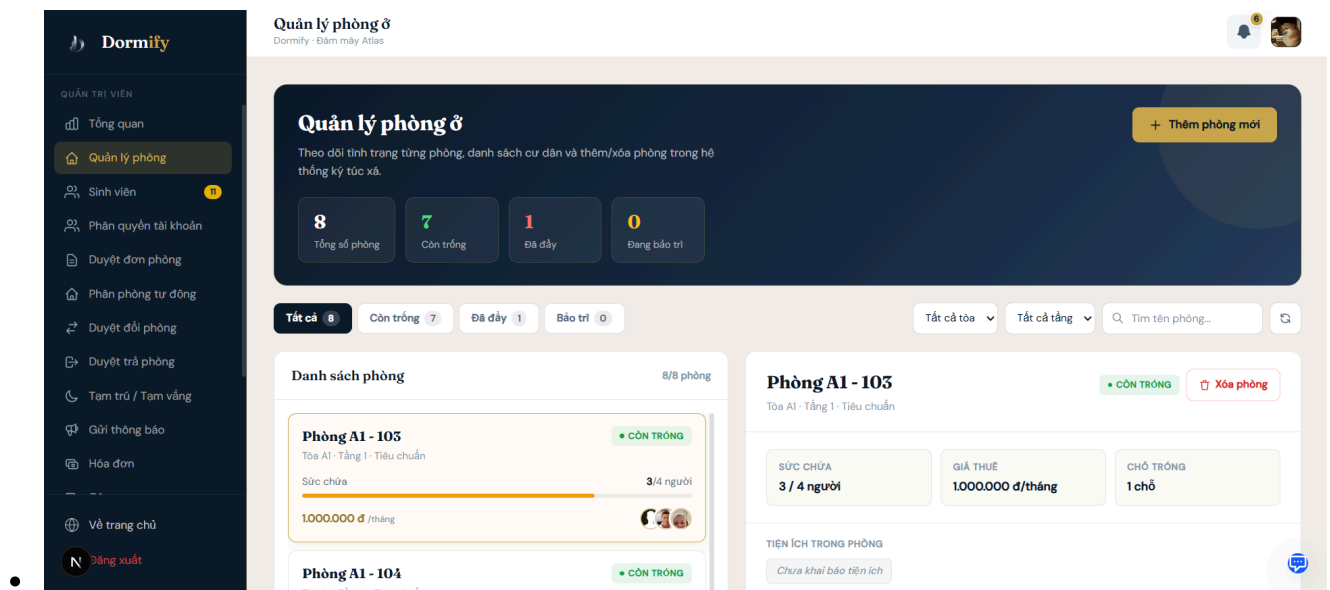
Postconditions:

- Room information and status are updated.

Special Requirements:

- Room lists should support building and floor filters.

Prototype Screens:





Thêm phòng mới

Phòng mới sẽ ở trạng thái "Còn trống" và sẵn sàng nhận đăng ký

Tên / Số phòng *

Ví dụ: P101

Tòa nhà *

Ví dụ: A1

Tầng *

1

Sức chứa tối đa *

4

Giá thuê (VNĐ/tháng) *

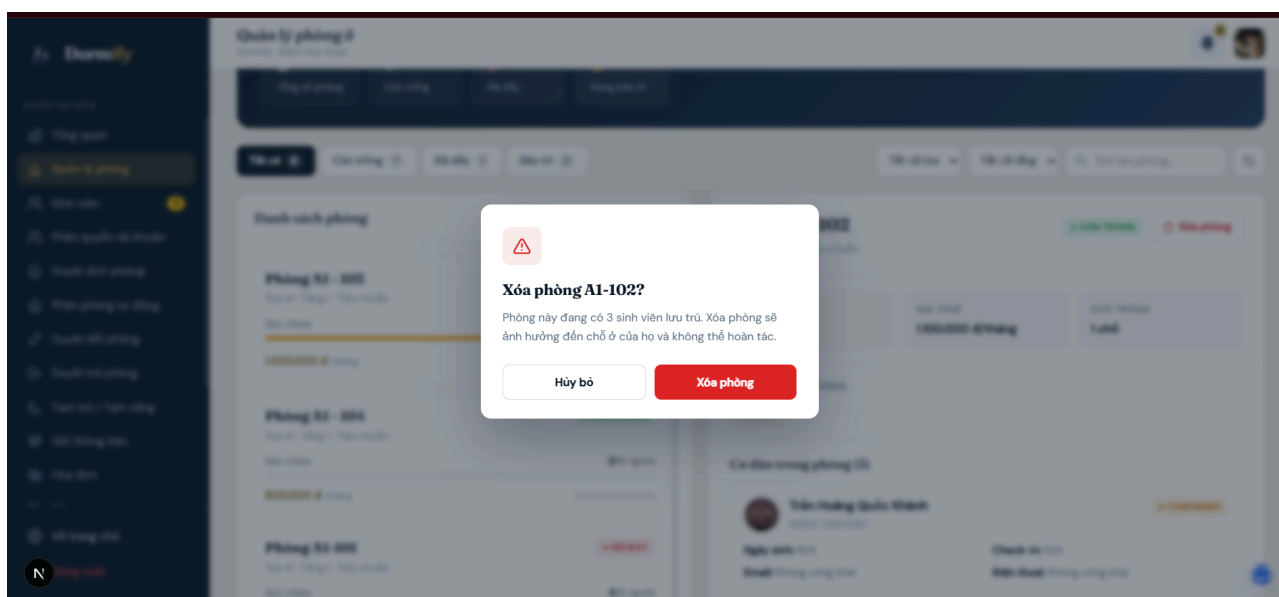
Ví dụ: 1200000

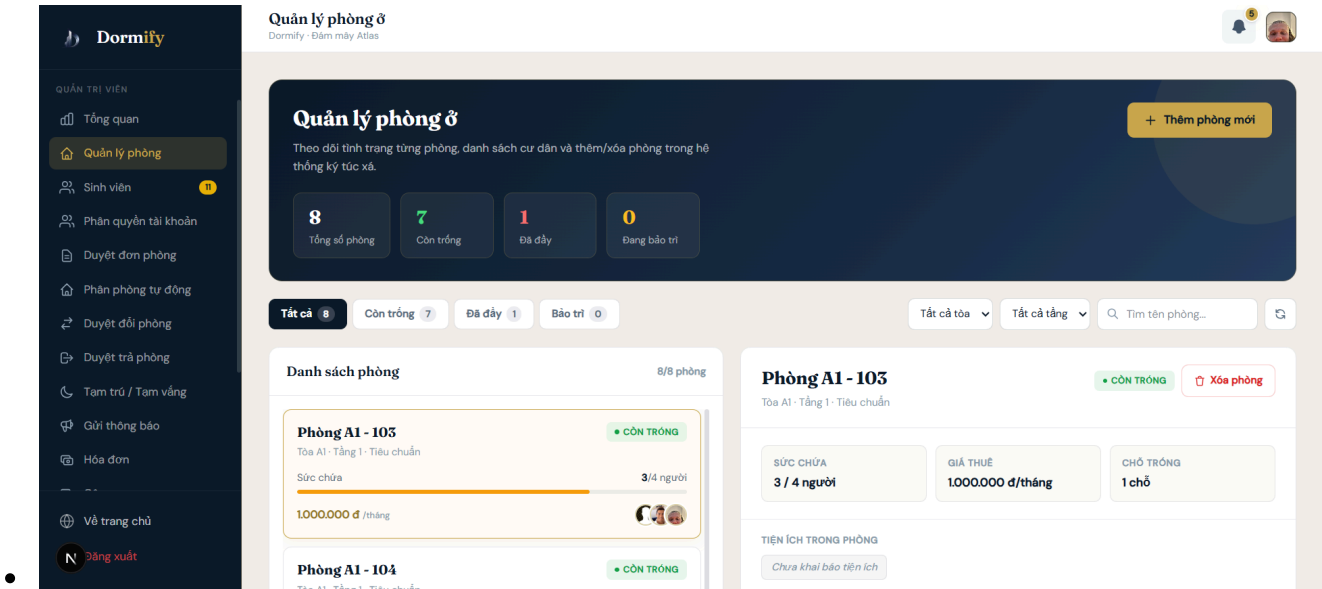
Tiện ích (cách nhau bằng dấu phẩy)

Ví dụ: Máy lạnh, Tủ lạnh, Tủ đồ cá nhân

Hủy bỏ

✓ Lưu phòng mới





- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ROOM-01.

Relationships:

- None.

Traceability: FR09, FM03

UC-STU-01 — Manage Student Records

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-STU-01

Use-case name: Manage Student Records

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager views, filters, and updates student records and reviews room-rental history.

Trigger: The manager opens Student Management.

Preconditions:

1. The actor is authenticated as Dormitory Manager.

Basic Flow (Main Success Scenario):

1. The system lists students with identity, contact, room, contract, and status summaries.
2. The manager searches or filters by building, floor, room, status, or student identifier.
3. The manager opens a student record and reviews residence history.
4. The manager edits authorized identity/contact fields and submits.
5. The system validates and saves the changes and records an audit entry.

Alternative Flows:

- A1. Duplicate citizen/student identifier: the update is rejected.
- A2. Student has historical records: deletion is blocked; deactivation is offered.
- A3. Unauthorized field or action: the system denies access.

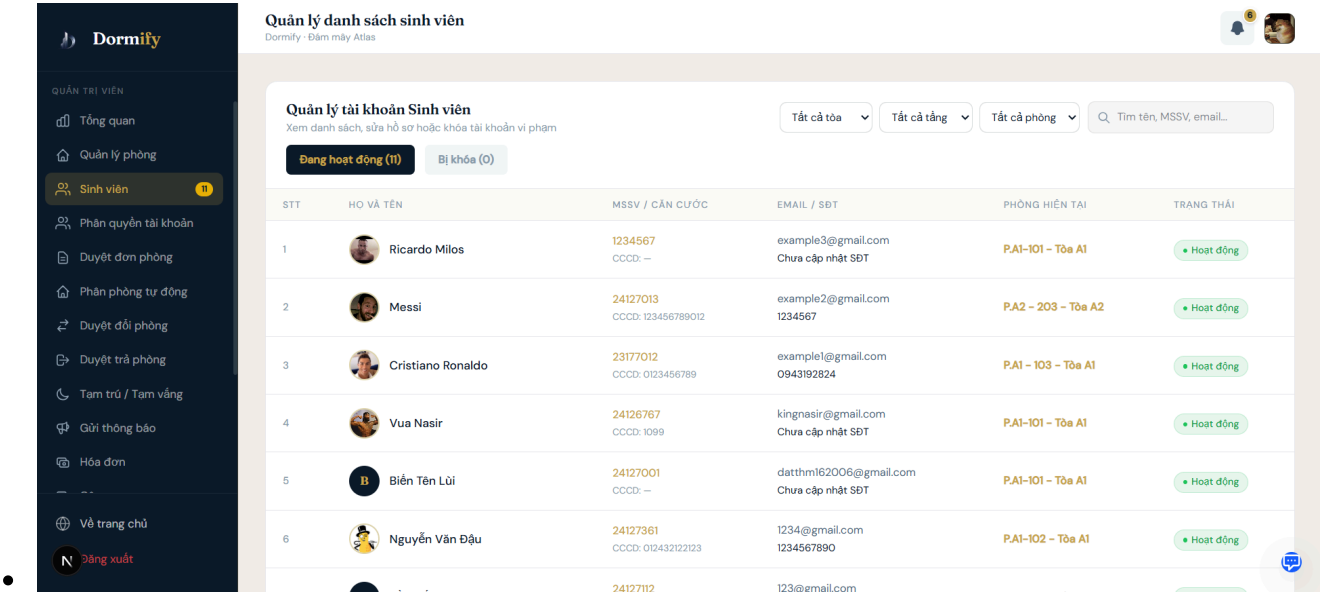
Postconditions:

- The student record is updated or remains unchanged after a failed validation.

Special Requirements:

- None.

Prototype Screens:



Hồ sơ Sinh viên

Thông tin cá nhân

Quản lý quyền & Lời nhắn

Điểm hành vi



Chọn Ảnh

Xóa Ảnh

Tài khoản Email

1234@gmail.com

Họ và Tên

Nguyễn Văn Đậu

Số điện thoại

1234567890

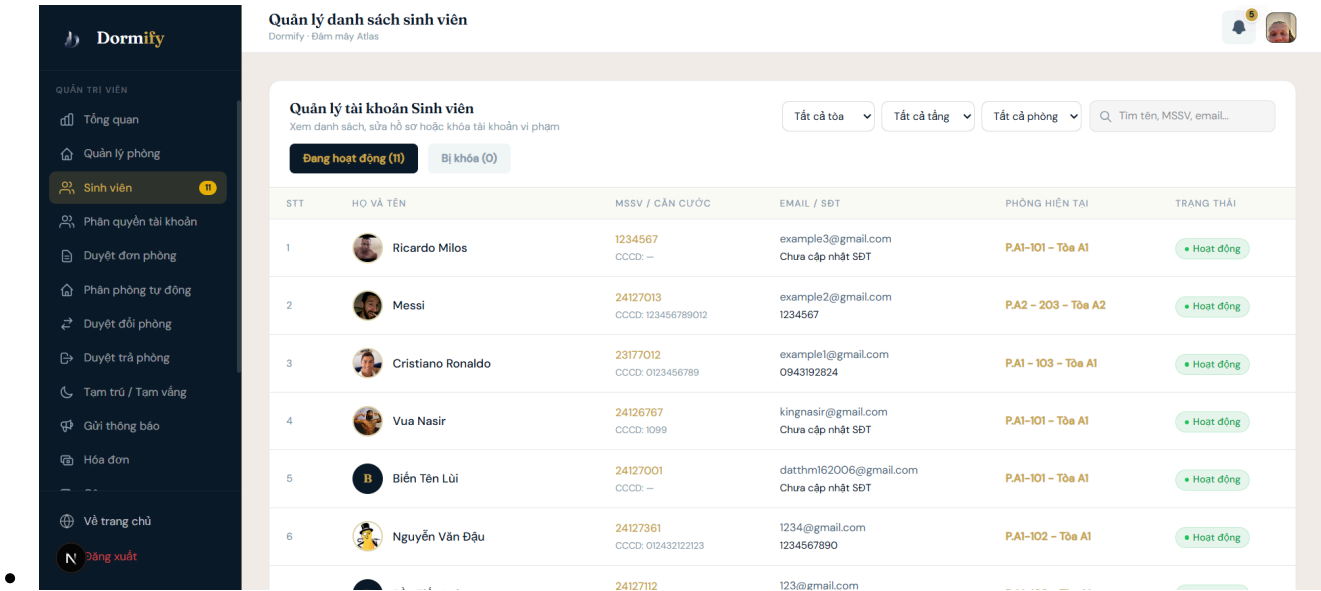
Số CCCD

012432122123

Xóa sinh viên

Hủy bỏ

Lưu thay đổi



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-STU-01.

Relationships:

- None.

Traceability: FR03, FR10, FM01, FM02

UC-ROOM-02 — Search Available Rooms

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-ROOM-02

Use-case name: Search Available Rooms

Actor(s): Student

Supporting Actor(s): None

Description: A student searches for rooms with available beds using accommodation filters.

Trigger: The student opens Room Search.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system displays available rooms.
2. The student filters by room type, building, price range, gender policy, and number of available beds.
3. The system refreshes the result list and shows matching rooms.

Alternative Flows:

- A1. No rooms match: the system displays an empty state and allows filters to be cleared.
- A2. Room availability changes during search: the system refreshes the latest capacity before application.

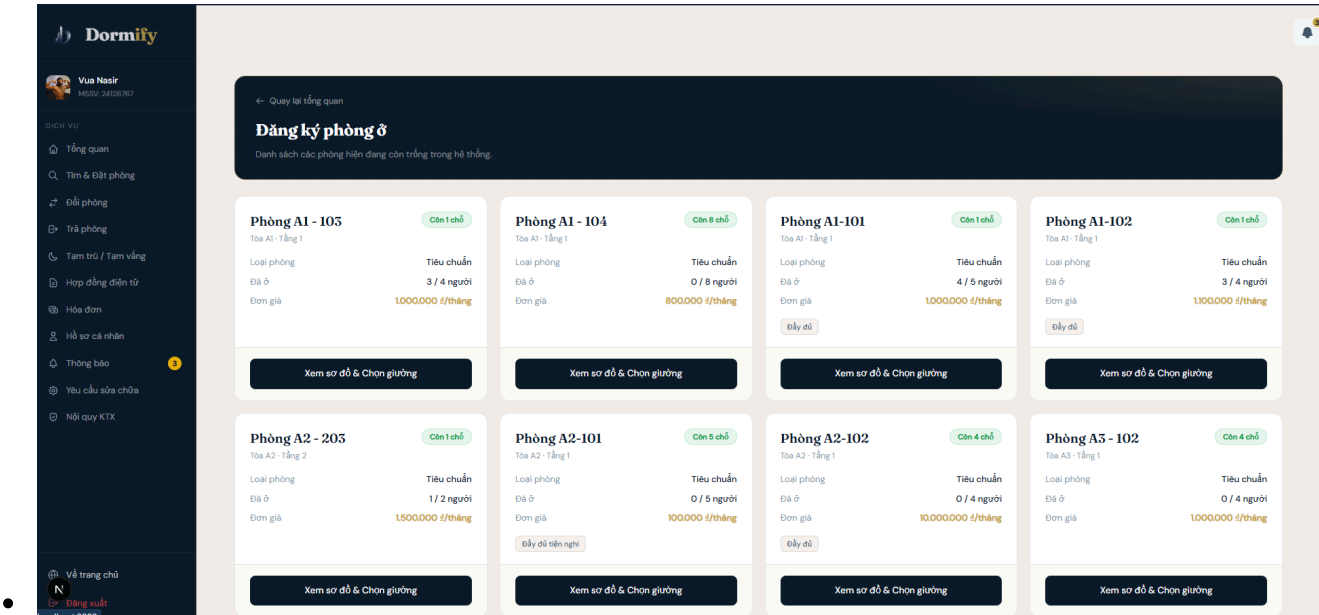
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ROOM-02.

Relationships:

- None.

Traceability: ST07

UC-ROOM-03 — View Room Details

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-ROOM-03

Use-case name: View Room Details

Actor(s): Student

Supporting Actor(s): None

Description: A student views the details of a room before submitting an application.

Trigger: The student selects a room from search results.

Preconditions:

1. The room exists.

Basic Flow (Main Success Scenario):

1. The system displays building, floor, room type, gender policy, price, facilities, capacity, occupancy, and available beds.
2. The system displays current availability and applicable rules.
3. The student may proceed to submit an application.

Alternative Flows:

- A1. Room no longer available: the system disables application and suggests returning to search.
- A2. Room removed or unavailable: the system displays an error and returns to the room list.

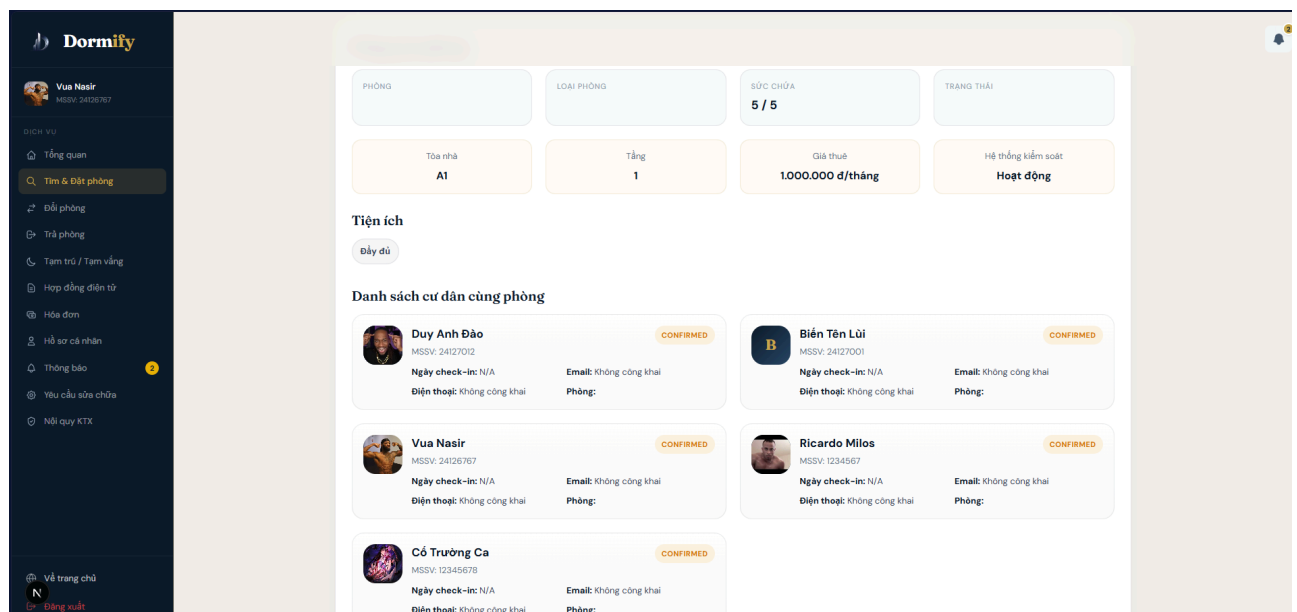
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ROOM-03.

Relationships:

- Included by UC-ROOM-04 Submit Room Application.

Traceability: ST08

UC-ROOM-04 — Submit Room Application

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-ROOM-04

Use-case name: Submit Room Application

Actor(s): Student

Supporting Actor(s): None

Description: A student applies to rent a selected available room.

Trigger: The student selects Apply for this room.

Preconditions:

1. The student is authenticated.

- 2. The student has viewed the room details.
- 3. The student has no active room assignment and no conflicting pending application.

Basic Flow (Main Success Scenario):

- 1. The system revalidates room availability and eligibility.
- 2. The student confirms accommodation preferences and application information.
- 3. The system creates a pending room application.
- 4. The system notifies the Dormitory Manager.
- 5. The system displays the application reference and status.

Alternative Flows:

- A1. Room becomes full: the application is not created and the student returns to search.
- A2. Existing room or pending application: the system rejects the duplicate/conflicting request.
- A3. Ineligible gender or room policy: the system explains the restriction.

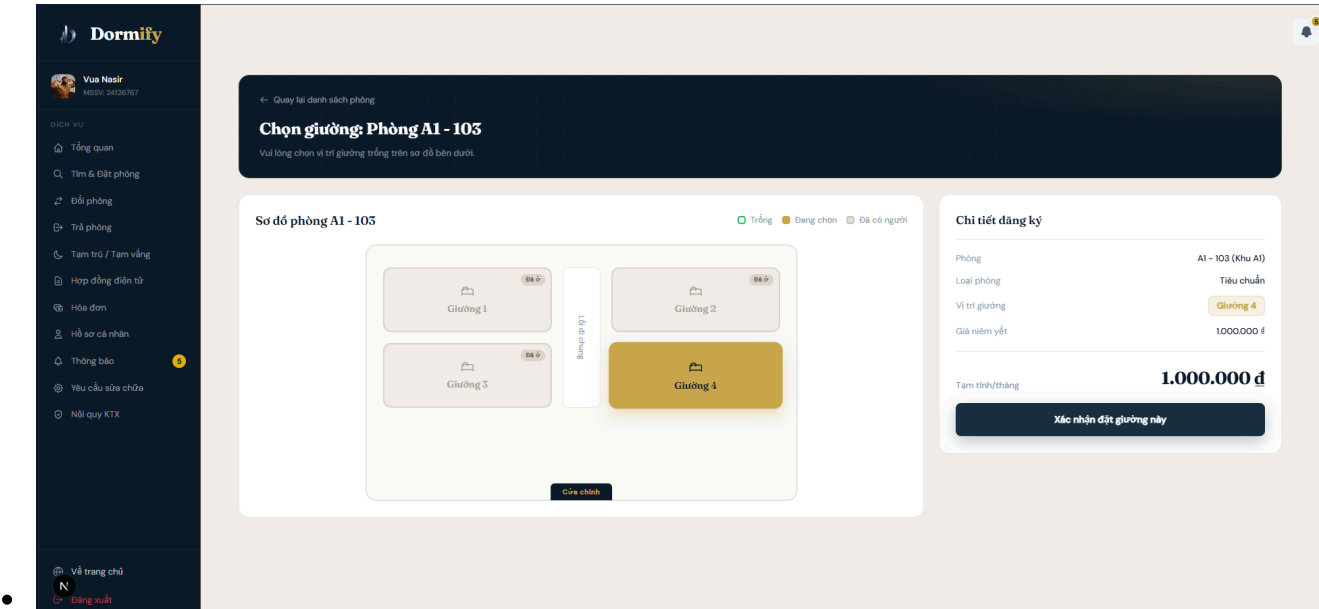
Postconditions:

- A pending room application exists.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ROOM-04.

Relationships:

- Includes UC-ROOM-03 View Room Details.

Traceability: FR11, ST09

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-ROOM-05

Use-case name: Track Room Application Status

Actor(s): Student

Supporting Actor(s): None

Description: A student reviews current and previous room application decisions.

Trigger: The student opens My Room Applications.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system retrieves the student's applications.
2. The system displays pending, approved, rejected, or cancelled status with submission and decision dates.
3. The student opens an application to view room and decision details.

Alternative Flows:

- A1. No applications: the system provides a link to room search.
- A2. Pending application cancellation is allowed: the student confirms cancellation and the system marks it cancelled.

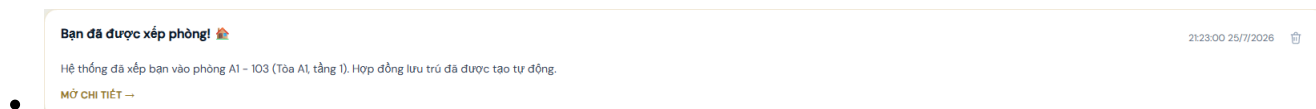
Postconditions:

- No data is changed unless a pending application is cancelled.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ROOM-05.

Relationships:

- None.

Traceability: FR11, ST10

UC-ROOM-06 — Review Room Application

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-ROOM-06

Use-case name: Review Room Application

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager reviews an applicant profile and approves or rejects a room application.

Trigger: A pending application appears in the manager queue.

Preconditions:

1. The actor is authenticated as Dormitory Manager.
2. The application is pending.

Basic Flow (Main Success Scenario):

1. The system displays the application, applicant profile, selected room, and current capacity.
2. The manager checks eligibility and supporting information.
3. The manager selects Approve or Reject and optionally enters a note.
4. For approval, the system reserves a bed, creates the room assignment and rental contract, and updates occupancy atomically.
5. The system updates the application status and notifies the student.

Alternative Flows:

- A1. Room is full at approval time: approval is rejected and the manager may select another room or reject the application.
- A2. Applicant becomes ineligible or already assigned: the system blocks approval.
- A3. Rejection: no room or contract is created.

Postconditions:

- The application is approved or rejected.
- Approved students receive an assignment and contract.

Special Requirements:

- Approval must prevent over-capacity under concurrent requests.

Prototype Screens:

Dormify

Duyệt đơn lưu trú
Dormify · Đám mây Atlas

Danh sách Đơn đăng ký lưu trú

Tất cả tòa | Tất cả tầng | Tất cả phòng

SINH VIÊN	PHÒNG ĐĂNG KÝ	NGÀY GỬI	TRẠNG THÁI	HÀNH ĐỘNG
Nguyễn Văn A MSSV: 123456789	A1 - 103 Tòa A1	16/7/2026	APPROVED	Đã xử lý xong
Em Linh Hà Lan MSSV: 24127022	A1 - 103 Tòa A1	16/7/2026	APPROVED	Đã xử lý xong
Cristiano Ronaldo MSSV: 23177012	A1 - 103 Tòa A1	16/7/2026	APPROVED	Đã xử lý xong
Cố Trường Ca MSSV: 12345678	A1-101 Tòa A1	9/7/2026	APPROVED	Đã xử lý xong
Ricardo Milos MSSV: 1234567	A1-101 Tòa A1	7/7/2026	APPROVED	Đã xử lý xong
Messi MSSV: 24127022	A2-101 Tòa A2	28/6/2026	APPROVED	Đã xử lý xong

- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ROOM-06.

Relationships:

- None.

Traceability: FR11

UC-ROOM-07 — Run Automatic Room Allocation

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-ROOM-07

Use-case name: Run Automatic Room Allocation

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager automatically allocates many eligible students to rooms based on preferences, gender, capacity, and availability.

Trigger: The manager starts an automatic allocation run.

Preconditions:

1. The actor is authenticated as Dormitory Manager.
2. Eligible unassigned students and available beds exist.

Basic Flow (Main Success Scenario):

1. The system previews candidates and available capacity.
2. The manager selects the candidate set and confirms allocation rules.
3. The system orders candidates according to the configured priority.
4. For each candidate, the system selects a compatible room and reserves a bed.
5. The system creates approved applications/assignments and rental contracts.
6. The system returns a result report and notifies assigned students.

Alternative Flows:

- A1. No compatible room for a student: the system skips the student and records the reason.
- A2. Capacity changes during the run: the system retries another compatible room or skips the candidate.
- A3. Manager cancels before confirmation: no assignment is made.

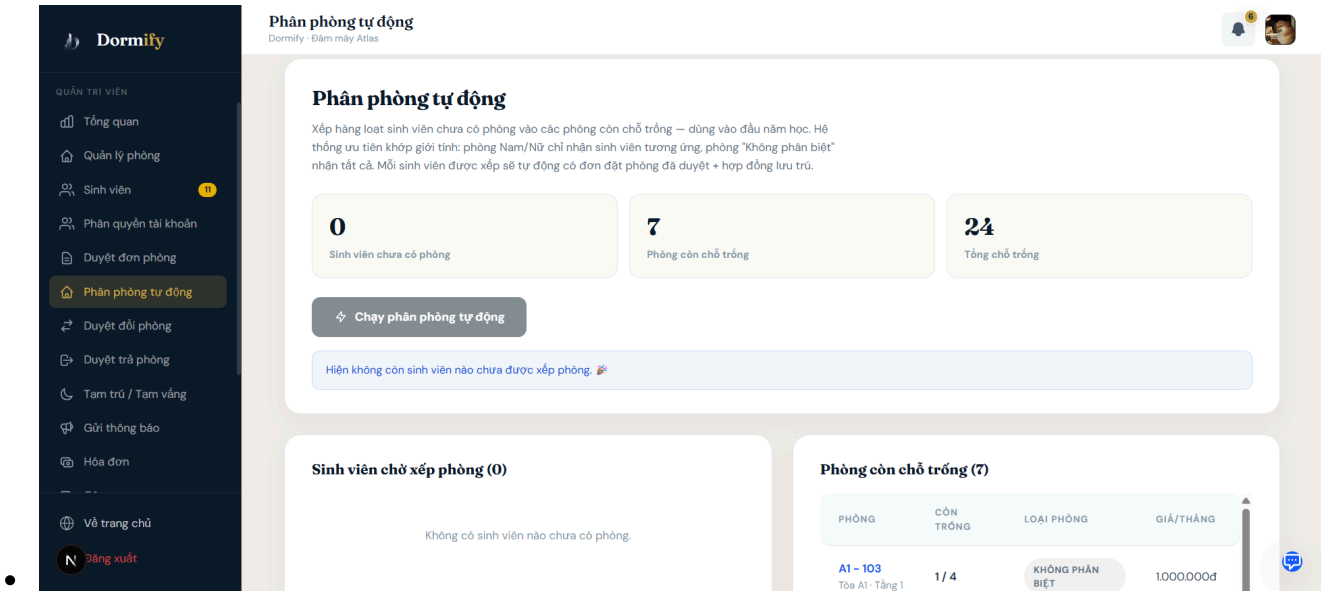
Postconditions:

- Zero or more students are assigned without exceeding capacity.
- An allocation report is stored or available for review.

Special Requirements:

- Allocation must be deterministic or auditable for the same inputs and rule set.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ROOM-07.

Relationships:

- None.

Traceability: FR12

UC-ROOM-08 — Submit Room Transfer Request

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-ROOM-08

Use-case name: Submit Room Transfer Request

Actor(s): Student

Supporting Actor(s): None

Description: A resident requests transfer from the current room to another suitable room.

Trigger: The student selects Request Room Transfer.

Preconditions:

1. The student is authenticated.
2. The student has an active room assignment and contract.
3. No conflicting pending transfer request exists.

Basic Flow (Main Success Scenario):

1. The student selects a preferred destination room or transfer preferences.
2. The student enters a reason and submits.
3. The system validates destination availability and policy constraints.
4. The system creates a pending transfer request and notifies the Dormitory Manager.

Alternative Flows:

- A1. Destination is the current room: the system rejects the request.
- A2. No available destination: the student may submit preferences without a fixed room if policy allows.

- A3. Existing pending transfer: the duplicate request is rejected.

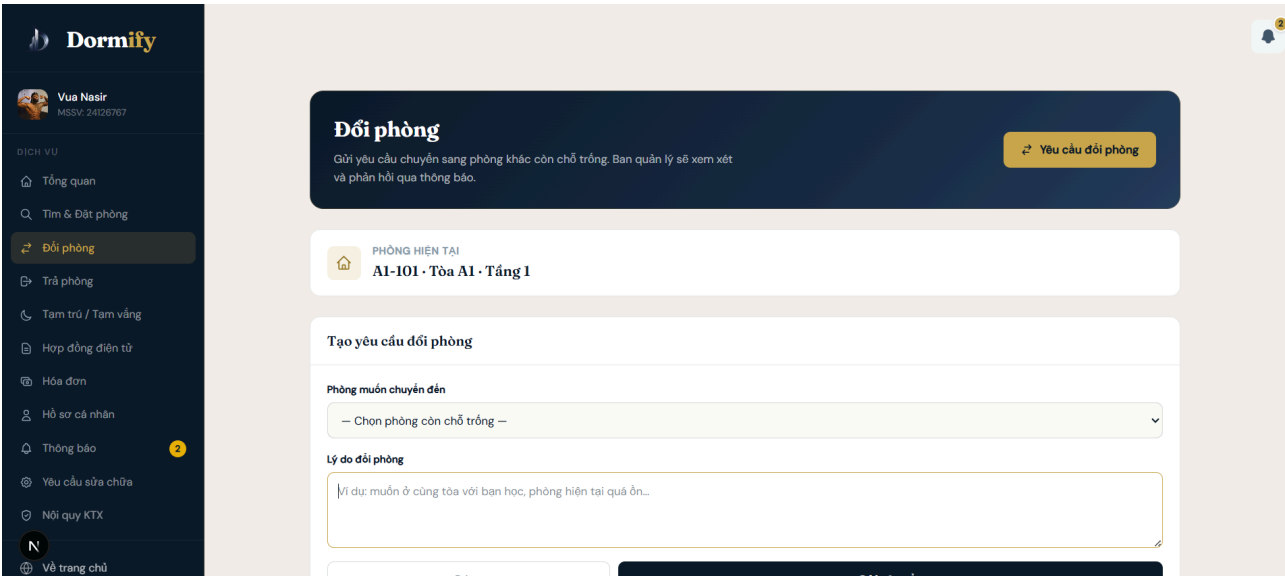
Postconditions:

- A pending transfer request exists.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ROOM-08.

Relationships:

- None.

Traceability: FR13

UC-ROOM-09 — Track Room Transfer Status

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-ROOM-09

Use-case name: Track Room Transfer Status

Actor(s): Student

Supporting Actor(s): None

Description: A student reviews the processing status and result of room transfer requests.

Trigger: The student opens My Transfer Requests.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system displays transfer requests and their pending, approved, rejected, or cancelled status.

2. The student opens a request to view source room, destination, reason, notes, and decision date.

Alternative Flows:

- A1. Pending request may be cancelled according to policy.
- A2. No transfer history: the system displays an empty state.

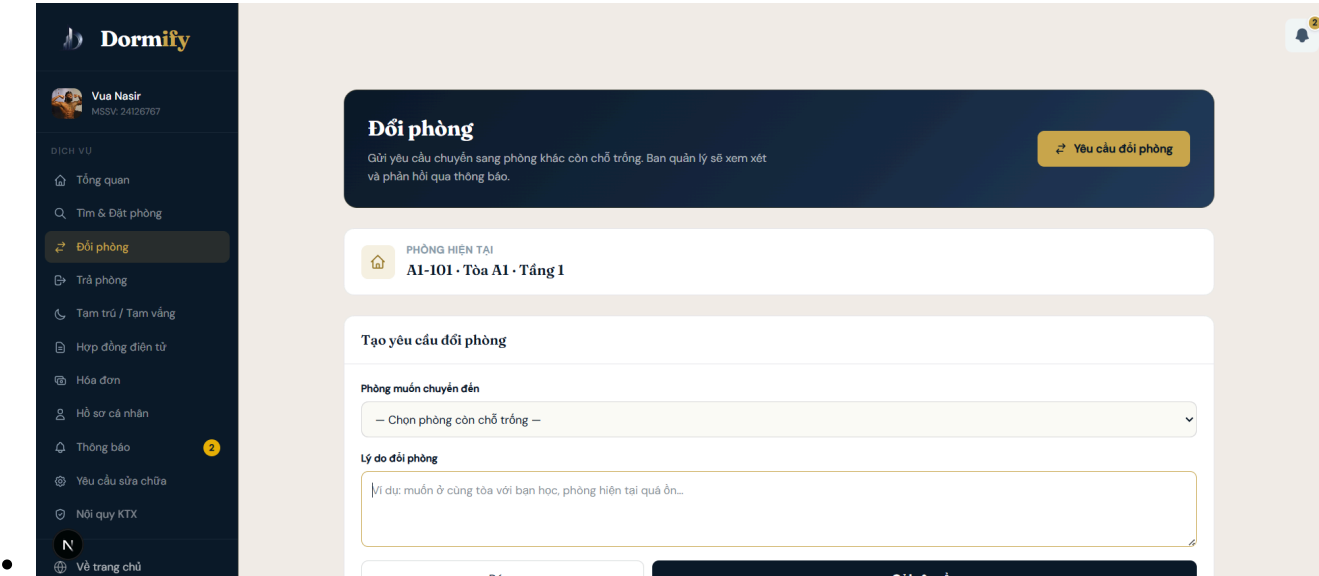
Postconditions:

- No data is changed unless cancellation is performed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ROOM-09.

Relationships:

- None.

Traceability: FR13

UC-ROOM-10 — Review Room Transfer Request

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-ROOM-10

Use-case name: Review Room Transfer Request

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager approves or rejects a transfer and records the transfer history.

Trigger: A pending transfer request appears in the manager queue.

Preconditions:

1. The actor is authenticated as Dormitory Manager.
2. The student has an active assignment.
3. The request is pending.

Basic Flow (Main Success Scenario):

1. The system displays the student, source room, requested destination/preferences, and reason.
2. The manager checks destination capacity and compatibility.
3. The manager approves with a destination room or rejects with a note.
4. On approval, the system releases the old bed, reserves the new bed, updates the assignment/contract as required, and records transfer history atomically.
5. The system updates the request and notifies the student.

Alternative Flows:

- A1. Destination fills before approval: the system requests another room.
- A2. Student no longer occupies the source room: the request is invalidated.
- A3. Rejection: existing assignment remains unchanged.

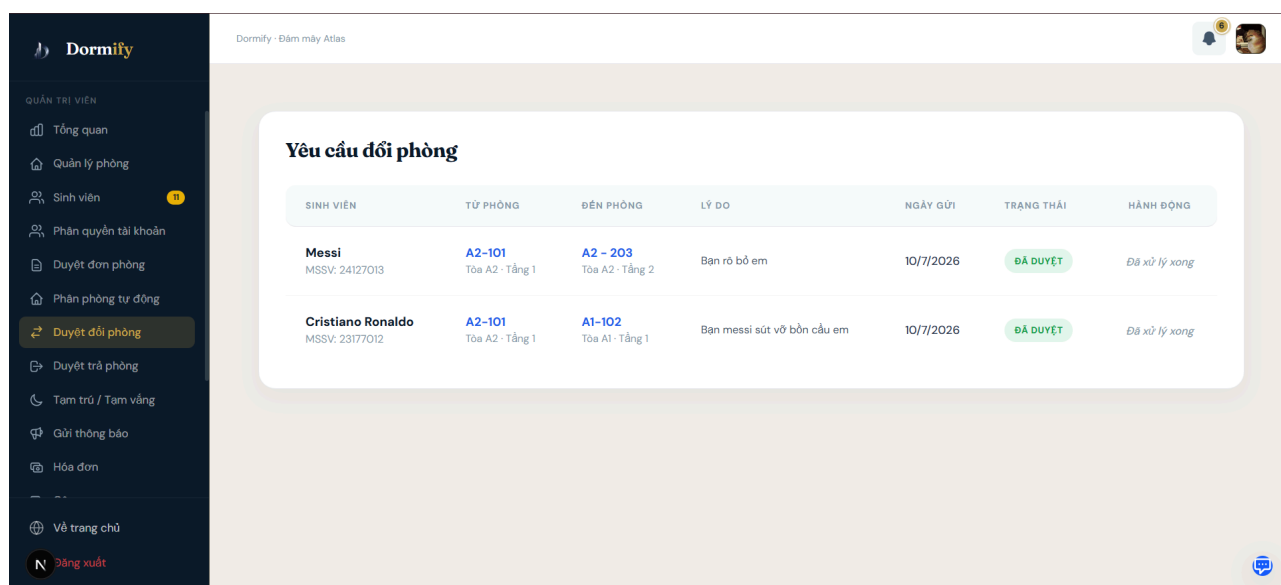
Postconditions:

- Approved transfer changes the active room and creates a history record; rejected transfer makes no room change.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-ROOM-10.

Relationships:

- None.

Traceability: FR13

5. Contract Management

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Creation, extension, liquidation, viewing, and PDF export of rental contracts.

ID	Use case	Primary actor
UC-CON-01	Create Rental Contract	Dormitory Manager
UC-CON-02	Extend Rental Contract	Dormitory Manager
UC-CON-03	Liquidate Rental Contract	Dormitory Manager
UC-CON-04	Export Contract PDF	Dormitory Manager
UC-CON-05	View Rental Contract	Student
UC-CON-06	Download Contract PDF	Student

UC-CON-01 — Create Rental Contract

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-CON-01

Use-case name: Create Rental Contract

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager creates a rental contract after a room application or automatic allocation has been approved.

Trigger: An approved assignment requires a contract.

Preconditions:

1. The student has an approved room assignment.
2. No conflicting active contract exists.

Basic Flow (Main Success Scenario):

1. The system pre-fills student, room, rental fee, deposit, start date, and standard terms.
2. The manager reviews and completes required contract data.
3. The system validates dates, fees, and uniqueness.
4. The manager confirms creation.
5. The system stores an active contract and associates it with the student and room.
6. The student is notified that the contract is available.

Alternative Flows:

- A1. Conflicting active contract: creation is rejected.
- A2. Invalid date range or fee: correction is required.
- A3. Manager cancels: no contract is created.

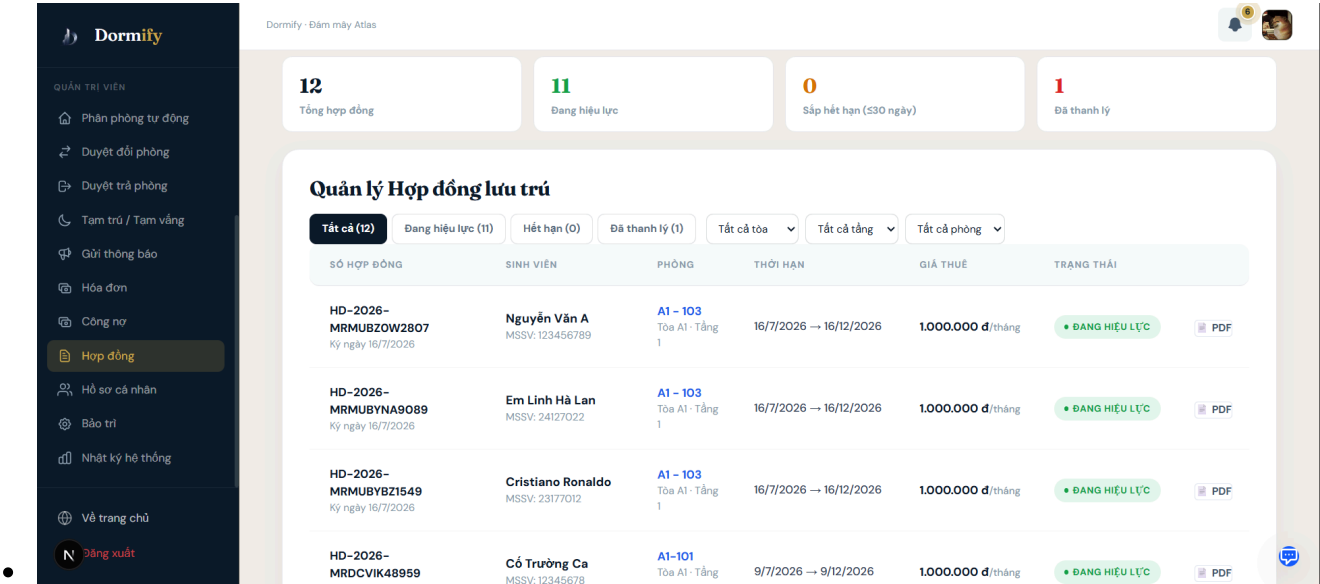
Postconditions:

- An active rental contract exists.

Special Requirements:

- Contract number must be unique and immutable.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-CON-01.

Relationships:

- None.

Traceability: FR14

UC-CON-02 — Extend Rental Contract

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-CON-02

Use-case name: Extend Rental Contract

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager extends the end date of an eligible active rental contract.

Trigger: The manager selects Extend contract.

Preconditions:

1. The actor is authenticated as Dormitory Manager.
2. The contract is active and eligible for extension.

Basic Flow (Main Success Scenario):

1. The system displays current contract dates and room information.
2. The manager enters the extension period or new end date.
3. The system checks policy, room availability, and date validity.

- 4. The manager confirms.
- 5. The system updates the contract and records the extension history.
- 6. The student is notified.

Alternative Flows:

- A1. Contract already terminated/expired beyond policy: extension is rejected.
- A2. Room unavailable for the extension period: the system requires another arrangement.
- A3. Invalid date: the system requests correction.

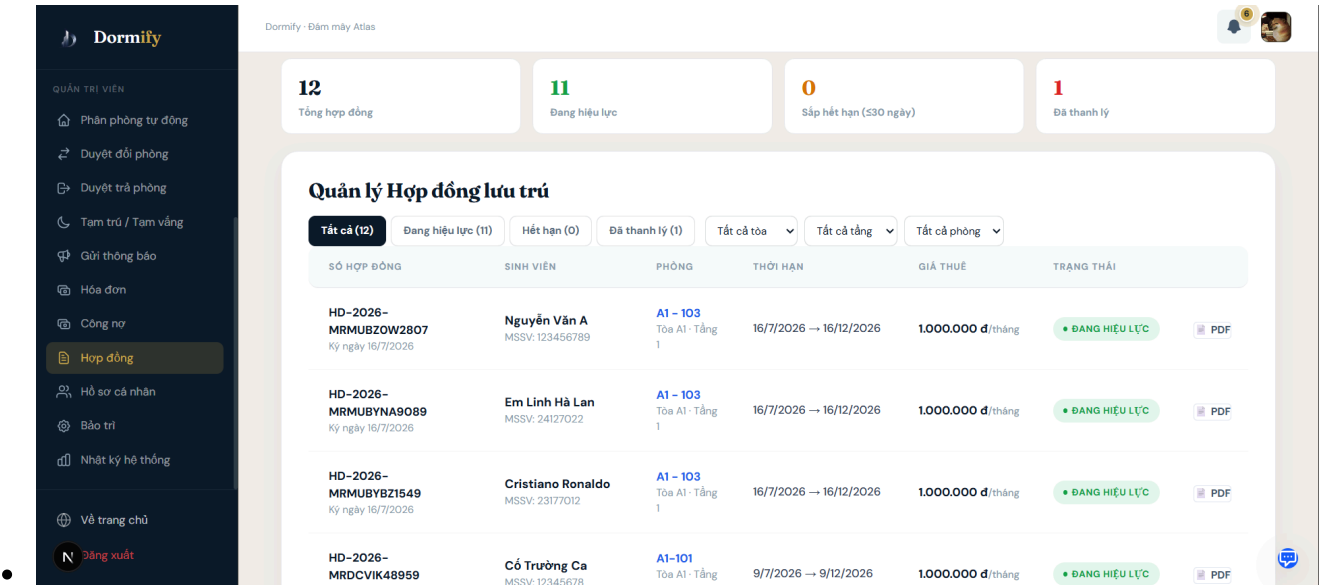
Postconditions:

- The contract end date and extension history are updated.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-CON-02.

Relationships:

- None.

Traceability: FR14, FR15

UC-CON-03 — Liquidate Rental Contract

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-CON-03
Use-case name: Liquidate Rental Contract
Actor(s): Dormitory Manager
Supporting Actor(s): None

Description: The manager formally terminates a rental contract and records the liquidation.

Trigger: The manager selects Liquidate contract or checkout completion invokes this use case.

Preconditions:

1. The actor is authenticated as Dormitory Manager or the use case is invoked by checkout completion.
2. An active contract exists.

Basic Flow (Main Success Scenario):

1. The system displays contract obligations, outstanding invoices, and related checkout data.
2. The manager confirms the effective termination date and reason.
3. The system validates that required checkout/settlement conditions are satisfied.
4. The system changes the contract status to liquidated/terminated and records the event.
5. The system releases the room assignment when appropriate and notifies the student.

Alternative Flows:

- A1. Outstanding mandatory settlement: liquidation is blocked or marked pending settlement according to policy.
- A2. Already terminated contract: the system returns the existing status without duplicate processing.

Postconditions:

- The contract is no longer active.
- The liquidation history is stored.

Special Requirements:

- None.

Prototype Screens:

The screenshot shows the Dormify web application interface. The top navigation bar includes the Dormify logo and the text 'Dormify - Đám mây Atlas'. The main content area is titled 'Quản lý Hợp đồng lưu trú' (Manage Rental Contracts). It features a sidebar with various navigation options, including 'Hợp đồng' (Contracts) which is currently selected. The main content area displays a table of rental contracts with the following columns: SỔ HỢP ĐỒNG (Contract ID), SINH VIÊN (Student), PHÒNG (Room), THỜI HẠN (Term), GIÁ THUÊ (Rent), and TRẠNG THÁI (Status). The table lists four active contracts, all with a status of 'ĐANG HIỆU LỰC' (Active). The contracts are as follows:

SỔ HỢP ĐỒNG	SINH VIÊN	PHÒNG	THỜI HẠN	GIÁ THUÊ	TRẠNG THÁI
HD-2026-MRMUBZOW2807 Ký ngày 16/7/2026	Nguyễn Văn A MSSV: 123456789	A1 - 103 Tòa A1 - Tầng 1	16/7/2026 → 16/12/2026	1.000.000 đ/tháng	ĐANG HIỆU LỰC
HD-2026-MRMUBYN9089 Ký ngày 16/7/2026	Em Linh Hà Lan MSSV: 24127022	A1 - 103 Tòa A1 - Tầng 1	16/7/2026 → 16/12/2026	1.000.000 đ/tháng	ĐANG HIỆU LỰC
HD-2026-MRMUBYBZ1549 Ký ngày 16/7/2026	Cristiano Ronaldo MSSV: 23177012	A1 - 103 Tòa A1 - Tầng 1	16/7/2026 → 16/12/2026	1.000.000 đ/tháng	ĐANG HIỆU LỰC
HD-2026-MRDCVIK48959	Cố Trường Ca MSSV: 12345678	A1-101 Tòa A1 - Tầng	9/7/2026 → 9/12/2026	1.000.000 đ/tháng	ĐANG HIỆU LỰC

- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-CON-03.

Relationships:

- Included by UC-CHK-04 Refund Deposit and Complete Checkout.

Traceability: FR14, FR16

UC-CON-04 — Export Contract PDF

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-CON-04

Use-case name: Export Contract PDF

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager exports a contract as a formatted PDF for administration or signing.

Trigger: The manager selects Export PDF on a contract.

Preconditions:

1. The contract exists.

Basic Flow (Main Success Scenario):

1. The system retrieves the current contract and related student/room data.
2. The system renders the contract using the approved template.
3. The system provides the PDF for download or printing.
4. The export event may be recorded.

Alternative Flows:

- A1. Required data is missing: the system identifies the incomplete fields and does not generate an invalid document.
- A2. PDF generation fails: the system reports the error and preserves the contract.

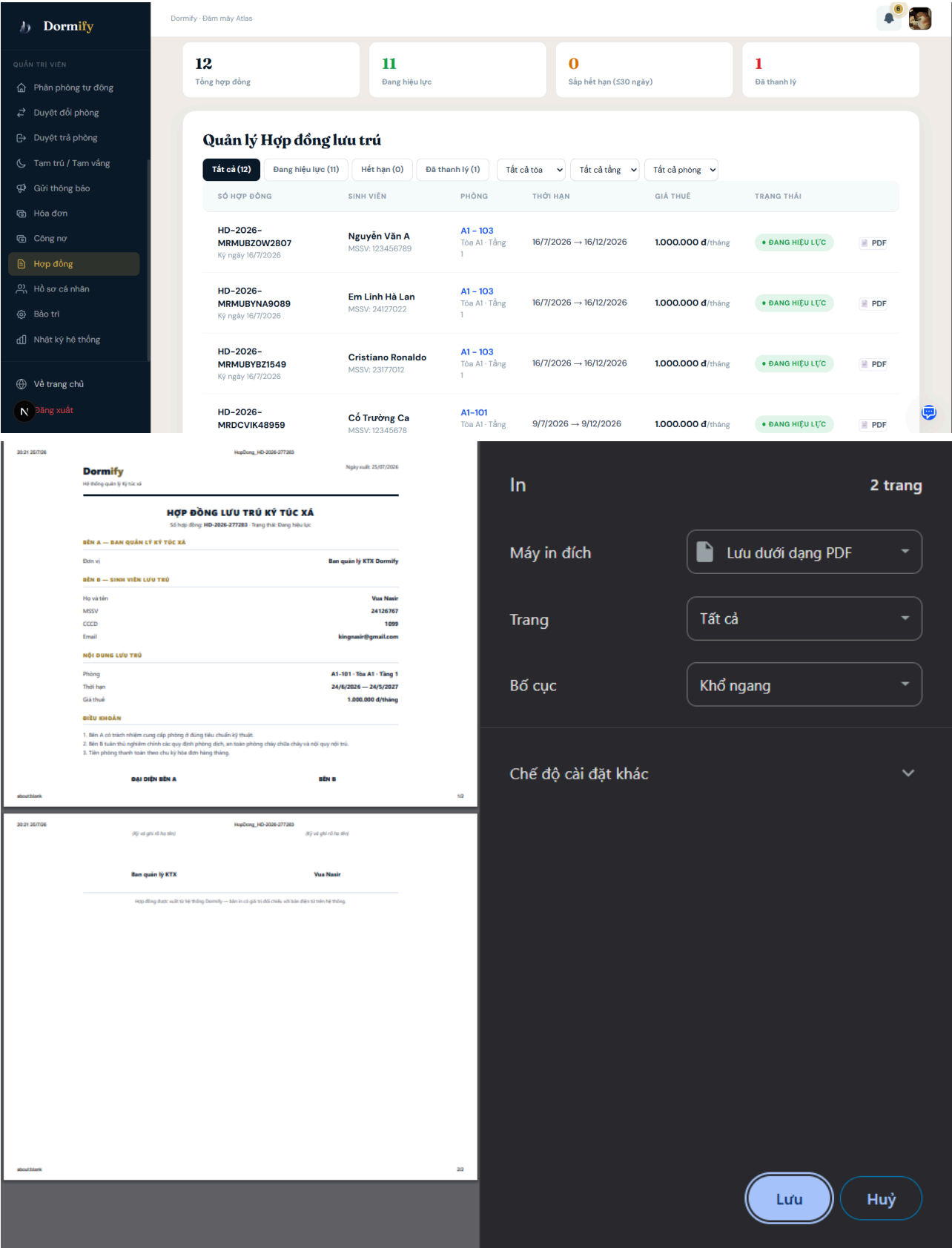
Postconditions:

- A PDF representation is generated; the contract data is unchanged.

Special Requirements:

- The PDF must preserve Vietnamese characters and display a contract number/version.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-CON-04.

Relationships:

- None.

Traceability: FR14, FR17

UC-CON-05 — View Rental Contract

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-CON-05

Use-case name: View Rental Contract

Actor(s): Student

Supporting Actor(s): None

Description: A student views the active and historical rental contract information.

Trigger: The student opens My Contract.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system retrieves contracts belonging to the student.
2. The system displays contract number, room, term, fees, deposit, status, and terms.
3. The student may select a historical contract.

Alternative Flows:

- A1. No contract exists: the system displays an empty state.
- A2. Only historical contracts exist: the system lists historical contracts and clearly indicates that no active contract is available.
- A3. Contract file unavailable: the system displays contract metadata and tells the student to contact dormitory management for the official document.

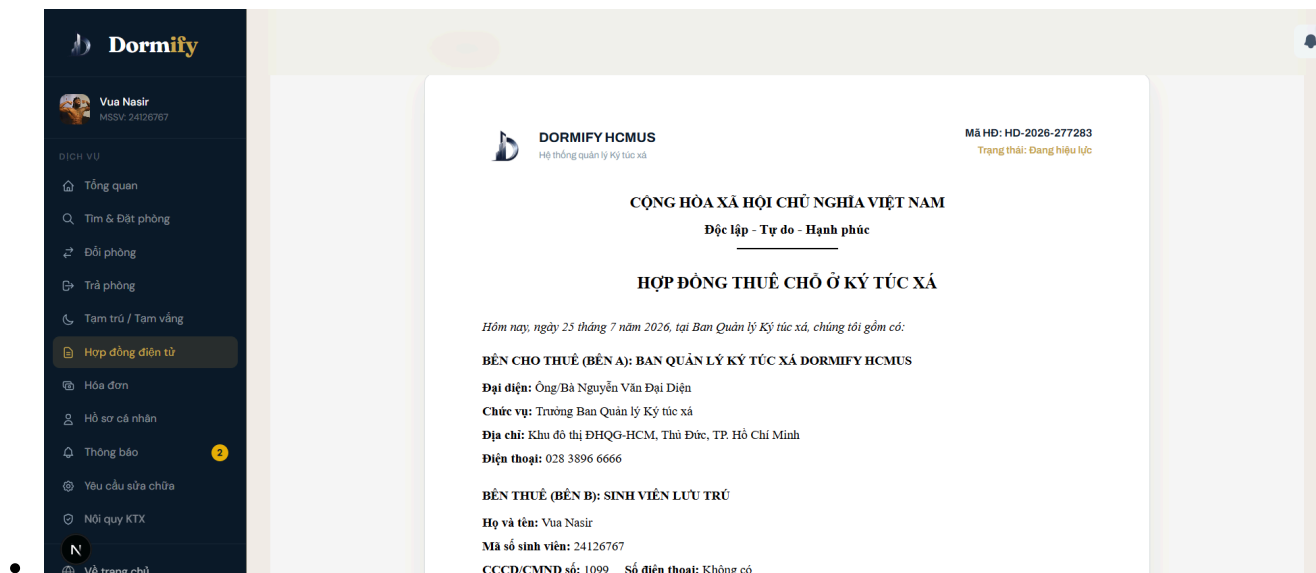
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-CON-05.

Relationships:

- None.

Traceability: ST11

UC-CON-06 — Download Contract PDF

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-CON-06

Use-case name: Download Contract PDF

Actor(s): Student

Supporting Actor(s): None

Description: A student downloads a PDF copy of a contract being viewed.

Trigger: The student selects Download PDF.

Preconditions:

1. The student is authenticated.
2. The selected contract belongs to the student.

Basic Flow (Main Success Scenario):

1. The system verifies ownership.
2. The system generates or retrieves the contract PDF.
3. The browser downloads the PDF.

Alternative Flows:

- A1. Unauthorized contract identifier: access is denied.
- A2. PDF generation unavailable: the system displays an error and keeps the contract view open.

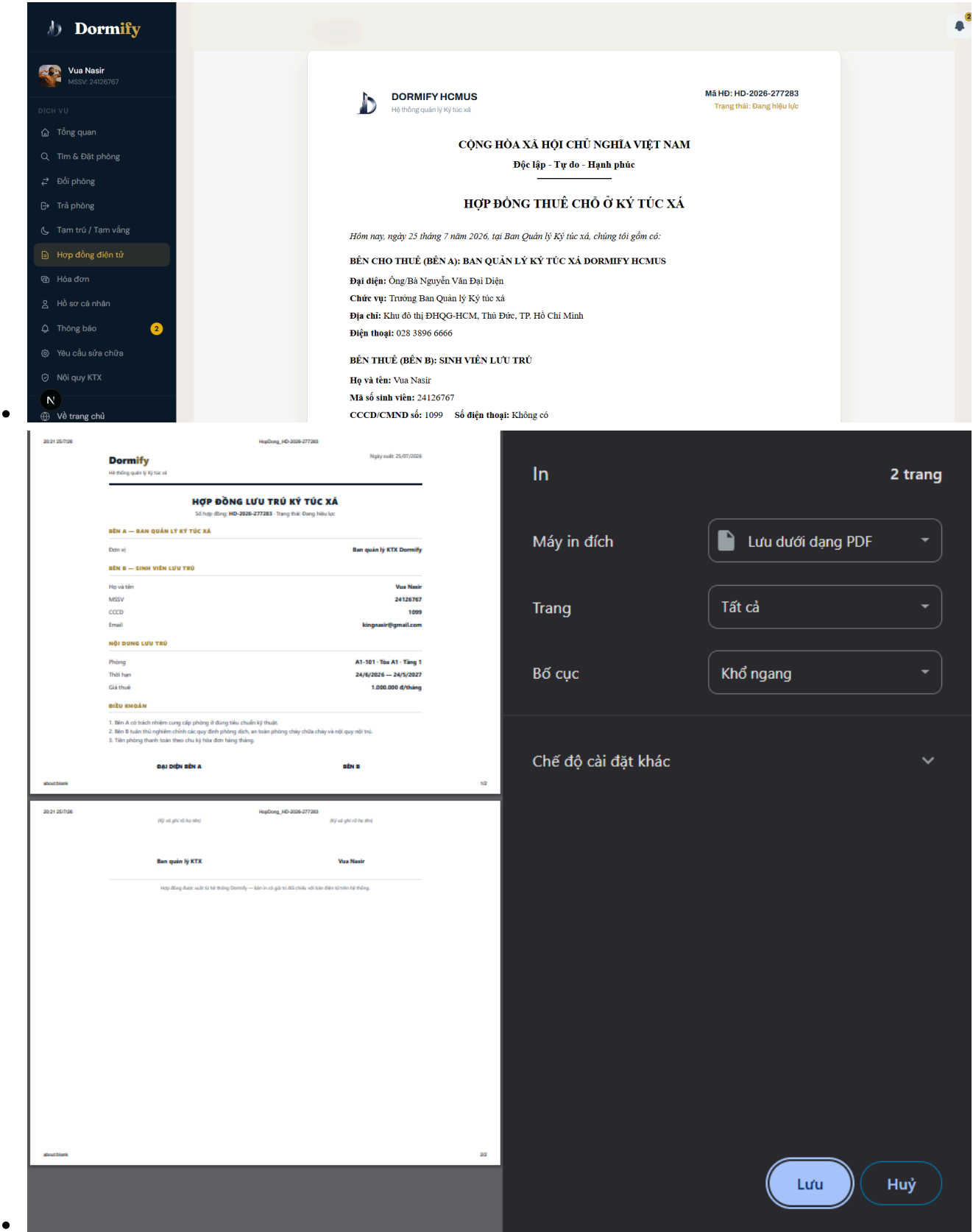
Postconditions:

- A PDF copy is delivered; no contract data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-CON-06.

Relationships:

- Extends UC-CON-05 View Rental Contract.

Traceability: FR17, ST12

6. Checkout and Deposit Refund

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Student checkout requests, inspection, compensation calculation, deposit refund, and stay closure.

ID	Use case	Primary actor
UC-CHK-01	Submit Checkout Request	Student
UC-CHK-02	Review Checkout Request	Dormitory Manager
UC-CHK-03	Calculate Compensation Fee	Dormitory Manager
UC-CHK-04	Refund Deposit and Complete Checkout	Dormitory Manager

UC-CHK-01 — Submit Checkout Request

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-CHK-01**Use-case name:** Submit Checkout Request**Actor(s):** Student**Supporting Actor(s):** None**Description:** A resident requests to leave the assigned room on an expected date.**Trigger:** The student selects Request Checkout.**Preconditions:**

1. The student is authenticated.
2. The student has an active room assignment and contract.
3. No pending checkout request exists.

Basic Flow (Main Success Scenario):

1. The student enters expected departure date and reason.
2. The system validates the date and active residence.
3. The system creates a pending checkout request.
4. The system notifies the Dormitory Manager.
5. The system displays the request reference and status.

Alternative Flows:

- A1. Departure date is in the past: the system requests a valid date.
- A2. No active room/contract: the request is rejected.
- A3. Existing pending checkout: the system directs the student to the existing request.

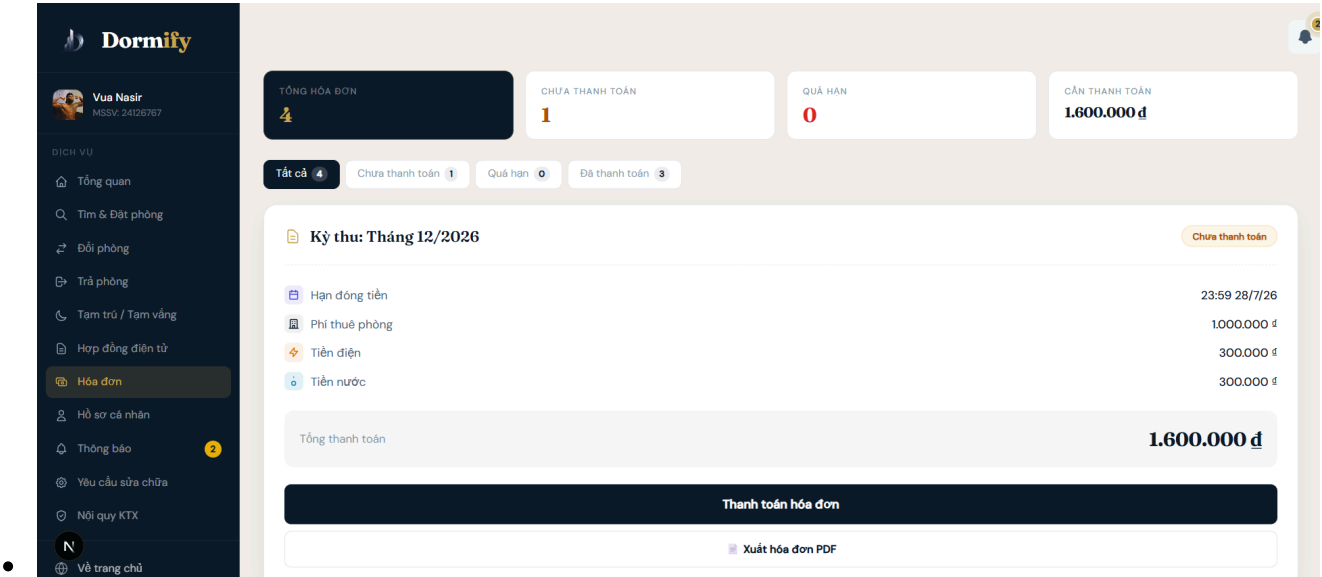
Postconditions:

- A pending checkout request exists.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-CHK-01.

Relationships:

- None.

Traceability: FR18

UC-CHK-02 — Review Checkout Request

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-CHK-02

Use-case name: Review Checkout Request

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager reviews the checkout request, plans inspection, and accepts or rejects processing.

Trigger: The manager opens a pending checkout request.

Preconditions:

1. The actor is authenticated as Dormitory Manager.
2. The checkout request is pending.

Basic Flow (Main Success Scenario):

1. The system displays the student, room, contract, departure date, and reason.

2. The manager verifies outstanding obligations and inspects room assets.
3. The manager records notes and decides to reject or proceed to completion.
4. If damage or missing assets are identified, UC-CHK-03 is invoked.
5. The system saves the review decision and notifies the student when rejected or scheduled.

Alternative Flows:

- A1. Reject request: status becomes rejected and no contract/room change occurs.
- A2. Inspection cannot be completed: the request remains pending with a scheduled follow-up.
- A3. No damage: the manager proceeds directly to UC-CHK-04 with zero compensation.

Postconditions:

- The request is rejected, remains pending for inspection, or is ready for completion.

Special Requirements:

- None.

Prototype Screens:

Hệ thống hóa đơn
Dormify - Đám mây Atlas

Quản lý Hệ thống Hóa đơn

PHÒNG	KỶ THU	TIỀN PHÒNG	ĐIỆN & NƯỚC	TỔNG CỘNG	TRẠNG THÁI	HÀNH ĐỘNG
A1-101 Tòa A1	Tháng 12/2026	1.000.000 ₫	⚡ 300.000 ₫ 💧 300.000 ₫	1.600.000 ₫	CHỜ THU	Xác nhận Thu PDF
A1 - 103 Tòa A1	Tháng 8/2026	1.000.000 ₫	⚡ 100.000 ₫ 💧 100.000 ₫	1.200.000 ₫	ĐÃ THU	Đã hoàn tất PDF
A1-101 Tòa A1	Tháng 8/2026	1.000.000 ₫	⚡ 3.000.000 ₫ 💧 3.000.000 ₫	7.000.000 ₫	ĐÃ THU	Đã hoàn tất PDF
A1-101 Tòa A1	Tháng 7/2026	1.000.000 ₫	⚡ 300.000 ₫ 💧 300.000 ₫	1.600.000 ₫	ĐÃ THU	Đã hoàn tất PDF
A1-102 Tòa A1	Tháng 6/2026	1.100.000 ₫	⚡ 100 ₫ 💧 100 ₫	1.100.200 ₫	ĐÃ THU	Đã hoàn tất PDF
A2-101 Tòa A2	Tháng 6/2026	100.000 ₫	⚡ 300.000 ₫ 💧 300.000 ₫	700.000 ₫	ĐÃ THU	Đã hoàn tất PDF

Duyệt yêu cầu trả phòng
Dormify - Đám mây Atlas

Yêu cầu trả phòng

SINH VIÊN	PHÒNG	LÝ DO	NGÀY RỜI DỰ KIẾN	TRẠNG THÁI	HOÀN CỤC	HÀNH ĐỘNG
Cristiano Ronaldo MSSV: 23177012	A1-102 Tòa A1 - Tầng 1	Bàn messi đá vào mắt em	16/7/2026	ĐÃ HOÀN TẤT	100.000đ	Chi tiết

- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-CHK-02.

Relationships:

- None.

Traceability: FR18, FR19

UC-CHK-03 — Calculate Compensation Fee

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-CHK-03**Use-case name:** Calculate Compensation Fee**Actor(s):** Dormitory Manager**Supporting Actor(s):** None**Description:** The manager records damaged or missing items and the system calculates deductions from the deposit.**Trigger:** Damage is found during checkout review.**Preconditions:**

1. UC-CHK-02 is in progress.
2. A deposit amount or settlement basis is known.

Basic Flow (Main Success Scenario):

1. The manager adds each damaged/missing item, quantity, condition, fee, and note.
2. The system validates non-negative values and calculates the compensation total.
3. The system calculates the refundable deposit as the greater of zero and deposit minus compensation.
4. The manager reviews and confirms the calculation for checkout completion.

Alternative Flows:

- A1. No damage item remains: compensation becomes zero.
- A2. Compensation exceeds deposit: refund becomes zero and the excess is recorded as an amount owed according to policy.
- A3. Invalid fee or incomplete item: confirmation is blocked.

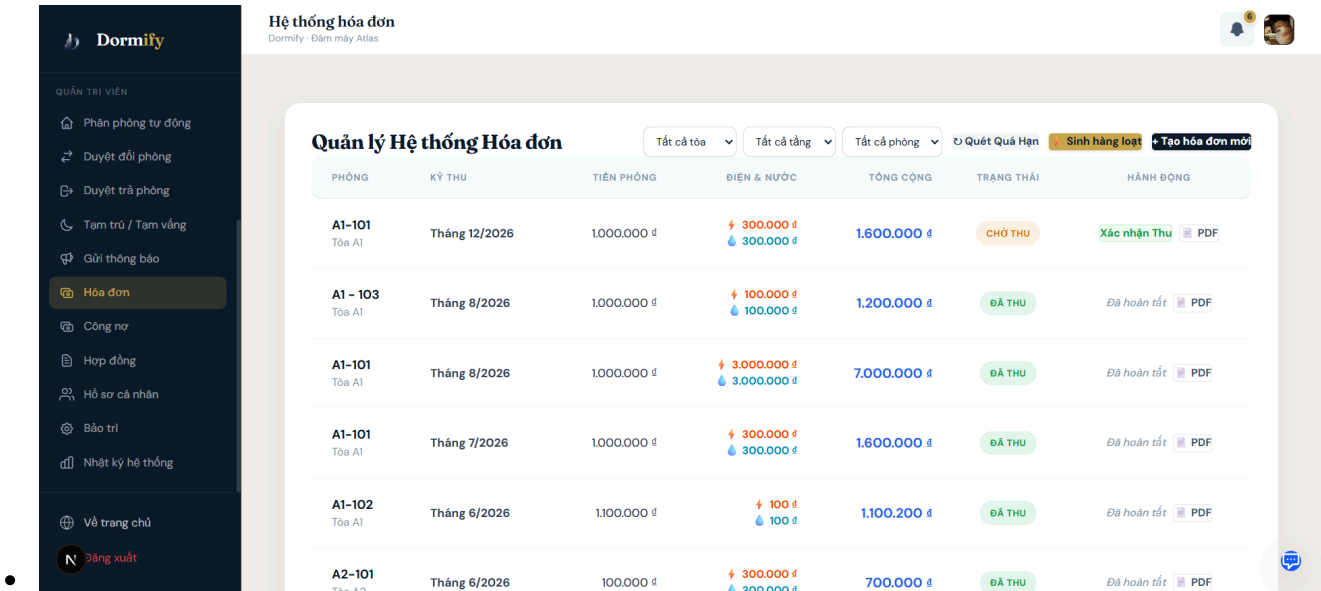
Postconditions:

- A compensation breakdown is prepared for UC-CHK-04.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-CHK-03.

Relationships:

- Extends UC-CHK-02 Review Checkout Request.

Traceability: FR19, FR20

UC-CHK-04 — Refund Deposit and Complete Checkout

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-CHK-04

Use-case name: Refund Deposit and Complete Checkout

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager finalizes checkout, determines the refund, liquidates the contract, and releases the room.

Trigger: The manager confirms checkout completion.

Preconditions:

1. The checkout request has been reviewed.
2. Compensation has been calculated or confirmed as zero.

Basic Flow (Main Success Scenario):

1. The system displays deposit, compensation, refund, outstanding balance, and final checkout details.
2. The manager confirms the final settlement and payment method/status.
3. The system records the refund amount and completion date.
4. The system invokes UC-CON-03 to liquidate the rental contract.
5. The system releases the bed, updates room occupancy, and closes the residence assignment atomically.
6. The system notifies the student and makes the completed checkout record available.

Alternative Flows:

- A1. Refund processing fails: operational checkout remains pending settlement and room release follows configured policy.
- A2. Concurrent room/contract change: the transaction is stopped and the manager refreshes the request.
- A3. Additional amount owed: the system creates or records the debt before completion according to policy.

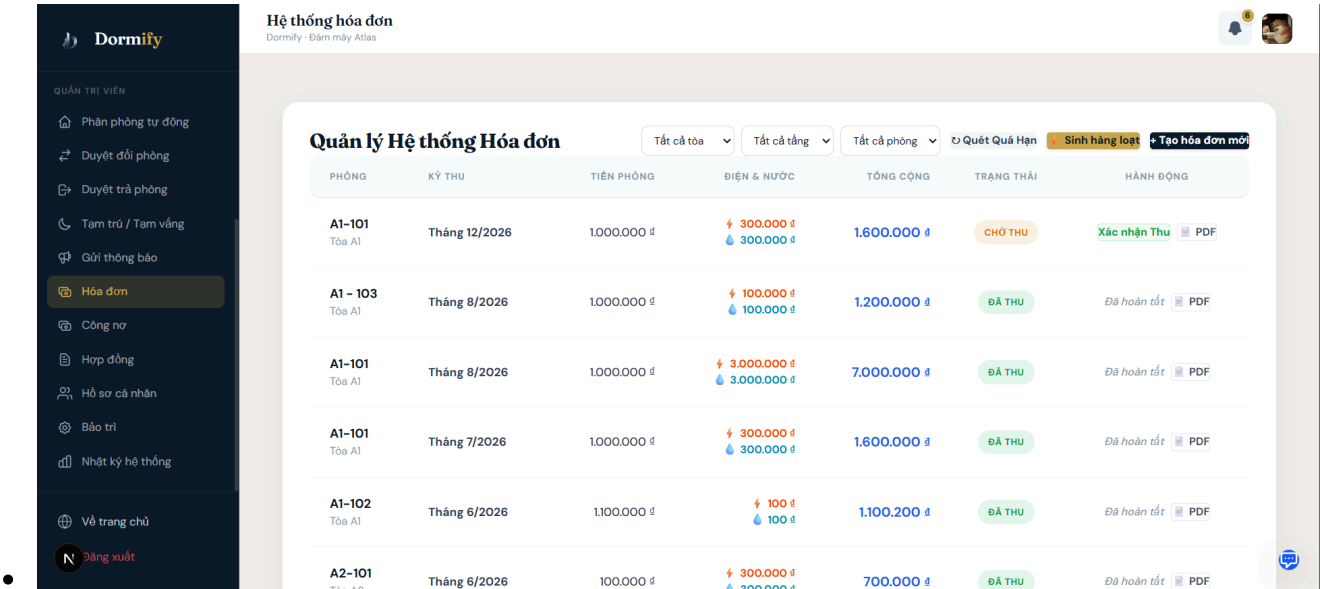
Postconditions:

- Checkout is completed.
- The contract is liquidated and the room/bed is released.
- Refund and compensation records are stored.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-CHK-04.

Relationships:

- Includes UC-CON-03 Liquidate Rental Contract.

Traceability: FR21

7. Finance and Meter Management

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Meter readings, invoices, payments, debts, reminders, overdue processing, and revenue reporting.

ID	Use case	Primary actor
UC-FIN-01	Manage Electricity and Water Meter Readings	Dormitory Manager

ID	Use case	Primary actor
UC-FIN-02	Create or Bulk-Generate Invoices	Dormitory Manager
UC-FIN-03	Mark Invoice as Paid Manually	Dormitory Manager
UC-FIN-04	View Debt Summary	Dormitory Manager
UC-FIN-05	Send Debt Reminders	Dormitory Manager, Scheduled Trigger
UC-FIN-06	Generate Revenue Report	Dormitory Manager
UC-FIN-07	Mark Overdue Invoices	Scheduled Trigger
UC-FIN-08	View Invoices	Student
UC-FIN-09	Pay Invoice	Student
UC-FIN-10	View Payment History	Student
UC-FIN-11	Download Invoice PDF	Student

UC-FIN-01 — Manage Electricity and Water Meter Readings

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-FIN-01

Use-case name: Manage Electricity and Water Meter Readings

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager records opening and closing electricity/water readings, uploads meter evidence, and reviews consumption history.

Trigger: The manager opens Meter Management for a billing period.

Preconditions:

1. The actor is authenticated as Dormitory Manager.
2. The room and billing period exist.

Basic Flow (Main Success Scenario):

1. The system displays rooms and the previous closing readings.
2. The manager enters opening/closing electricity and water readings and uploads meter photos.
3. The system validates that closing readings are not lower than opening readings.
4. The system calculates consumption.
5. The manager confirms and the system stores the readings and evidence.
6. The manager may view historical readings and consumption trends.

Alternative Flows:

- A1. Duplicate room/period reading: the system requests editing the existing record.
- A2. Closing reading lower than opening: submission is rejected.
- A3. Photo upload fails: the system follows the configured evidence requirement and reports the failure.

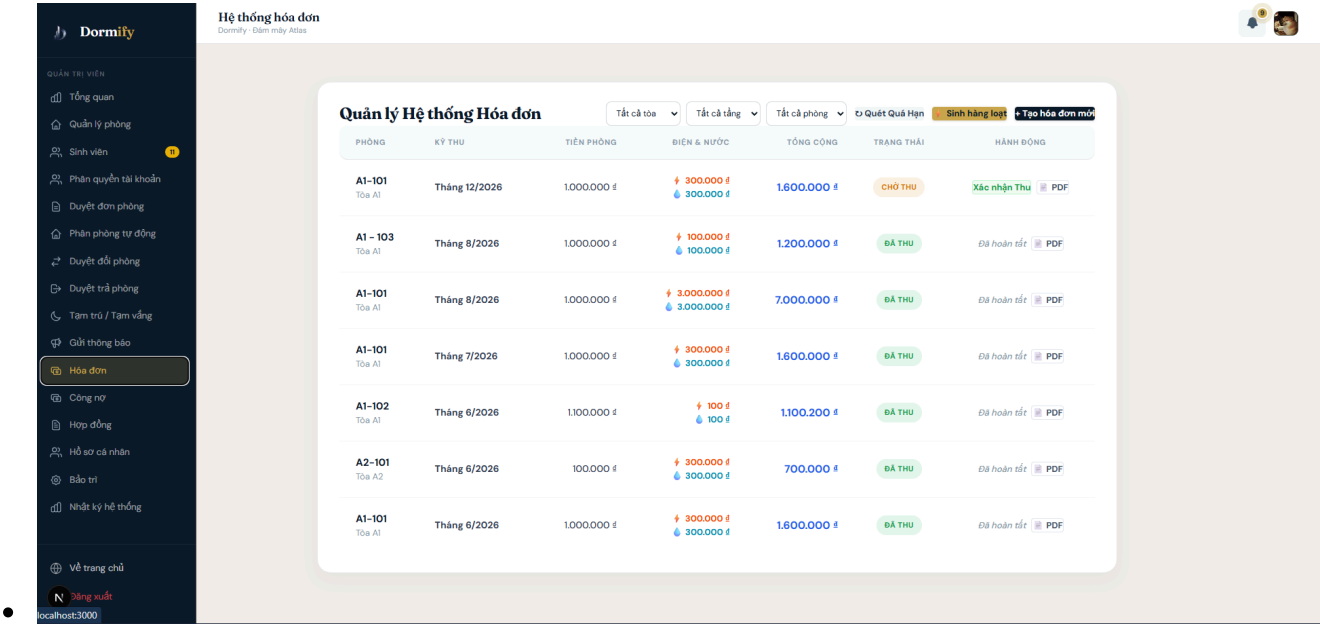
Postconditions:

- Meter readings and consumption history are stored.

Special Requirements:

- Photo files must be restricted by type and size.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FIN-01.

Relationships:

- None.

Traceability: FR22, FM04, FM05, FM06, FM07, FM08, FM09

UC-FIN-02 — Create or Bulk-Generate Invoices

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-FIN-02

Use-case name: Create or Bulk-Generate Invoices

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager creates individual or bulk invoices for accommodation, electricity, water, repairs, or other approved charges.

Trigger: The manager starts invoice generation for a billing period.

Preconditions:

1. The actor is authenticated as Dormitory Manager.
2. Required room, contract, tariff, and meter data exist.

Basic Flow (Main Success Scenario):

1. The manager selects billing period and invoice type.
2. The system loads eligible rooms/students and calculates applicable charges.
3. The manager reviews generated amounts and due dates.
4. The system validates that duplicate invoices are not created for the same charge and period.
5. The manager confirms creation.
6. The system stores pending invoices and notifies affected students.

Alternative Flows:

- A1. Missing meter/tariff data: affected invoices are skipped and listed in an error report.
- A2. Duplicate invoice: the system excludes it or requires explicit correction, not duplication.
- A3. Single invoice: the manager selects one student/room and enters the charge manually.

Postconditions:

- One or more pending invoices exist.

Special Requirements:

- Currency amounts must use integer VND or a defined decimal policy.

Prototype Screens:

Dormify

QUẢN TRỊ VIỆN

Tổng quan

Quản lý phòng

Sinh viên

Phân quyền tài khoản

Duyệt đơn phòng

Phân phòng tự động

Duyệt đổi phòng

Duyệt trả phòng

Tạm trú / Tạm vắng

Gửi thông báo

Hóa đơn

Công nợ

Hợp đồng

Hồ sơ cá nhân

Bảo trì

Nhật ký hệ thống

Về trang chủ

Năng suất

localhost:3000

Hệ thống hóa đơn

Dormify - Giải pháp Atlas

Quản lý Hệ thống Hóa đơn

Tất cả tòa

Tất cả tầng

Tất cả phòng

Quét QR Hạn

Sinh hàng loạt

Tạo hóa đơn mới

PHÒNG	KÝ THU	TIỀN PHÒNG	ĐIỆN & NƯỚC	TỔNG CỘNG	TRANG THÁI	HÀNH ĐỘNG
A1-101 Tòa A1	Tháng 12/2026	1,000,000 ₫	300,000 ₫ 300,000 ₫	1,600,000 ₫	CHỜ THU	Xác nhận Thu PDF
A1 - 103 Tòa A1	Tháng 8/2026	1,000,000 ₫	100,000 ₫ 100,000 ₫	1,200,000 ₫	ĐÃ THU	Đã hoàn tất PDF
A1-101 Tòa A1	Tháng 8/2026	1,000,000 ₫	3,000,000 ₫ 3,000,000 ₫	7,000,000 ₫	ĐÃ THU	Đã hoàn tất PDF
A1-101 Tòa A1	Tháng 7/2026	1,000,000 ₫	300,000 ₫ 300,000 ₫	1,600,000 ₫	ĐÃ THU	Đã hoàn tất PDF
A1-102 Tòa A1	Tháng 6/2026	1,100,000 ₫	100 ₫ 100 ₫	1,100,200 ₫	ĐÃ THU	Đã hoàn tất PDF
A2-101 Tòa A2	Tháng 6/2026	100,000 ₫	300,000 ₫ 300,000 ₫	700,000 ₫	ĐÃ THU	Đã hoàn tất PDF
A1-101 Tòa A1	Tháng 6/2026	1,000,000 ₫	300,000 ₫ 300,000 ₫	1,600,000 ₫	ĐÃ THU	Đã hoàn tất PDF

- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FIN-02.

Relationships:

- None.

Traceability: FR22

UC-FIN-03 — Mark Invoice as Paid Manually

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

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Use-case ID: UC-FIN-03

Use-case name: Mark Invoice as Paid Manually

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager confirms an offline payment such as cash or verified bank transfer.

Trigger: The manager selects Mark as paid.

Preconditions:

- 1. The invoice exists and is pending or overdue.
- 2. The actor is authenticated as Dormitory Manager.

Basic Flow (Main Success Scenario):

- 1. The system displays invoice and payer details.
- 2. The manager enters payment date, method, reference, and optional evidence.
- 3. The system validates the payment amount.
- 4. The manager confirms.
- 5. The system marks the invoice paid, records payment history, and notifies the student.

Alternative Flows:

- A1. Invoice already paid: duplicate confirmation is blocked.
- A2. Partial amount: the system follows configured partial-payment policy or rejects the action.
- A3. Invalid reference/evidence: the system requests correction.

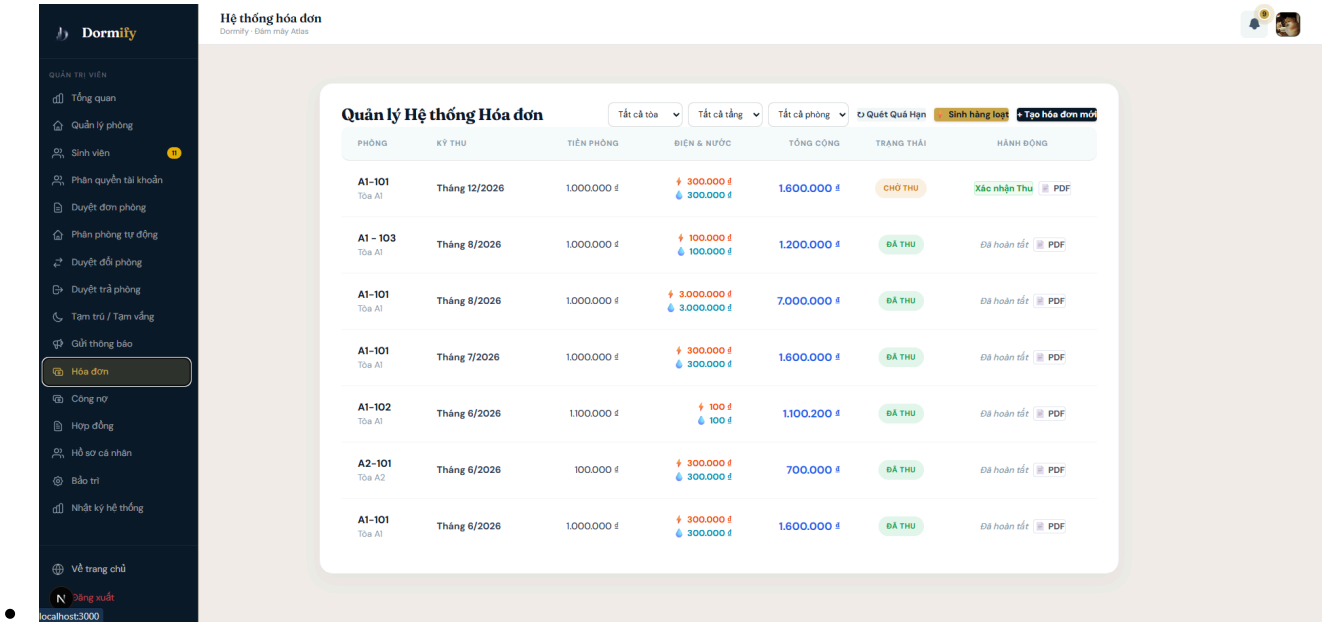
Postconditions:

- The invoice is paid and a payment record exists.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FIN-03.

Relationships:

- None.

Traceability: FR22

UC-FIN-04 — View Debt Summary

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-FIN-04

Use-case name: View Debt Summary

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager reviews unpaid and overdue amounts by student, room, building, or period.

Trigger: The manager opens Debt Management.

Preconditions:

1. The actor is authenticated as Dormitory Manager.

Basic Flow (Main Success Scenario):

1. The system aggregates pending and overdue invoices.
2. The system displays total debt, overdue count, oldest due date, students/rooms involved, and filters.
3. The manager opens a debtor entry to view invoice details and available actions.

Alternative Flows:

- A1. No debt exists: the system displays a positive empty state.
- A2. Data changes during review: totals refresh to the latest values.

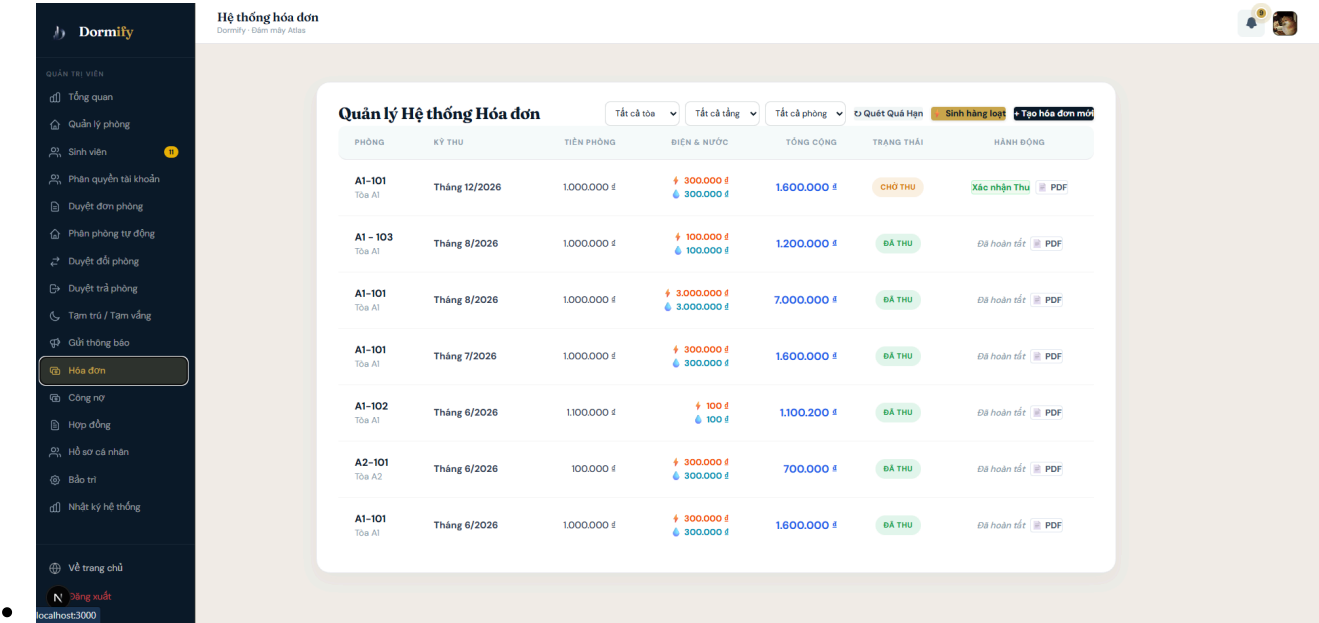
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FIN-04.

Relationships:

- None.

Traceability: FR23

UC-FIN-05 — Send Debt Reminders

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-FIN-05
Use-case name: Send Debt Reminders
Actor(s): Dormitory Manager, Scheduled Trigger
Supporting Actor(s): None
Description: The system sends manual or scheduled reminders for unpaid or overdue invoices.
Trigger: A manager selects Send reminder, or the configured schedule is reached.

Preconditions:

1. At least one eligible unpaid invoice exists.

Basic Flow (Main Success Scenario):

1. The system identifies eligible recipients and invoices.
2. The system composes a reminder containing amount, due date, and payment guidance.
3. For a manual reminder, the manager confirms the target set.
4. The system sends notifications/messages and records reminder time and result.

Alternative Flows:

- A1. Invoice was paid before sending: the system excludes it.
- A2. Recipient delivery fails: the failure is recorded and other reminders continue.
- A3. Reminder frequency limit reached: the system postpones or blocks duplicate reminders.

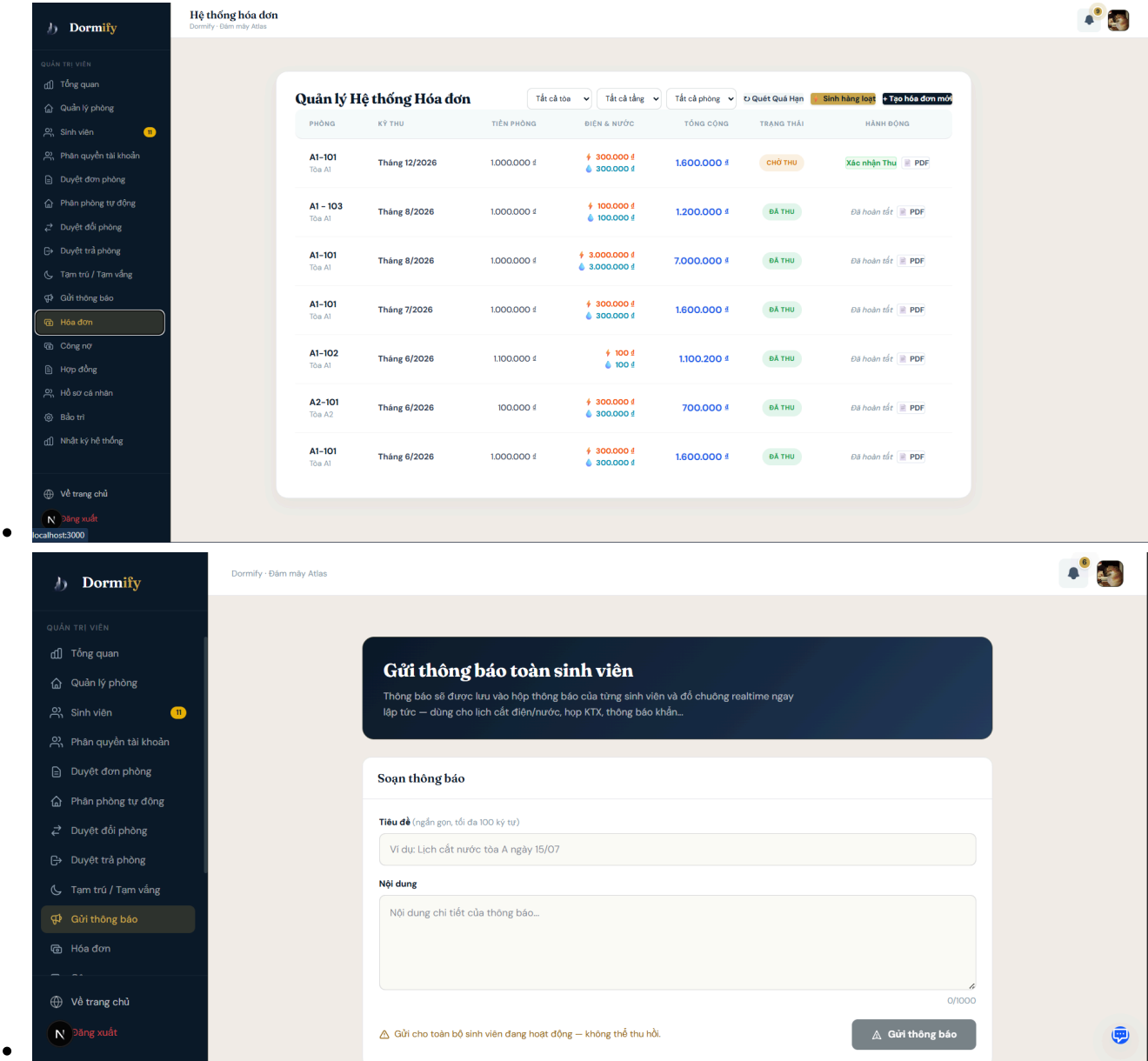
Postconditions:

- Reminder records are created and messages are sent or queued.

Special Requirements:

- Scheduled reminders must avoid excessive or duplicate messaging.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FIN-05.

Relationships:

- None.

Traceability: FR23

UC-FIN-06 — Generate Revenue Report

Use-case ID: UC-FIN-06

Use-case name: Generate Revenue Report

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager generates collected-revenue reports for monthly, quarterly, or yearly periods.

Trigger: The manager opens Revenue Reports and selects a period.

Preconditions:

1. The actor is authenticated as Dormitory Manager.

Basic Flow (Main Success Scenario):

1. The manager chooses period, grouping, and optional building/charge filters.
2. The system aggregates successfully paid invoices and refunds/adjustments according to accounting rules.
3. The system displays totals and breakdowns by accommodation, electricity, water, repair, and other charges.
4. The manager may export or print the report.

Alternative Flows:

- A1. No paid transactions: the report displays zero values.
- A2. Invalid period: the system requests a valid range.
- A3. Export failure: the on-screen report remains available.

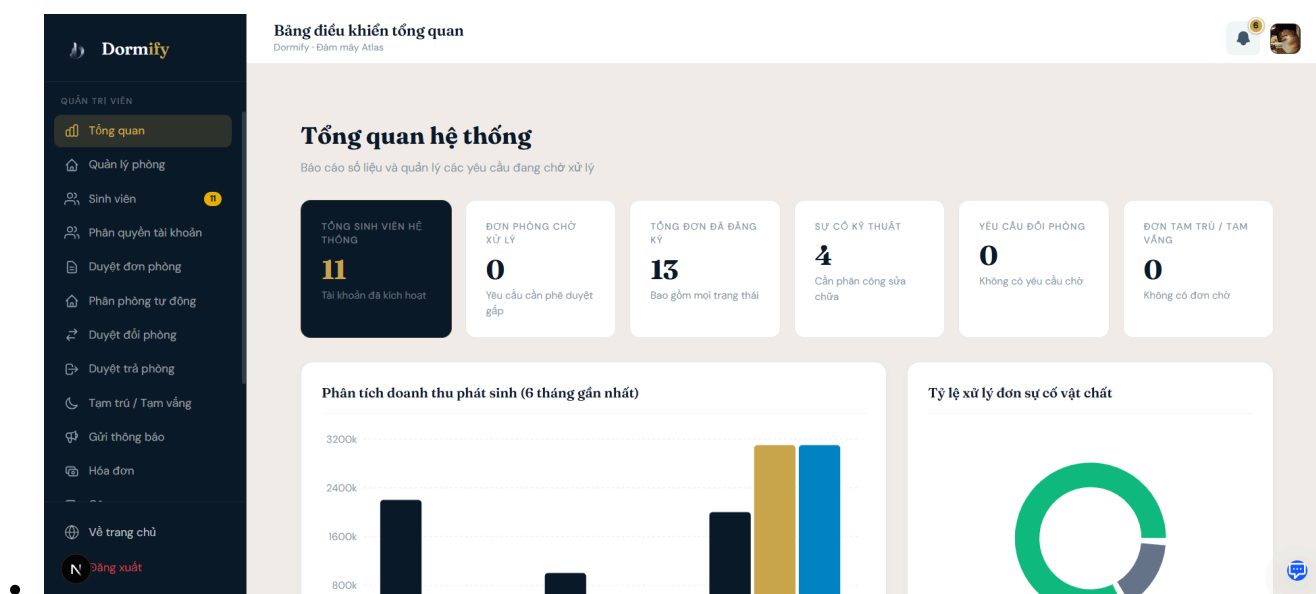
Postconditions:

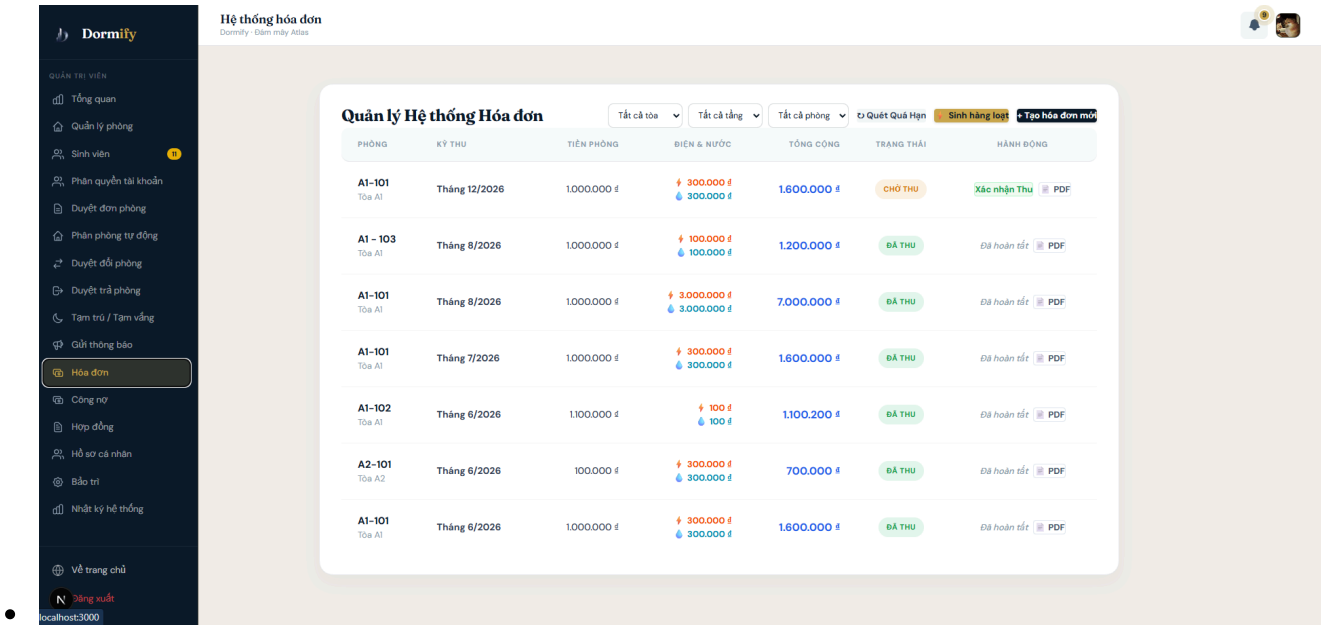
- A report is displayed or exported; financial records are unchanged.

Special Requirements:

- The report must clearly state whether figures are gross or net of refunds.

Prototype Screens:





- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FIN-06.

Relationships:

- None.

Traceability: FR24

UC-FIN-07 — Mark Overdue Invoices

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-FIN-07

Use-case name: Mark Overdue Invoices

Actor(s): Scheduled Trigger

Supporting Actor(s): Dormitory Manager

Description: The system automatically changes eligible unpaid invoices from pending to overdue.

Trigger: The scheduled overdue check runs or a manager triggers it manually.

Preconditions:

- 1. An unpaid invoice has passed its due date.

Basic Flow (Main Success Scenario):

- 1. The system finds pending invoices with due dates earlier than the current time.
- 2. The system marks each eligible invoice overdue and records the overdue timestamp.
- 3. The system sends an overdue notification to affected students.
- 4. The system records processing results.

Alternative Flows:

- A1. Invoice paid concurrently: it is not marked overdue.
- A2. No eligible invoices: the run completes with zero updates.
- A3. Notification failure: invoice status remains updated and delivery is retried/logged.

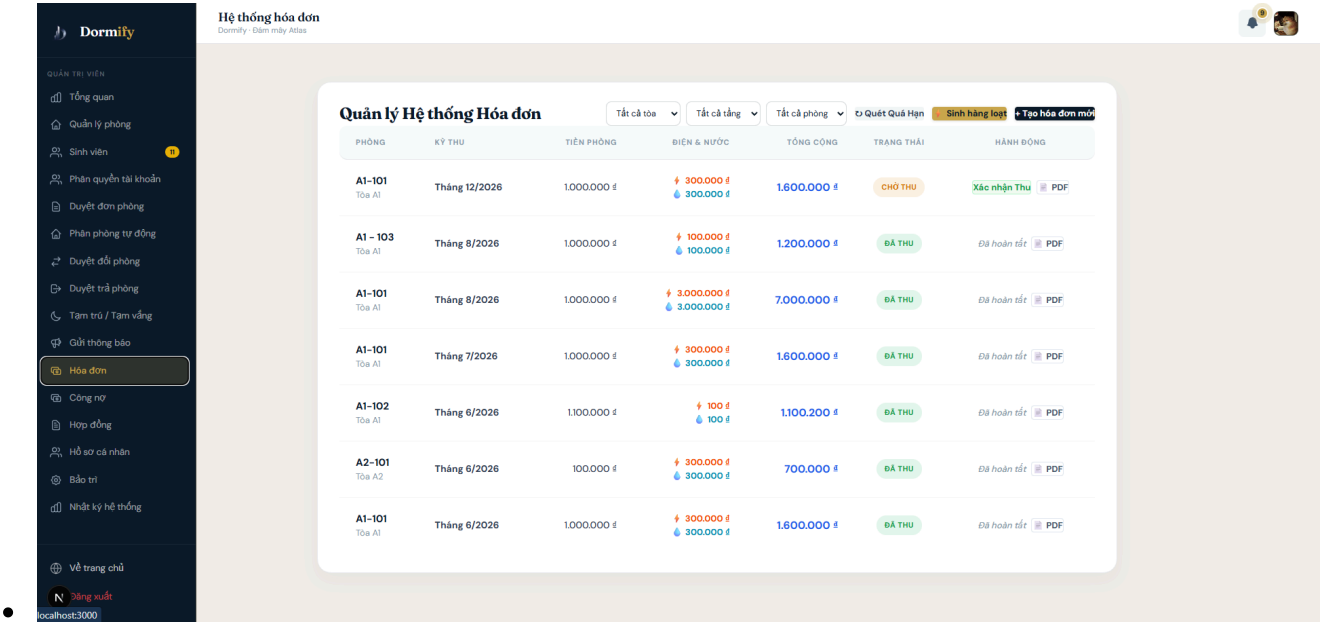
Postconditions:

- Eligible invoices are overdue and appear in the debt summary.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FIN-07.

Relationships:

- None.

Traceability: FR22, FR23

UC-FIN-08 — View Invoices

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-FIN-08

Use-case name: View Invoices

Actor(s): Student

Supporting Actor(s): None

Description: A student views invoices related to the current or previous residence.

Trigger: The student opens My Invoices.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system retrieves invoices belonging to the student or assigned room according to billing policy.

- 2. The system displays amount, charge breakdown, due date, status, and billing period.
- 3. The student opens an invoice to view full details and payment actions.

Alternative Flows:

- A1. No invoices: the system displays an empty state.
- A2. Invoice ownership mismatch: access is denied.

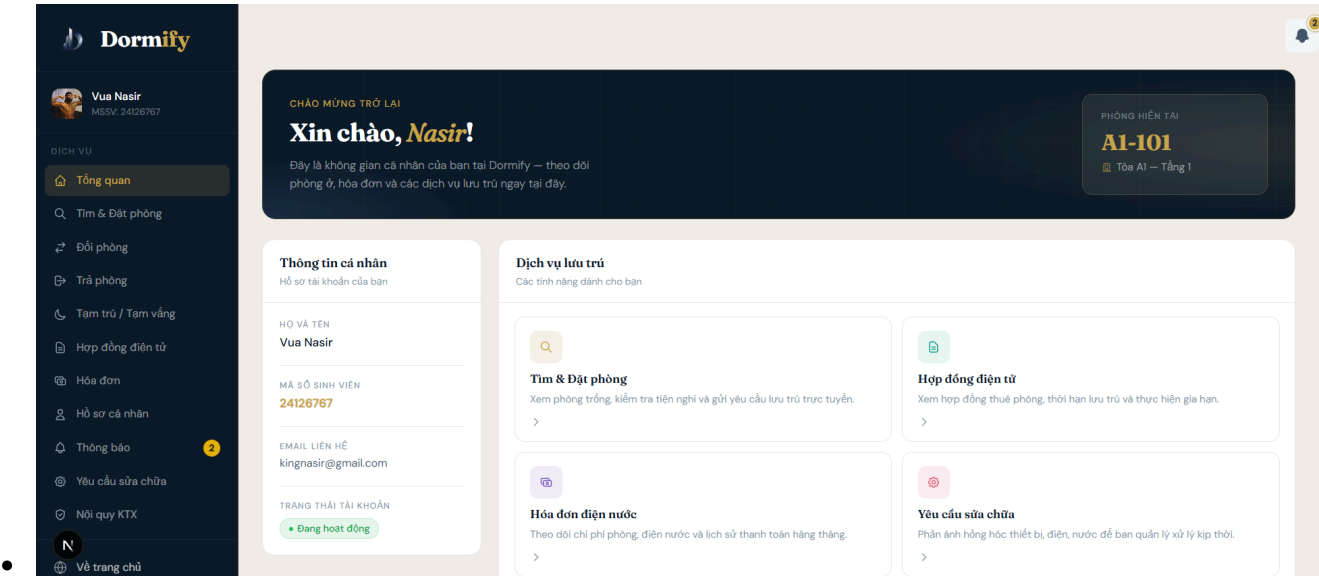
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FIN-08.

Relationships:

- None.

Traceability: FR22, ST13

UC-FIN-09 — Pay Invoice

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-FIN-09

Use-case name: Pay Invoice

Actor(s): Student

Supporting Actor(s): Payment Gateway

Description: A student pays an invoice using a supported bank or electronic payment provider.

Trigger: The student selects Pay on an unpaid invoice.

Preconditions:

1. The student is authenticated.
2. The invoice belongs to the student and is unpaid.

Basic Flow (Main Success Scenario):

1. The system displays the payable amount and supported methods.
2. The student selects bank, VNPay, MoMo, ZaloPay, or another configured provider.
3. The system creates a payment transaction and redirects or opens the Payment Gateway.
4. The Payment Gateway processes the payment and returns a signed result.
5. The system verifies the result, records the transaction, updates the invoice when successful, and notifies the student.

Alternative Flows:

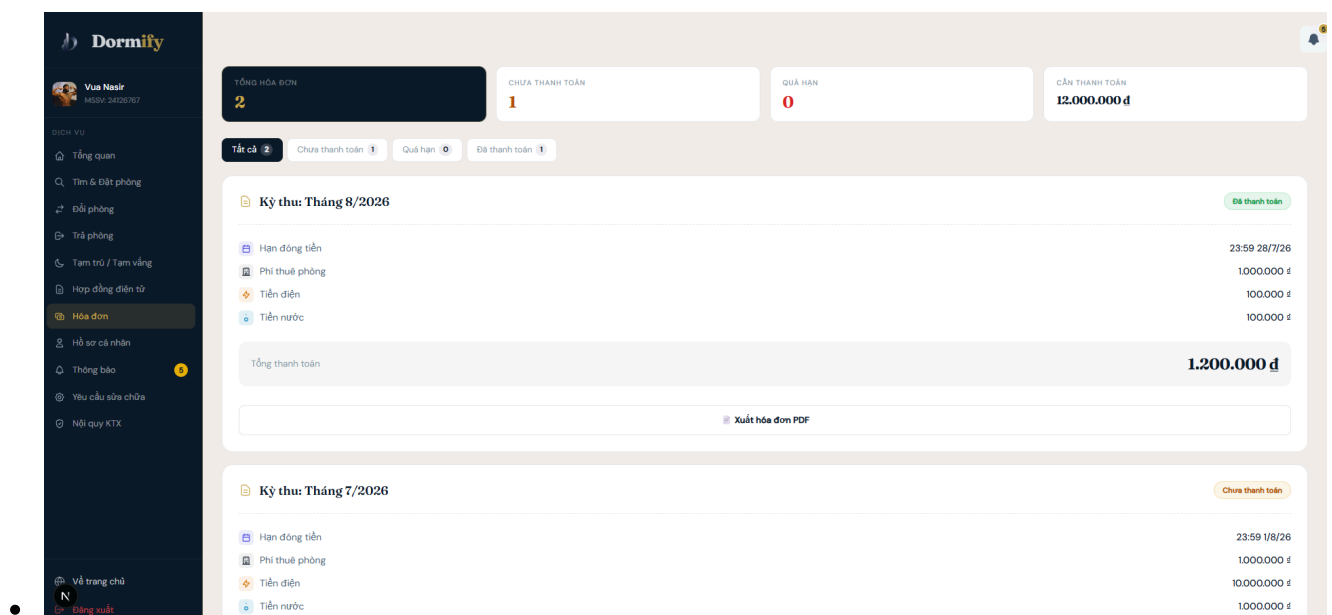
- A1. Payment cancelled: transaction is recorded as cancelled and invoice remains unpaid.
- A2. Payment failed: transaction is recorded as failed and retry is allowed.
- A3. Callback delayed: transaction remains pending until verified.
- A4. Duplicate callback: the system processes it idempotently.

Postconditions:

- Successful payment marks the invoice paid; otherwise it remains unpaid with transaction history.

Special Requirements:

- Gateway signatures and transaction identifiers must be verified.

Prototype Screens:

- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FIN-09.

Relationships:

- None.

Traceability: FR22, ST14

UC-FIN-10 — View Payment History

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-FIN-10

Use-case name: View Payment History

Actor(s): Student

Supporting Actor(s): None

Description: A student reviews successful, pending, failed, and cancelled payment attempts.

Trigger: The student opens Payment History.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system retrieves the student's payment transactions.
2. The system displays invoice, amount, method, status, transaction reference, and time.
3. The student may open a transaction to view details.

Alternative Flows:

- A1. No payment history: the system displays an empty state.
- A2. Filters return no transactions: the system displays an empty filtered state and offers a clear-filter action.
- A3. Payment gateway reference unavailable: the system shows the local payment record and marks the external reference status as unavailable.

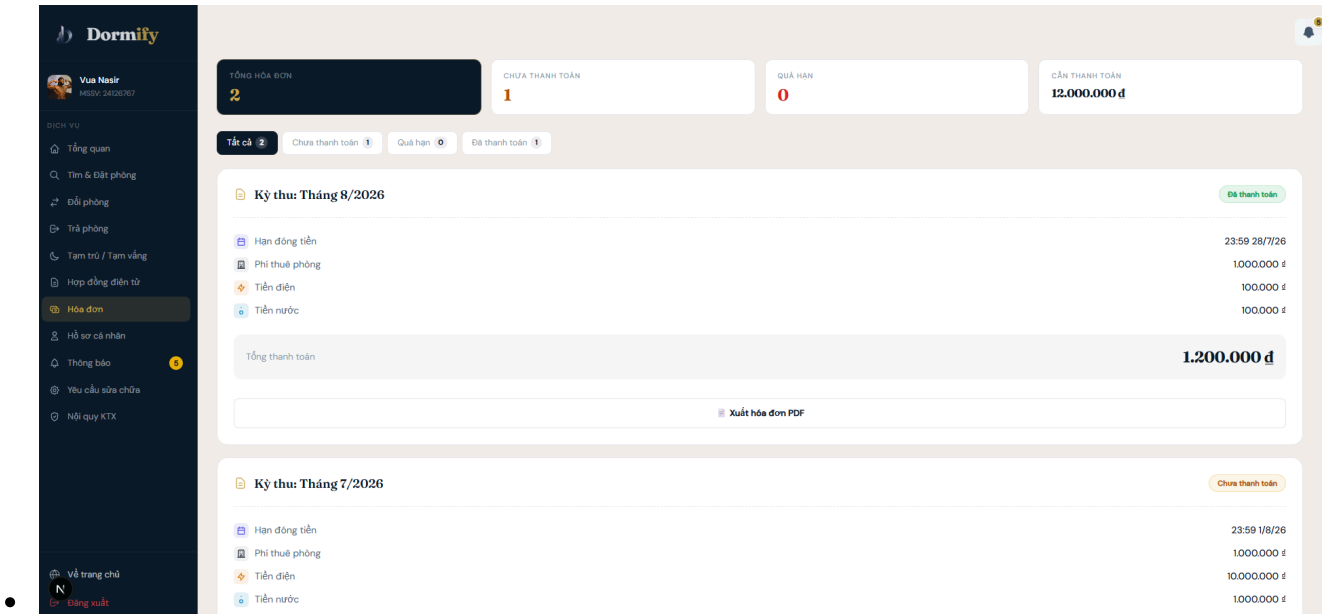
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FIN-10.

Relationships:

- None.

Traceability: FR22, ST15

UC-FIN-11 — Download Invoice PDF

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-FIN-11

Use-case name: Download Invoice PDF

Actor(s): Student

Supporting Actor(s): None

Description: A student downloads a formatted PDF copy of an invoice.

Trigger: The student selects Download PDF while viewing an invoice.

Preconditions:

1. The student is authenticated.
2. The invoice belongs to the student.

Basic Flow (Main Success Scenario):

1. The system verifies invoice ownership.
2. The system renders the invoice using the approved template.
3. The system provides the PDF for download.

Alternative Flows:

- A1. Unauthorized invoice: access is denied.
- A2. PDF generation fails: the invoice view remains available with an error message.

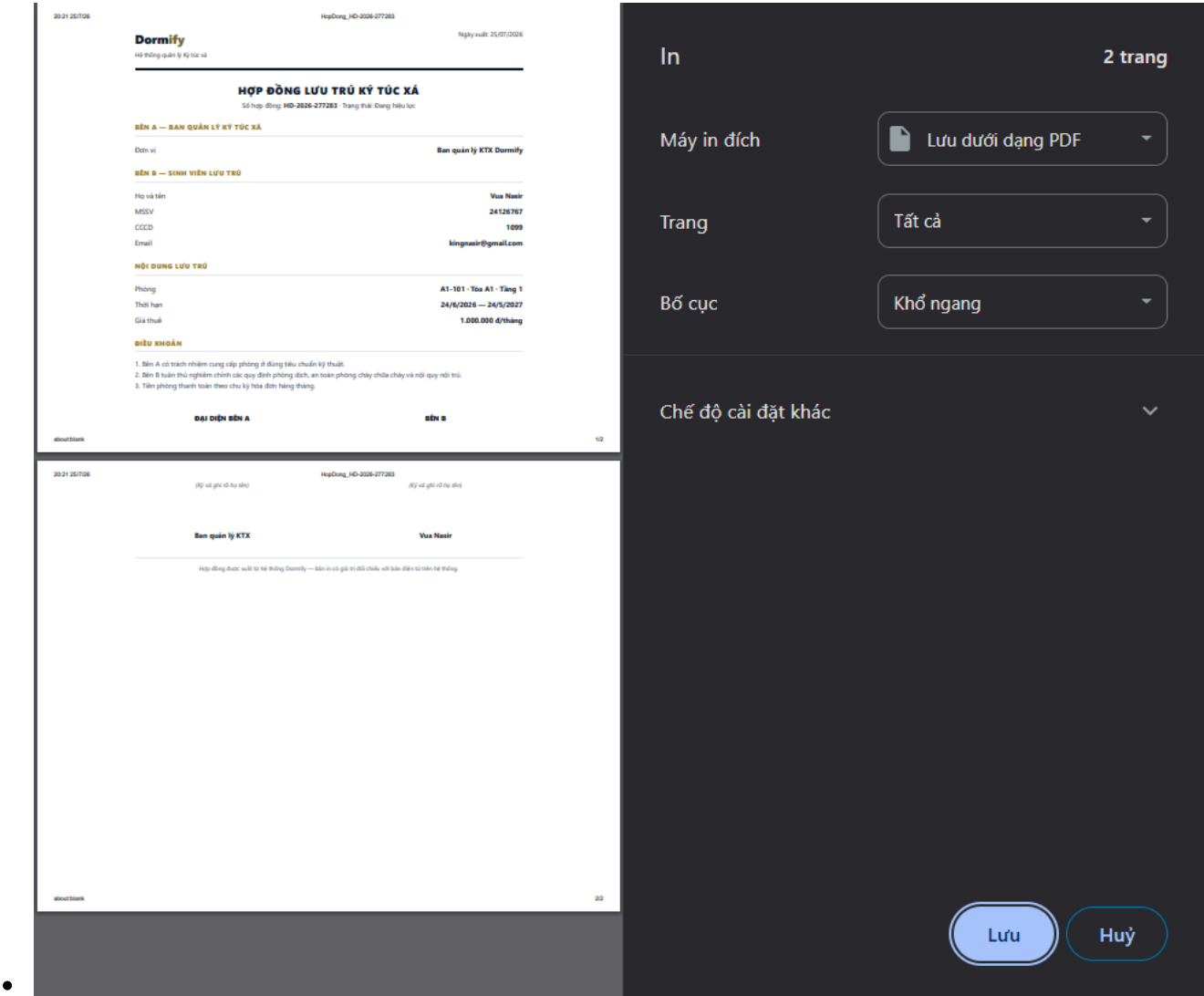
Postconditions:

- A PDF copy is generated; invoice data is unchanged.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FIN-11.

Relationships:

- Extends UC-FIN-08 View Invoices.

Traceability: FR22, ST16

8. Residence Management

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Overnight absence, temporary residence, long-term absence, visitor registration, and management review.

ID	Use case	Primary actor
UC-RES-01	Submit Overnight Absence Declaration	Student
UC-RES-02	Register Temporary Residence	Student
UC-RES-03	Register Long-Term Temporary Absence	Student
UC-RES-04	Register a Visitor	Student
UC-RES-05	Review and Track Residence Declarations	Dormitory Manager

UC-RES-01 — Submit Overnight Absence Declaration

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-RES-01

Use-case name: Submit Overnight Absence Declaration

Actor(s): Student

Supporting Actor(s): None

Description: A resident reports an overnight absence from the dormitory.

Trigger: The student selects Overnight Absence.

Preconditions:

1. The student is authenticated.
2. The student has an active room assignment.

Basic Flow (Main Success Scenario):

1. The student enters departure time/date, expected return time/date, destination or contact information where required, and reason.
2. The system validates the date range and policy.
3. The system creates a residence declaration and notifies the Dormitory Manager.
4. The system displays the declaration status.

Alternative Flows:

- A1. Start time is in the past beyond the allowed tolerance: the system requests correction.
- A2. End precedes start: submission is rejected.
- A3. Conflicting pending declaration: the system warns or rejects according to policy.

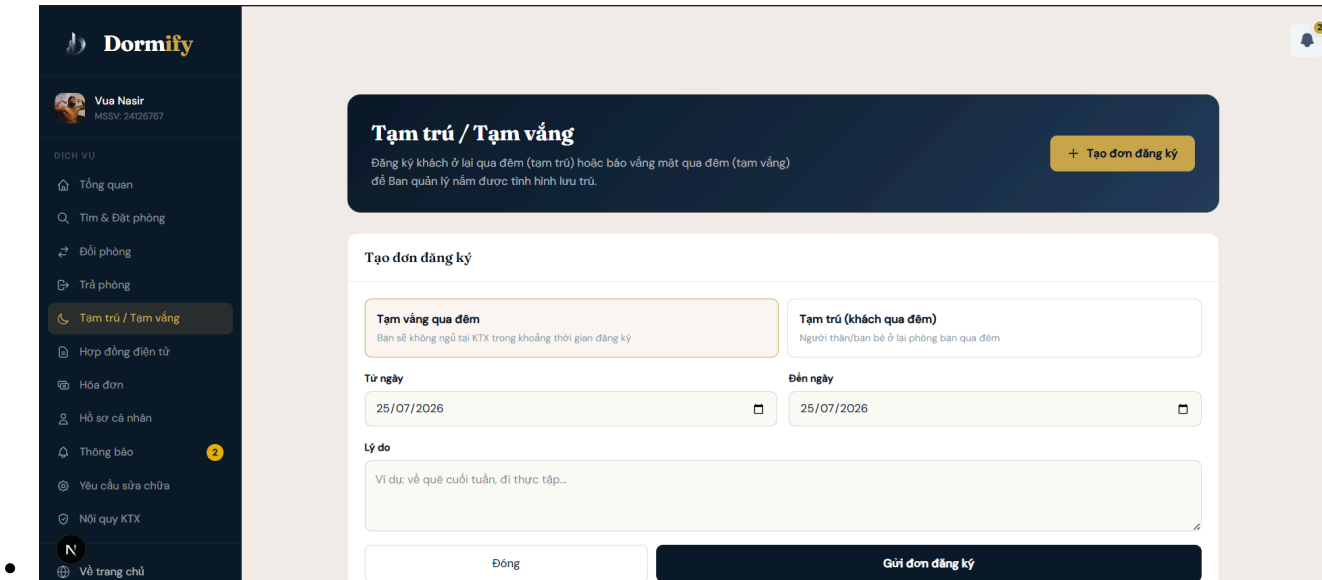
Postconditions:

- An overnight absence declaration is stored.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-RES-01.

Relationships:

- None.

Traceability: FR25, ST21

UC-RES-02 — Register Temporary Residence

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-RES-02

Use-case name: Register Temporary Residence

Actor(s): Student

Supporting Actor(s): None

Description: A student registers a person temporarily staying overnight in the assigned room.

Trigger: The student selects Temporary Residence Registration.

Preconditions:

1. The student is authenticated.
2. The student has an active room assignment.

Basic Flow (Main Success Scenario):

1. The student enters guest name, identification number, contact details, relationship, and stay period.
2. The system validates required guest information and dormitory rules.
3. The system creates a pending or registered temporary-residence declaration.
4. The system notifies the Dormitory Manager and displays the status.

Alternative Flows:

- A1. Missing guest identification: submission is rejected.
- A2. Stay exceeds the permitted duration: manager approval or a different process is required.
- A3. Room/visitor limit exceeded: submission is rejected.

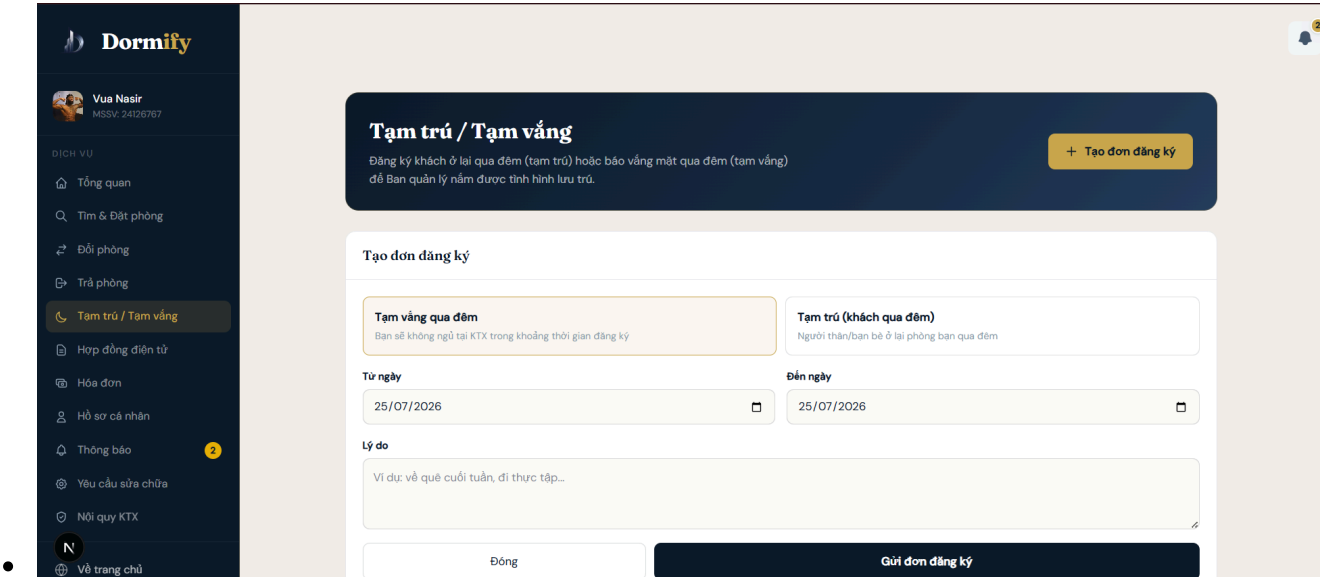
Postconditions:

- A temporary-residence declaration is stored.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-RES-02.

Relationships:

- None.

Traceability: ST22, FM11

UC-RES-03 — Register Long-Term Temporary Absence

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-RES-03

Use-case name: Register Long-Term Temporary Absence

Actor(s): Student

Supporting Actor(s): None

Description: A student reports an absence longer than the ordinary overnight period.

Trigger: The student selects Long-Term Absence.

Preconditions:

1. The student is authenticated.
2. The student has an active room assignment.

Basic Flow (Main Success Scenario):

1. The student enters the extended absence period, reason, destination/contact information, and optional evidence.
2. The system validates the duration and any additional requirements.
3. The system creates a declaration requiring manager review.
4. The system notifies the Dormitory Manager and displays status.

Alternative Flows:

- A1. Duration does not meet the long-term threshold: the system directs the student to UC-RES-01.
- A2. Missing required evidence: submission remains incomplete or is rejected.
- A3. Date conflict: the system requests correction.

Postconditions:

- A long-term temporary-absence declaration is stored.

Special Requirements:

- None.

Prototype Screens:

- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-RES-03.

Relationships:

- None.

Traceability: ST23, FM12

UC-RES-04 — Register a Visitor

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-RES-04

Use-case name: Register a Visitor

Actor(s): Student

Supporting Actor(s): None

Description: A student registers a day visitor and the intended visit time.

Trigger: The student selects Visitor Registration.

Preconditions:

1. The student is authenticated.
2. The student has an active room assignment.

Basic Flow (Main Success Scenario):

1. The student enters visitor name, identification/contact details, visit date, arrival/departure time, and purpose.
2. The system validates visiting hours and restrictions.
3. The system creates a visitor registration and notifies or exposes it to authorized management/security.
4. The system displays confirmation or pending approval status.

Alternative Flows:

- A1. Visit outside allowed hours: the system rejects or requires special approval.
- A2. Visitor/room limit exceeded: registration is rejected.
- A3. Visitor restricted by dormitory policy: registration is denied.

Postconditions:

- A visitor registration is stored.

Special Requirements:

- None.

Prototype Screens:

- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-RES-04.

Relationships:

- None.

Traceability: ST24

UC-RES-05 — Review and Track Residence Declarations

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-RES-05

Use-case name: Review and Track Residence Declarations

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager views and processes overnight absence, temporary residence, long-term absence, and visitor declarations.

Trigger: A declaration is submitted or the manager opens Residence Management.

Preconditions:

1. The actor is authenticated as Dormitory Manager.

Basic Flow (Main Success Scenario):

1. The system lists declarations with type, student, room, dates, and status.
2. The manager filters by building, floor, room, type, date, or status.
3. The manager opens a declaration and verifies its details.
4. The manager approves, rejects, acknowledges, or records notes according to declaration type.
5. The system updates status and notifies the student.

Alternative Flows:

- A1. Declaration period has already ended: the system marks it expired/closed according to policy.
- A2. Invalid or conflicting information: the manager rejects with a reason.
- A3. No approval required: the manager records acknowledgment only.

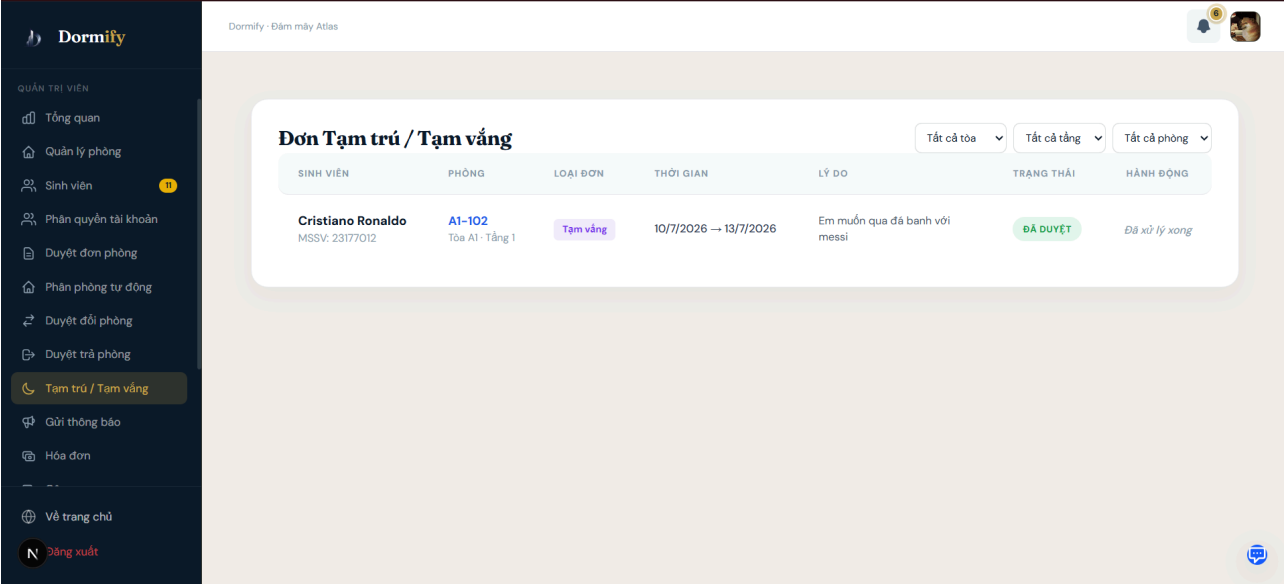
Postconditions:

- The declaration status and review history are updated.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-RES-05.

Relationships:

- None.

Traceability: FR25, FM10, FM11, FM12

9. Maintenance Management

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Repair reporting, assignment, progress tracking, evidence, staff history, and dashboard.

PA3 focus item	Coverage
Student flow	Submit and track repair requests.
Dormitory Manager flow	Review, assign, and monitor maintenance progress.
Maintenance Staff flow	View assigned jobs, update repair status, upload evidence, review history, and view dashboard metrics.
Evidence focus	Status timeline, image evidence, repair result, assignment history, and traceability to FR26-FR28 / MT01-MT08.

ID	Use case	Primary actor
UC-MNT-01	Submit Repair Request	Student
UC-MNT-02	Track Repair Request	Student
UC-MNT-03	Review and Assign Maintenance Request	Dormitory Manager
UC-MNT-04	Track Maintenance Progress	Dormitory Manager
UC-MNT-05	View Assigned Maintenance Jobs	Maintenance Staff

ID	Use case	Primary actor
UC-MNT-06	View Maintenance Job Details	Maintenance Staff
UC-MNT-07	Update Repair Status and Result	Maintenance Staff
UC-MNT-08	Upload Before and After Repair Photos	Maintenance Staff
UC-MNT-09	View Maintenance History	Maintenance Staff
UC-MNT-10	View Maintenance Dashboard	Maintenance Staff

UC-MNT-01 — Submit Repair Request

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-MNT-01

Use-case name: Submit Repair Request

Actor(s): Student

Supporting Actor(s): None

Description: A student reports a room or facility incident and may attach photographs.

Trigger: The student selects New Repair Request.

Preconditions:

1. The student is authenticated.
2. The student has an active room assignment or selects an authorized common-area location.

Basic Flow (Main Success Scenario):

1. The student enters title, description, location, category, priority, and optional photos.
2. The system validates required data and file restrictions.
3. The system creates a pending maintenance request.
4. The system notifies the Dormitory Manager.
5. The system displays the request reference and status.

Alternative Flows:

- A1. Invalid file: the system rejects the attachment and allows correction.
- A2. Urgent safety issue: the system highlights emergency guidance and raises priority.
- A3. Duplicate suspected: the system warns the student but allows confirmation according to policy.

Postconditions:

- A pending maintenance request exists.

Special Requirements:

- None.

Prototype Screens:



Báo cáo sự cố



Mô tả rõ để kỹ thuật viên chuẩn bị đúng vật tư

Tóm tắt sự cố

Một câu ngắn gọn mô tả vấn đề

VD: Hỏng bóng đèn nhà vệ sinh, Vòi nước rò rỉ...

Mô tả chi tiết

Vị trí cụ thể và tình trạng hư hỏng để kỹ thuật viên chuẩn bị trước

VD: Bóng đèn ở góc trái nhà vệ sinh bị cháy hoàn toàn từ tối qua, gây tối trong khu vực...

Mức độ ảnh hưởng

Chọn mức độ phù hợp để ban quản lý ưu tiên xử lý

Thấp

Bình thường

Ưu tiên cao

Khẩn cấp

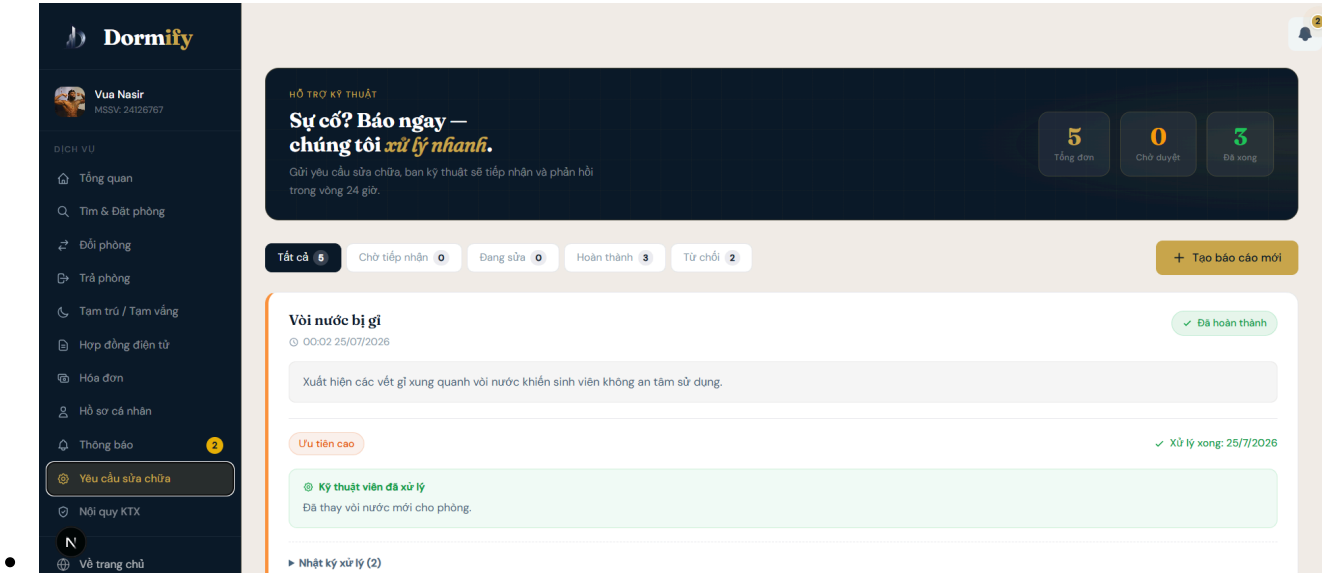
Ảnh thực tế

Đính kèm ảnh rõ khu vực hư hỏng để ban quản lý đánh giá nhanh hơn



Hủy bỏ

+ Gửi yêu cầu



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-MNT-01.

Relationships:

- None.

Traceability: FR26, ST17

UC-MNT-02 — Track Repair Request

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-MNT-02

Use-case name: Track Repair Request

Actor(s): Student

Supporting Actor(s): None

Description: A student views repair request status and recorded results.

Trigger: The student opens My Repair Requests.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system displays the student’s requests with pending, in-progress, completed, or rejected status.
2. The student opens a request to view assignment, progress notes, result, photos, and timestamps.
3. When permitted, the student acknowledges or rates completed work.

Alternative Flows:

- A1. No requests: the system displays an empty state.
- A2. Request rejected: the system displays the reason.

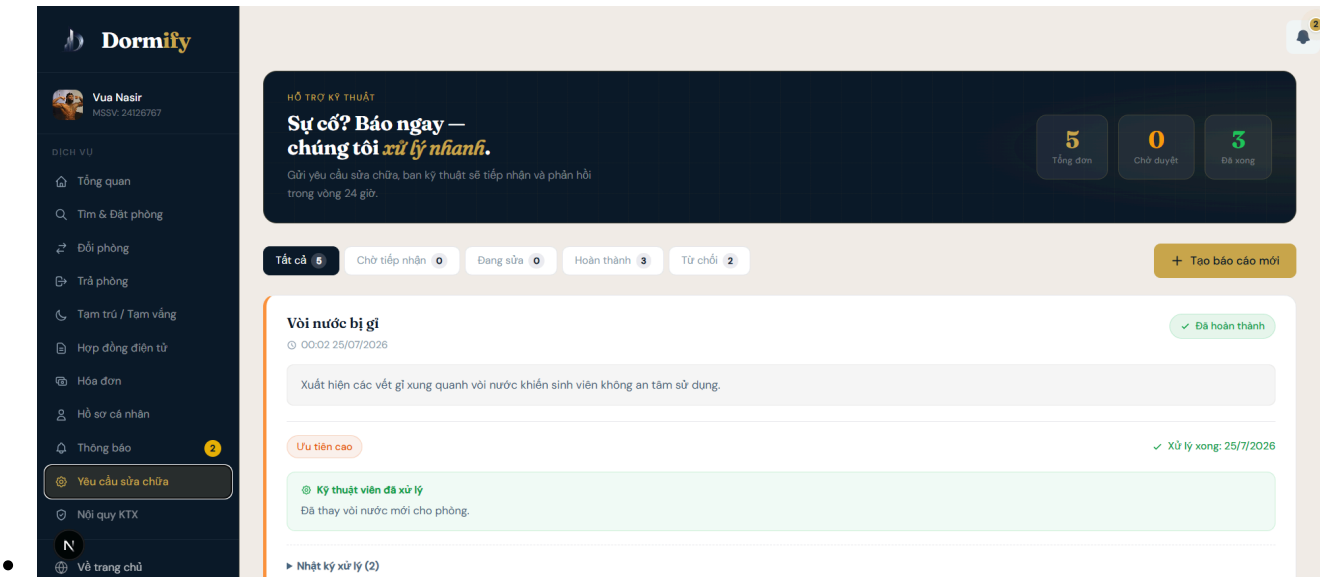
Postconditions:

- No data is changed unless an acknowledgment/rating is submitted.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-MNT-02.

Relationships:

- None.

Traceability: FR28, ST18

UC-MNT-03 — Review and Assign Maintenance Request

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-MNT-03

Use-case name: Review and Assign Maintenance Request

Actor(s): Dormitory Manager

Supporting Actor(s): Maintenance Staff

Description: The manager reviews a repair request and assigns it to an internal employee or approved third-party provider.

Trigger: A new pending maintenance request is received.

Preconditions:

1. The actor is authenticated as Dormitory Manager.

Basic Flow (Main Success Scenario):

1. The system displays request details, location, priority, attachments, and reporter contact.
2. The manager validates and prioritizes the request.
3. The manager selects an available Maintenance Staff member/provider and adds instructions.
4. The system records the assignment and notifies the assignee and student.

Alternative Flows:

- A1. Invalid/duplicate request: the manager rejects it with a reason.
- A2. No staff available: the request remains pending/unassigned or is assigned to an external provider.
- A3. Reassignment: the system records both old and new assignees and notifies affected parties.

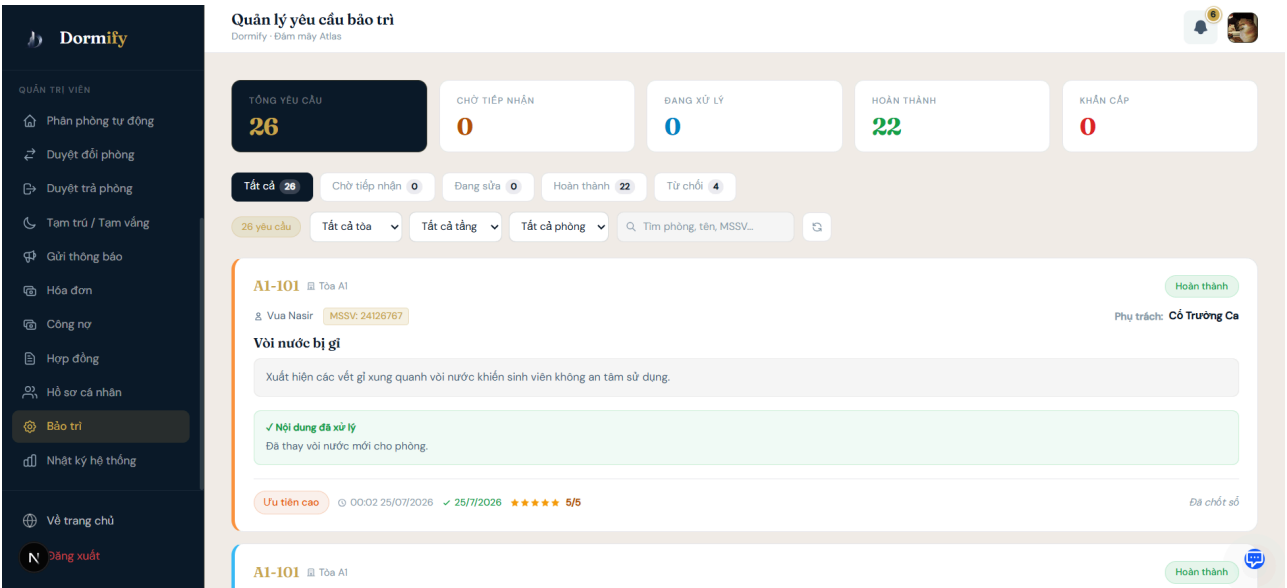
Postconditions:

- The request is assigned or rejected.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-MNT-03.

Relationships:

- None.

Traceability: FR26, FR27

UC-MNT-04 — Track Maintenance Progress

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-MNT-04

Use-case name: Track Maintenance Progress

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager monitors pending, in-progress, and completed maintenance requests.

Trigger: The manager opens Maintenance Management.

Preconditions:

- 1. The actor is authenticated as Dormitory Manager.

Basic Flow (Main Success Scenario):

- 1. The system lists requests with priority, location, assignee, status, and age.
- 2. The manager filters/searches and opens a request.
- 3. The manager reviews status changes, notes, photos, and completion result.
- 4. The manager may follow up, reassign, or escalate according to permissions.

Alternative Flows:

- A1. Overdue request: the system highlights it for escalation.
- A2. No results: the system displays an empty state.

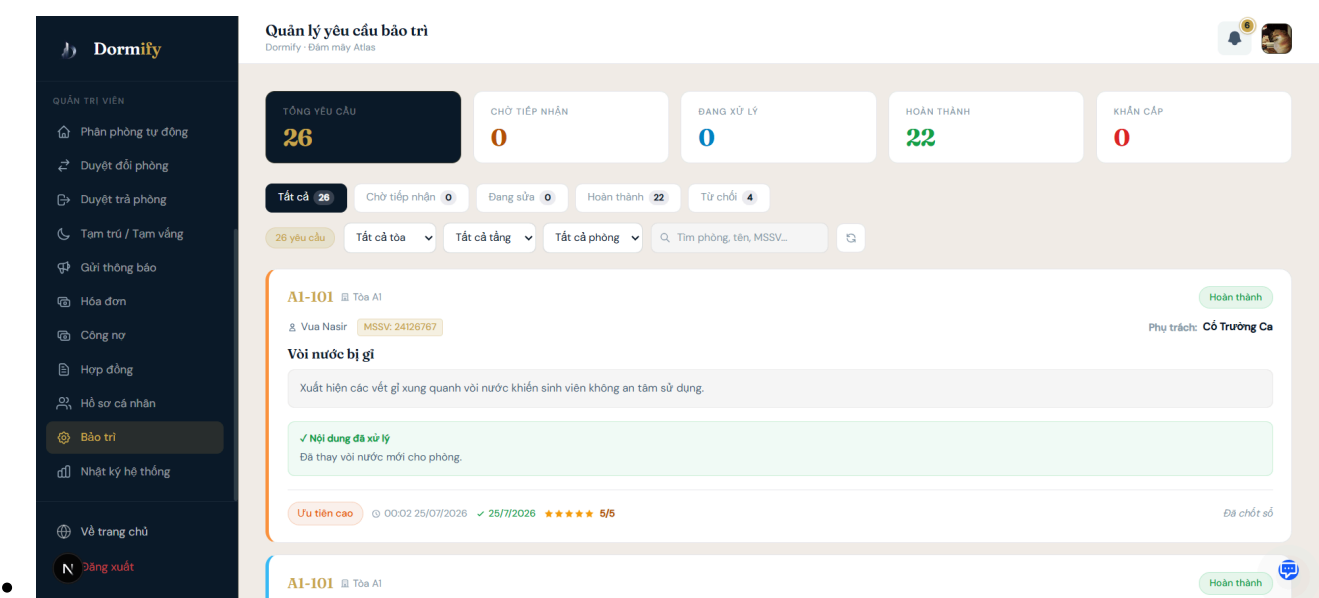
Postconditions:

- No data is changed unless the manager performs an allowed follow-up action.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-MNT-04.

Relationships:

- None.

Traceability: FR28, FM16

UC-MNT-05 — View Assigned Maintenance Jobs

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-MNT-05
Use-case name: View Assigned Maintenance Jobs

Actor(s): Maintenance Staff

Supporting Actor(s): None

Description: A maintenance staff member views work assigned to them.

Trigger: The staff member enters the maintenance workspace.

Preconditions:

1. The actor is authenticated as Maintenance Staff.

Basic Flow (Main Success Scenario):

1. The system lists assigned jobs grouped or filterable by status and priority.
2. The staff member searches by room, issue, date, or keyword.
3. The staff member selects a job to view details.

Alternative Flows:

- A1. No assigned jobs: the system displays an empty state.
- A2. Job reassigned while displayed: the system refreshes access and removes it from the list.

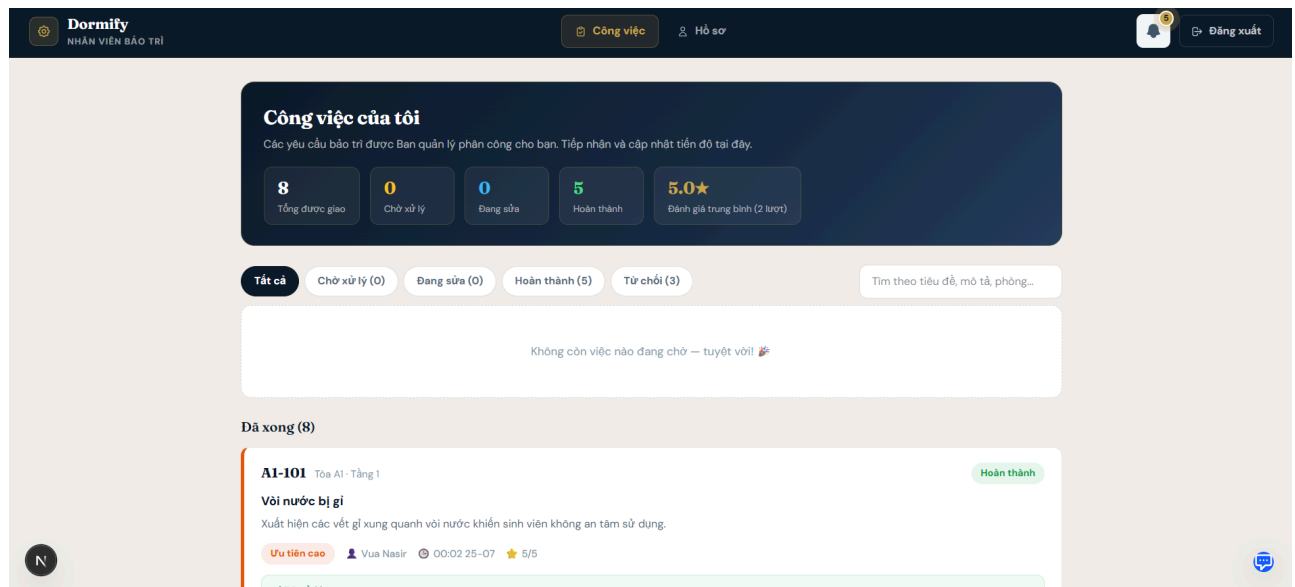
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-MNT-05.

Relationships:

- None.

Traceability: FR28, MT01

UC-MNT-06 — View Maintenance Job Details

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-MNT-06

Use-case name: View Maintenance Job Details

Actor(s): Maintenance Staff

Supporting Actor(s): None

Description: A staff member views the issue, location, priority, reporter information, instructions, and attachments for an assigned job.

Trigger: The staff member selects a job from the assigned list.

Preconditions:

- 1. The job is assigned to the staff member or access is otherwise authorized.

Basic Flow (Main Success Scenario):

- 1. The system verifies assignment.
- 2. The system displays full job details, timeline, and available status actions.
- 3. The staff member reviews attached evidence and manager instructions.

Alternative Flows:

- A1. Job is no longer assigned to the staff member: access is denied and the list refreshes.
- A2. Attachment unavailable: the system displays the remaining details and an attachment error.

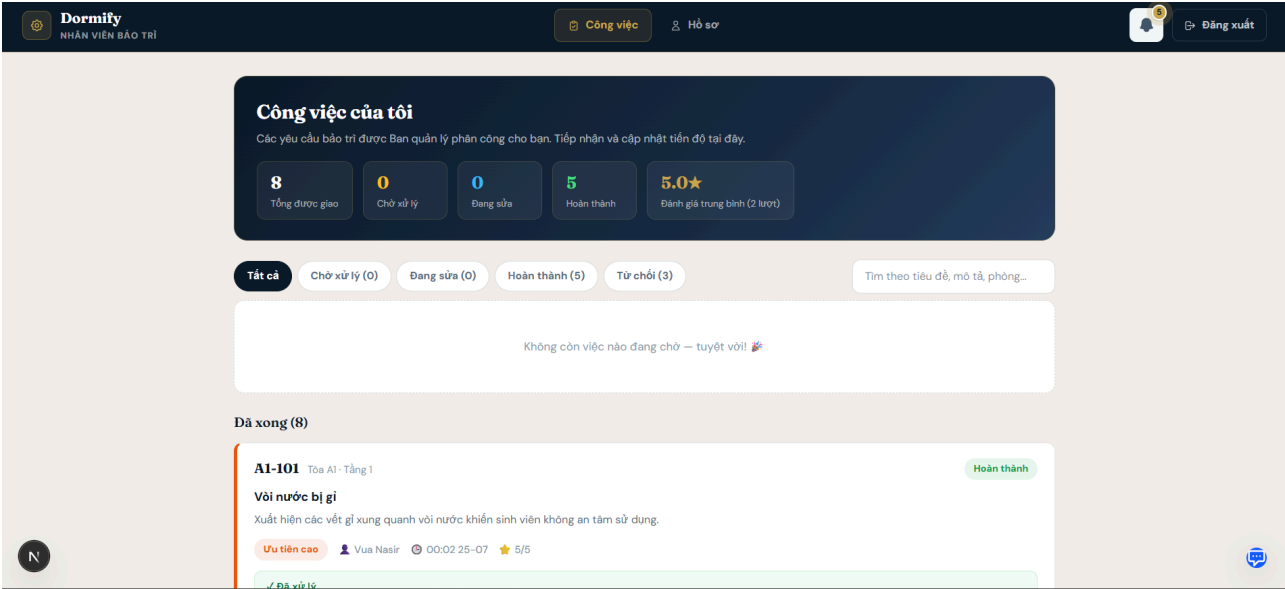
Postconditions:

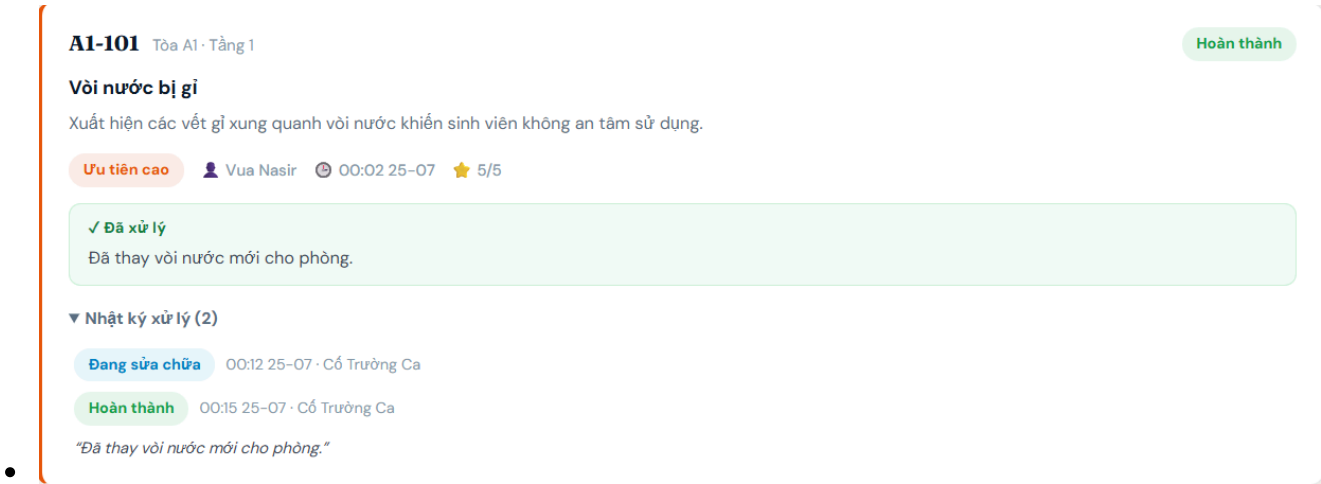
- No data is changed.

Special Requirements:

- None.

Prototype Screens:





- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-MNT-06.

Relationships:

- Accessed from UC-MNT-05 View Assigned Maintenance Jobs (include relationship in the model).

Traceability: FR28, MT02

UC-MNT-07 — Update Repair Status and Result

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-MNT-07
Use-case name: Update Repair Status and Result
Actor(s): Maintenance Staff
Supporting Actor(s): Dormitory Manager
Description: The assigned staff member updates a repair from pending to in progress to completed and records the result.
Trigger: The staff member selects a status action on an assigned job.

Preconditions:

1. The actor is assigned to the job.
2. The request is not closed in a conflicting state.

Basic Flow (Main Success Scenario):

1. The staff member changes status to In progress when work begins.
2. The system records the start time.
3. The staff member enters work notes, materials, and result.
4. When work is finished, the staff member changes status to Completed.
5. The system records completion time and notifies the student and manager.

Alternative Flows:

- A1. Invalid status transition: the system rejects the update.
- A2. Work cannot be completed: the staff member records a blocker and the request remains in progress or is escalated.
- A3. Manager rejects the completion after review: the request returns to in progress with instructions.

Postconditions:

- The status timeline and repair result are updated.

Special Requirements:

- None.

Prototype Screens:

Hoàn thành sửa chữa

Ghi ngắn gọn nội dung đã xử lý cho "Huw" (tùy chọn).

VD: Đã thay bóng đèn LED 9W, kiểm tra lại công tắc...

0/500

Hủy bỏ

Xác nhận hoàn thành

Từ chối yêu cầu

Nêu rõ lý do để sinh viên hiểu vì sao "Huw" bị từ chối.

VD: Báo sai, thiết bị vẫn hoạt động bình thường...

0/500

Hủy bỏ

Xác nhận từ chối

- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-MNT-07.

Relationships:

- None.

Traceability: FR28, MT03, MT04

UC-MNT-08 — Upload Before and After Repair Photos

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-MNT-08

Use-case name: Upload Before and After Repair Photos

Actor(s): Maintenance Staff

Supporting Actor(s): None

Description: A staff member uploads visual evidence before and/or after repair work.

Trigger: The staff member updates a repair status or result.

Preconditions:

1. The job is assigned to the staff member.

Basic Flow (Main Success Scenario):

1. The staff member selects Before or After photo type.
2. The system validates file type, size, and count.
3. The system uploads and associates the photo with the maintenance request.
4. The system displays the evidence in the job timeline.

Alternative Flows:

- A1. Upload fails: the status update is handled according to whether photos are mandatory.
- A2. Invalid file: the system rejects it.
- A3. Replacement/removal: the system records the change according to audit policy.

Postconditions:

- Photo evidence is associated with the job.

Special Requirements:

- None.

Prototype Screens:

Hoàn thành sửa chữa

Ghi ngắn gọn nội dung đã xử lý cho "Huw" (tùy chọn).

VD: Đã thay bóng đèn LED 9W, kiểm tra lại công tắc...

0/500

Hủy bỏ

Xác nhận hoàn thành



Báo cáo sự cố



Mô tả rõ để kỹ thuật viên chuẩn bị đúng vật tư

Tóm tắt sự cố

Một câu ngắn gọn mô tả vấn đề

VD: Hỏng bóng đèn nhà vệ sinh, Vòi nước rò rỉ...

Mô tả chi tiết

Vị trí cụ thể và tình trạng hư hỏng để kỹ thuật viên chuẩn bị trước

VD: Bóng đèn ở góc trái nhà vệ sinh bị cháy hoàn toàn từ tối qua, gây tối trong khu vực...

Mức độ ảnh hưởng

Chọn mức độ phù hợp để ban quản lý ưu tiên xử lý

Thấp

Bình thường

Ưu tiên cao

Khẩn cấp

Ảnh thực tế

Đính kèm ảnh rõ khu vực hư hỏng để ban quản lý đánh giá nhanh hơn

Hủy bỏ

+ Gửi yêu cầu

-
- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-MNT-08.

Relationships:

- Extends UC-MNT-07 Update Repair Status and Result.

Traceability: FR28, MT05, MT06

UC-MNT-09 — View Maintenance History

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-MNT-09

Use-case name: View Maintenance History

Actor(s): Maintenance Staff

Supporting Actor(s): None

Description: A staff member reviews previously processed maintenance work.

Trigger: The staff member opens Maintenance History.

Preconditions:

1. The actor is authenticated as Maintenance Staff.

Basic Flow (Main Success Scenario):

1. The system retrieves completed/rejected jobs previously assigned to the staff member.
2. The system displays filters for period, status, room, and issue type.
3. The staff member opens a historical job to view the final result and evidence.

Alternative Flows:

- A1. No history: the system displays an empty state.
- A2. Filters return no maintenance jobs: the system displays an empty filtered state and offers a clear-filter action.
- A3. Evidence file unavailable: the system still displays job history and marks the missing attachment as unavailable.

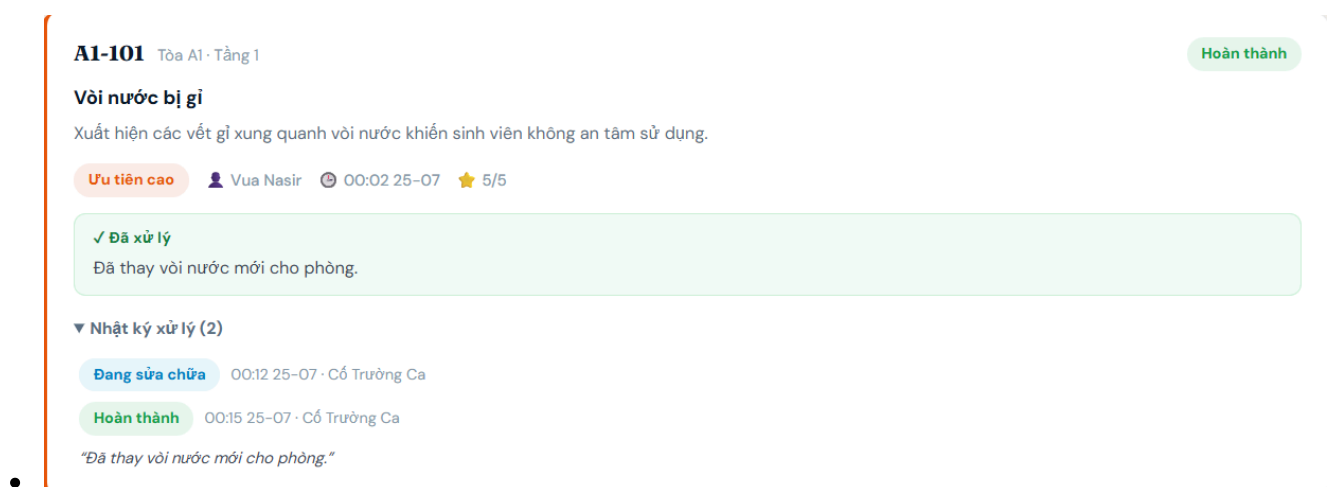
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-MNT-09.

Relationships:

- None.

Traceability: FR28, MT07

UC-MNT-10 — View Maintenance Dashboard

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-MNT-10

Use-case name: View Maintenance Dashboard

Actor(s): Maintenance Staff

Supporting Actor(s): None

Description: A staff member views workload and job-status statistics.

Trigger: The staff member opens the maintenance dashboard.

Preconditions:

1. The actor is authenticated as Maintenance Staff.

Basic Flow (Main Success Scenario):

1. The system aggregates assigned, pending, in-progress, completed, overdue, and priority counts.
2. The system displays recent assignments and performance summaries.
3. The staff member selects a metric to filter the job list.

Alternative Flows:

- A1. No jobs in the period: metrics display zero.
- A2. Metric refresh failure: the affected widget is marked unavailable.

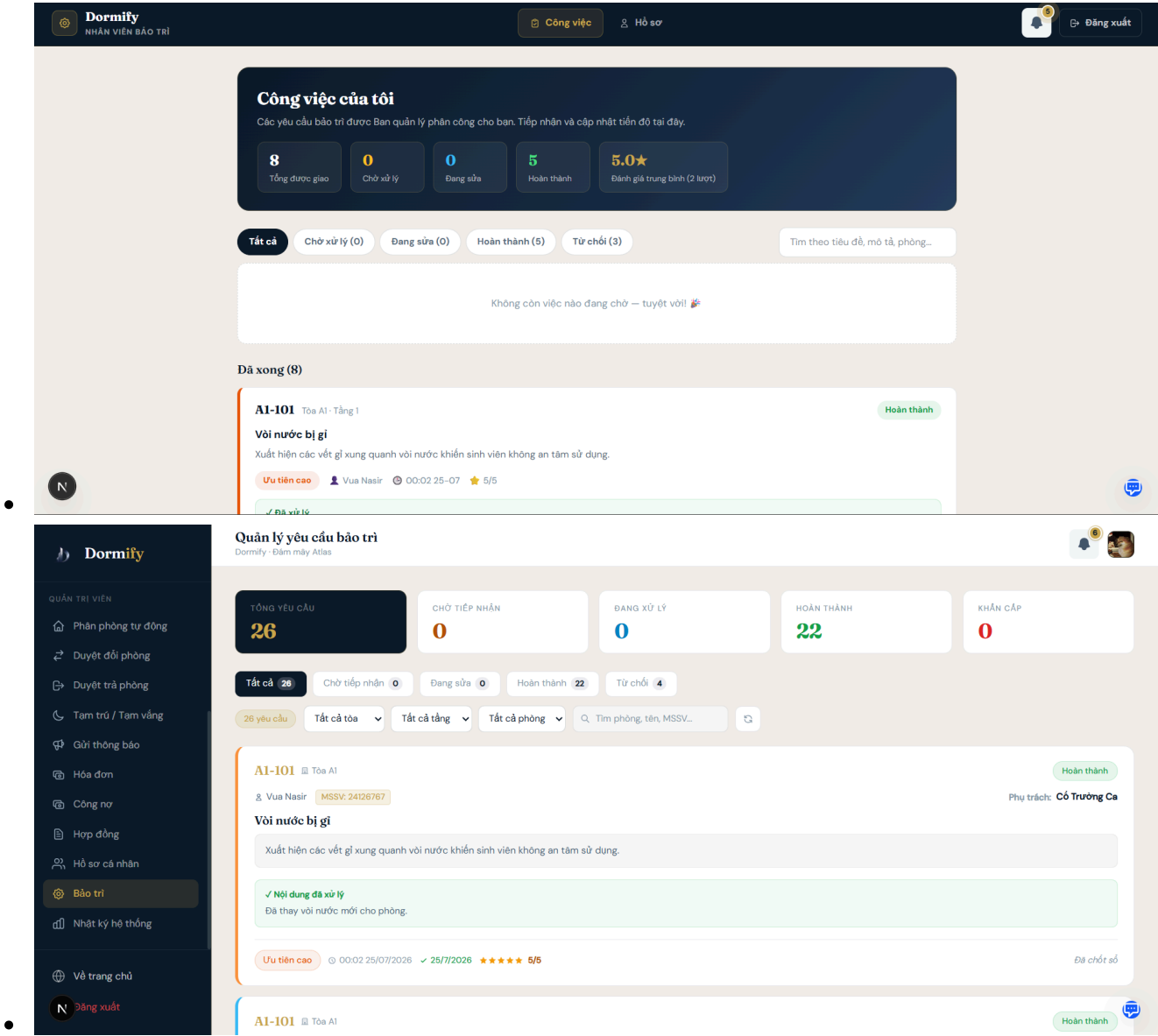
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-MNT-10.

Relationships:

- None.

Traceability: FR28, MT08

10. Feedback and Suggestions

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Student-initiated complaints, feedback, suggestions, and management responses.

ID	Use case	Primary actor
UC-FBK-01	Submit Complaint or Feedback	Student
UC-FBK-02	Submit Suggestion	Student
UC-FBK-03	Review and Respond to Feedback	Dormitory Manager

UC-FBK-01 — Submit Complaint or Feedback

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-FBK-01

Use-case name: Submit Complaint or Feedback

Actor(s): Student

Supporting Actor(s): None

Description: A student submits a complaint or feedback about dormitory facilities, services, billing, rules, or staff conduct.

Trigger: The student selects Submit Complaint / Feedback.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The student selects a category, enters a subject and detailed message, and optionally attaches evidence.
2. The student chooses identified or anonymous presentation where policy permits.
3. The system validates content and creates a pending feedback item.
4. The system notifies the Dormitory Manager and displays the tracking reference.

Alternative Flows:

- A1. Anonymous submissions not allowed for the category: identity is required.
- A2. Invalid attachment or empty message: submission is rejected.
- A3. Urgent safety complaint: the system highlights emergency contact guidance.

Postconditions:

- A pending complaint/feedback item exists.

Special Requirements:

- None.

Prototype Screens:



Báo cáo sự cố

Mô tả rõ để kỹ thuật viên chuẩn bị đúng vật tư



Tóm tắt sự cố

Một câu ngắn gọn mô tả vấn đề

VD: Hỏng bóng đèn nhà vệ sinh, Vòi nước rò rỉ...

Mô tả chi tiết

Vị trí cụ thể và tình trạng hư hỏng để kỹ thuật viên chuẩn bị trước

VD: Bóng đèn ở góc trái nhà vệ sinh bị cháy hoàn toàn từ tối qua, gây tối trong khu vực...

Mức độ ảnh hưởng

Chọn mức độ phù hợp để ban quản lý ưu tiên xử lý

Thấp

Bình thường

Ưu tiên cao

Khẩn cấp

Ảnh thực tế

Đính kèm ảnh rõ khu vực hư hỏng để ban quản lý đánh giá nhanh hơn

Hủy bỏ

+ Gửi yêu cầu

-
- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FBK-01.

Relationships:

- None.

Traceability: ST19, FM15

UC-FBK-02 — Submit Suggestion

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

95 / 115

Use-case ID: UC-FBK-02

Use-case name: Submit Suggestion

Actor(s): Student

Supporting Actor(s): None

Description: A student proposes an improvement to dormitory life or operations.

Trigger: The student selects Submit Suggestion.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The student selects a suggestion category and enters the proposal and expected benefit.
2. The system validates and stores the suggestion.
3. The system notifies the Dormitory Manager and displays the tracking reference.

Alternative Flows:

- A1. Duplicate/similar suggestion detected: the system may show existing items and allow confirmation.
- A2. Empty or inappropriate content: submission is rejected according to moderation rules.

Postconditions:

- A pending suggestion exists.

Special Requirements:

- None.

Prototype Screens:



Báo cáo sự cố

Mô tả rõ để kỹ thuật viên chuẩn bị đúng vật tư

X

Tóm tắt sự cố

Một câu ngắn gọn mô tả vấn đề

VD: Hỏng bóng đèn nhà vệ sinh, Vòi nước rò rỉ...

Mô tả chi tiết

Vị trí cụ thể và tình trạng hư hỏng để kỹ thuật viên chuẩn bị trước

VD: Bóng đèn ở góc trái nhà vệ sinh bị cháy hoàn toàn từ tối qua, gây tối trong khu vực...

Mức độ ảnh hưởng

Chọn mức độ phù hợp để ban quản lý ưu tiên xử lý

Thấp

Bình thường

Ưu tiên cao

Khẩn cấp

Ảnh thực tế

Đính kèm ảnh rõ khu vực hư hỏng để ban quản lý đánh giá nhanh hơn

Hủy bỏ

+ Gửi yêu cầu

-
- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FBK-02.

Relationships:

- None.

Traceability: ST20

UC-FBK-03 — Review and Respond to Feedback

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-FBK-03**Use-case name:** Review and Respond to Feedback**Actor(s):** Dormitory Manager**Supporting Actor(s):** None**Description:** The manager reviews complaints and suggestions, responds, and closes or resolves them.**Trigger:** A feedback item is submitted or the manager opens Feedback Management.**Preconditions:**

1. The actor is authenticated as Dormitory Manager.

Basic Flow (Main Success Scenario):

1. The system lists feedback by type, category, priority, status, and date.
2. The manager opens an item and reviews its content and evidence.
3. The manager assigns or investigates the item as needed.
4. The manager enters a response and marks it in progress, resolved, rejected, or closed.
5. The system stores the response and notifies the student when identity is available.

Alternative Flows:

- A1. Anonymous item: the response is stored for reporting but cannot be delivered directly.
- A2. Duplicate/spam: the manager closes it with a reason.
- A3. More information required: the manager requests clarification and keeps it open.

Postconditions:

- Feedback status and response history are updated.

Special Requirements:

- None.

Prototype Screens:

The screenshot displays the Dormify web application interface. On the left is a dark sidebar with the Dormify logo and a list of navigation items: QUẢN TRỊ VIÊN, Tổng quan, Quản lý phòng, Sinh viên (highlighted with a yellow badge), Phân quyền tài khoản, Duyệt đơn phòng, Phân phòng tự động, Duyệt đổi phòng, Duyệt trả phòng, Tam trú / Tam vắng, Gửi thông báo (highlighted with a yellow badge), Hóa đơn, and Vê trang chủ. The main content area has a light beige background. At the top, there's a header with the Dormify logo, the text 'Dormify - Đảm bảo Atlas', and user avatars. Below the header is a dark blue banner with the title 'Gửi thông báo toàn sinh viên' and a description: 'Thông báo sẽ được lưu vào hộp thông báo của từng sinh viên và đồ chuông realtime ngay lập tức — dùng cho lịch cắt điện/nước, họp KTX, thông báo khẩn...'. Underneath is a white form titled 'Soạn thông báo'. It contains two text input fields: 'Tiêu đề (ngắn gọn, tối đa 100 ký tự)' with the example 'Ví dụ: Lịch cắt nước tòa A ngày 15/07', and 'Nội dung' with the placeholder 'Nội dung chi tiết của thông báo...'. A character count '0/1000' is shown at the bottom right of the content field. At the bottom of the form, there's a warning icon and text: 'Gửi cho toàn bộ sinh viên đang hoạt động — không thể thu hồi.' and a 'Gửi thông báo' button.

- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-FBK-03.

Relationships:

- None.

Traceability: FM15

11. Conduct and Student Evaluation

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Violation records, conditional conduct-score deductions, periodic evaluations, and student history views.

ID	Use case	Primary actor
UC-COND-01	Record Student Violation	Dormitory Manager
UC-COND-02	Apply Conduct-Score Deduction	Dormitory Manager / System
UC-COND-03	Perform Periodic Student Evaluation	Dormitory Manager
UC-COND-04	View Evaluation History	Student
UC-COND-05	View Violation History	Student

UC-COND-01 — Record Student Violation

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-COND-01

Use-case name: Record Student Violation

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager creates a record when a student violates dormitory rules.

Trigger: The manager selects Record Violation for a student.

Preconditions:

1. The actor is authenticated as Dormitory Manager.
2. The target is a student.

Basic Flow (Main Success Scenario):

1. The manager selects the student and applicable rule.
2. The manager enters date, location, description, evidence, and optional witnesses.
3. The system displays the configured penalty and whether a score deduction applies.
4. The manager confirms the violation record.
5. The system stores the record and notifies the student.

Alternative Flows:

- A1. Warning-only incident: the manager records the violation without invoking a score deduction.
- A2. Duplicate incident detected: the system warns before confirmation.

- A3. Invalid/missing evidence where required: submission is blocked.

Postconditions:

- A violation record exists.

Special Requirements:

- None.

Prototype Screens:

Hồ sơ Sinh viên

Thông tin cá nhân

Quản lý quyền & Lời nhắc

Điểm hành vi



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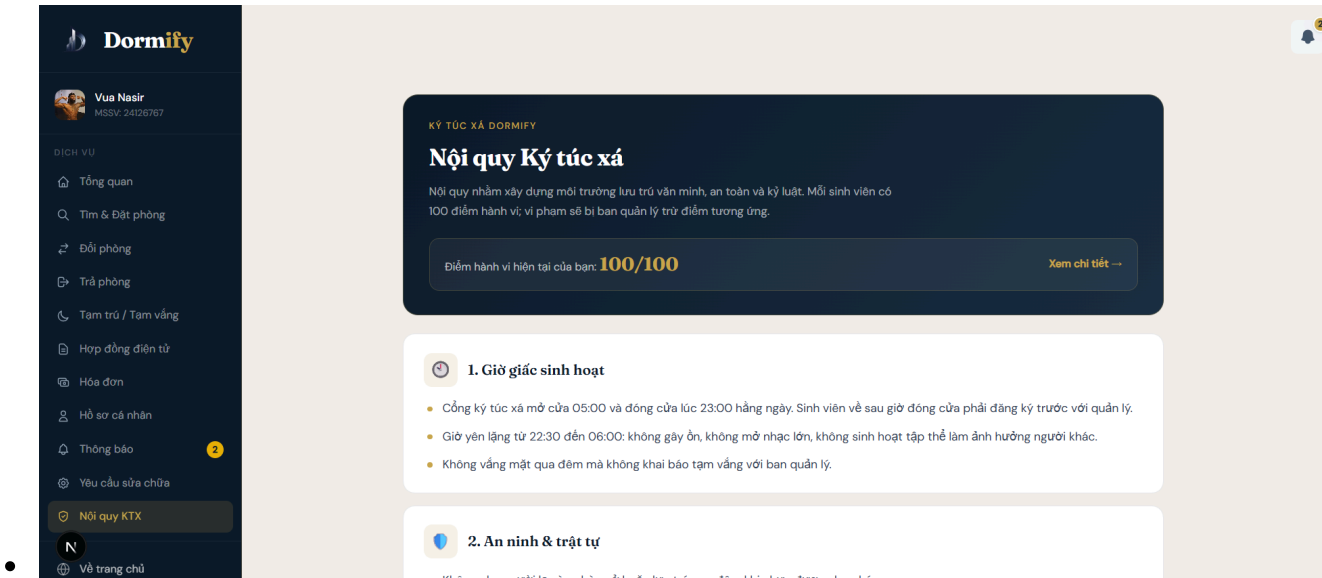
Số CCCD

012432122123

Xóa sinh viên

Hủy bỏ

Lưu thay đổi



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-COND-01.

Relationships:

- None.

Traceability: FR29, FM13

UC-COND-02 — Apply Conduct-Score Deduction

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-COND-02

Use-case name: Apply Conduct-Score Deduction

Actor(s): Dormitory Manager / System

Supporting Actor(s): None

Description: The system applies a configured conduct-score deduction when a recorded violation requires one.

Trigger: The manager confirms a point-bearing violation.

Preconditions:

1. UC-COND-01 is in progress.
2. The selected rule defines a deduction or the manager has authorized a permitted adjustment.

Basic Flow (Main Success Scenario):

1. The system reads the student’s current conduct score.
2. The system calculates the deduction according to the rule and configured minimum score.
3. The manager reviews the resulting score.
4. The system updates the score and stores before/after values with the violation.
5. The system includes the deduction in the student notification.

Alternative Flows:

- A1. Warning-only incident: this use case is not invoked.

- A2. Deduction would fall below minimum; the score is limited to the configured minimum.
- A3. Rule configuration missing; the system requires an authorized manual value or blocks completion.

Postconditions:

- The conduct score and deduction history are updated.

Special Requirements:

- None.

Prototype Screens:

Hồ sơ Sinh viên

Thông tin cá nhân

Quản lý quyền & Lời nhắc

Điểm hành vi



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Số CCCD

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Xóa sinh viên

Hủy bỏ

Lưu thay đổi

- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-COND-02.

Relationships:

- Extends UC-COND-01 Record Student Violation.

Traceability: FR30

UC-COND-03 — Perform Periodic Student Evaluation

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-COND-03**Use-case name:** Perform Periodic Student Evaluation**Actor(s):** Dormitory Manager**Supporting Actor(s):** None**Description:** The manager records a periodic conduct evaluation for students by month, term, or academic year.**Trigger:** The manager starts an evaluation period.**Preconditions:**

1. The actor is authenticated as Dormitory Manager.
2. An evaluation period is defined.

Basic Flow (Main Success Scenario):

1. The system lists eligible students and relevant violation/residence history.
2. The manager selects a student or bulk group and enters qualitative rating, comments, and permitted score adjustment.
3. The system validates that one final evaluation per student/period is maintained.
4. The manager saves draft or publishes the evaluation.
5. The system stores the evaluation and notifies the student when published.

Alternative Flows:

- A1. Existing evaluation: the manager edits the draft or creates a controlled revision.
- A2. Bulk evaluation: a baseline is applied and individual exceptions are edited.
- A3. Incomplete evaluation: it remains draft and is not visible to the student.

Postconditions:

- A periodic evaluation record exists in draft or published state.

Special Requirements:

- None.

Prototype Screens:

Hồ sơ Sinh viên

Thông tin cá nhân

Quản lý quyền & Lời nhắn

Điểm hành vi



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012432122123

Xóa sinh viên

Hủy bỏ

Lưu thay đổi

-
- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-COND-03.

Relationships:

- None.

Traceability: FR30, FM14

UC-COND-04 — View Evaluation History

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-COND-04

Use-case name: View Evaluation History

Actor(s): Student

Supporting Actor(s): None

Description: A student views published periodic conduct evaluations and score changes.

Trigger: The student opens Evaluation History.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system retrieves published evaluations belonging to the student.
2. The system displays period, rating, comments, score change, and publication date.
3. The student opens an evaluation to view details.

Alternative Flows:

- A1. No published evaluations: the system displays an empty state.
- A2. Draft evaluation: it is not visible.

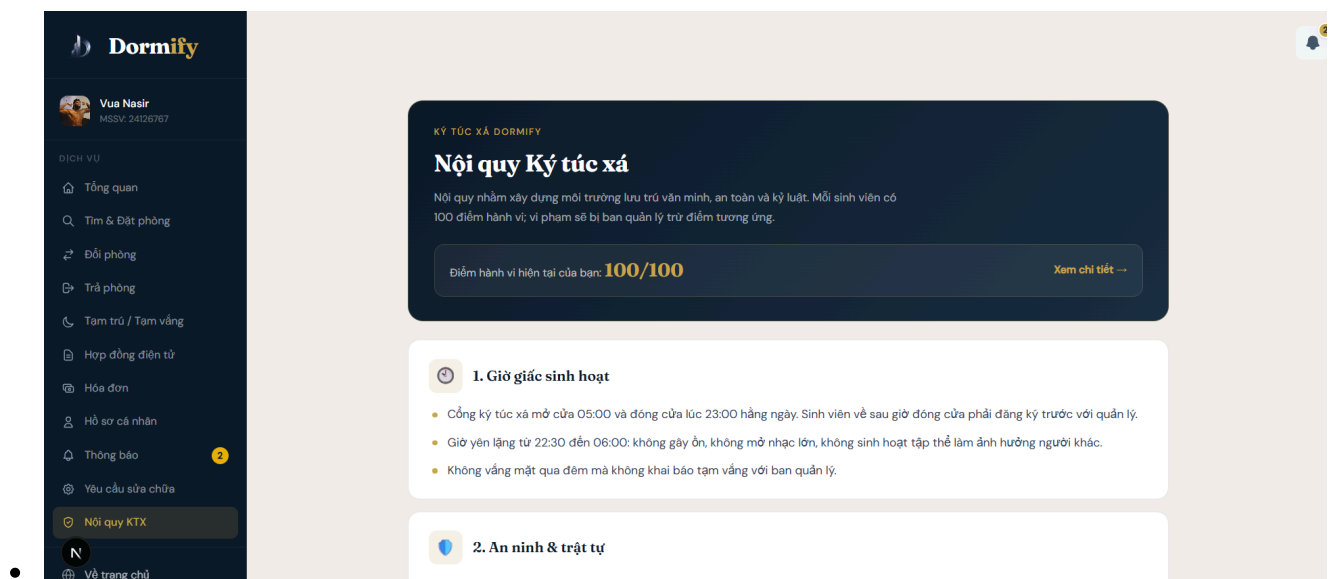
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-COND-04.

Relationships:

- None.

Traceability: FR30, ST27

UC-COND-05 — View Violation History

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-COND-05
Use-case name: View Violation History
Actor(s): Student
Supporting Actor(s): None
Description: A student views recorded violations and related penalties.
Trigger: The student opens Violation History.

Preconditions:

- 1. The student is authenticated.

Basic Flow (Main Success Scenario):

- 1. The system retrieves the student’s violation records.
- 2. The system displays rule, date, description, evidence availability, penalty, score deduction, and resulting score.
- 3. The student opens a record to view details.

Alternative Flows:

- A1. No violations: the system displays a positive empty state.
- A2. Restricted evidence: the system displays only permitted information.

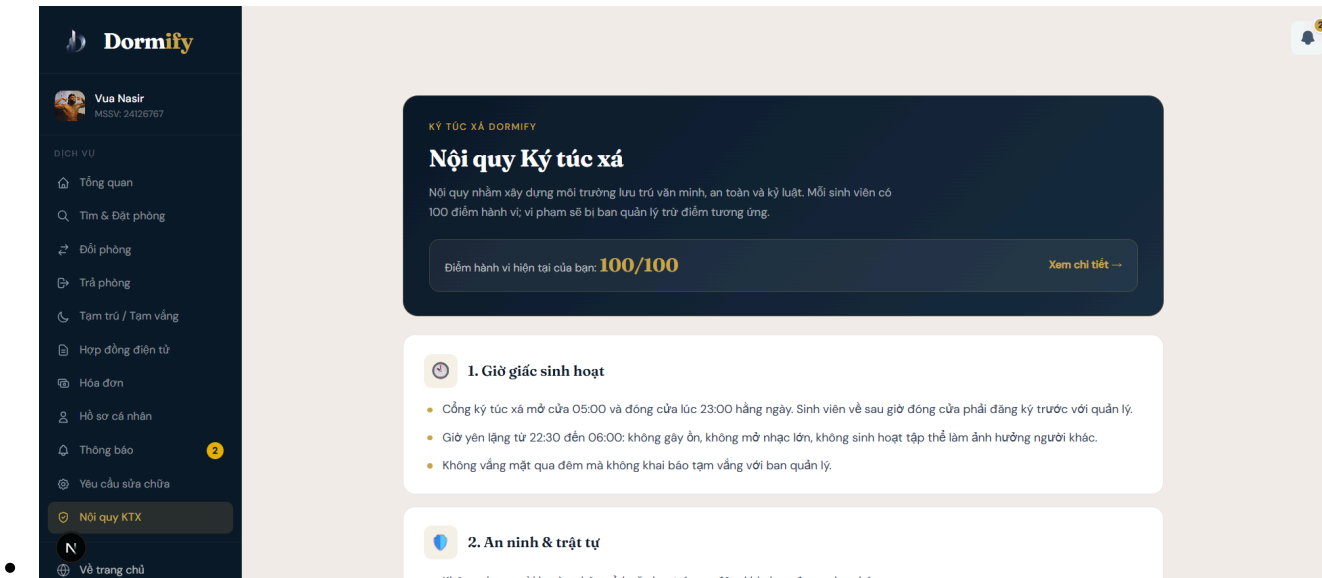
Postconditions:

- No data is changed.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-COND-05.

Relationships:

- None.

Traceability: FR29, ST28

12. Notifications and Message Center

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Notifications, communication history, and targeted management announcements.

ID	Use case	Primary actor
UC-NOT-01	View Notifications	Student
UC-NOT-02	Use Message Center	Student
UC-NOT-03	Send Announcement or Message	Dormitory Manager

UC-NOT-01 — View Notifications

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Use-case ID: UC-NOT-01**Use-case name:** View Notifications**Actor(s):** Student**Supporting Actor(s):** None**Description:** A student views system-generated and management notifications.**Trigger:** The student opens Notifications or selects the notification bell.**Preconditions:**

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system retrieves notifications ordered newest first.
2. The system displays type, title, content, time, source, and read status.
3. The student opens a notification and follows an internal link where available.
4. The student marks one or all notifications as read.

Alternative Flows:

- A1. No notifications: the system displays an empty state.
- A2. Real-time connection unavailable: the page continues to show stored notifications and refreshes through normal requests.

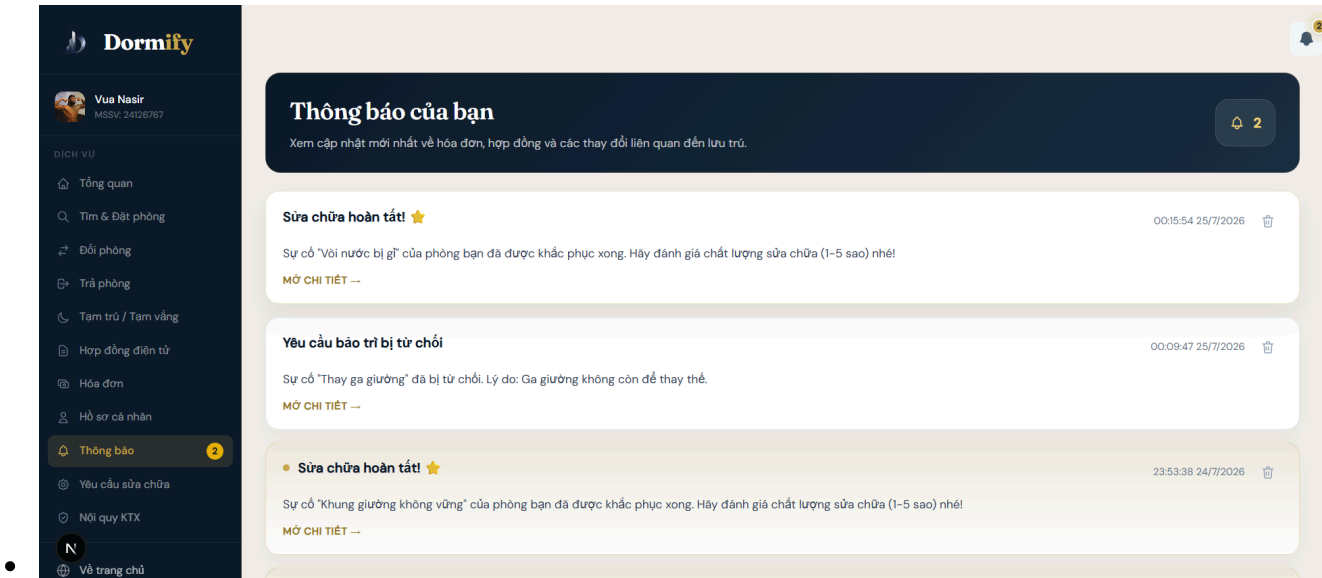
Postconditions:

- Selected notification read status is updated.

Special Requirements:

- Automatic notifications are postconditions of related business use cases.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-NOT-01.

Relationships:

- None.

Traceability: ST25

UC-NOT-02 — Use Message Center

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-NOT-02

Use-case name: Use Message Center

Actor(s): Student

Supporting Actor(s): None

Description: A student views received messages and communication history in a centralized inbox.

Trigger: The student opens Message Center.

Preconditions:

1. The student is authenticated.

Basic Flow (Main Success Scenario):

1. The system lists management messages and supported conversation threads.
2. The student filters by unread, sender, category, or date.
3. The student opens a message/thread and reads its history.
4. Where two-way communication is enabled, the student writes and sends a reply.

Alternative Flows:

- A1. Reply is not allowed for a broadcast/system message: the composer is disabled.
- A2. No messages: the system displays an empty state.
- A3. Send failure: the draft is preserved for retry.

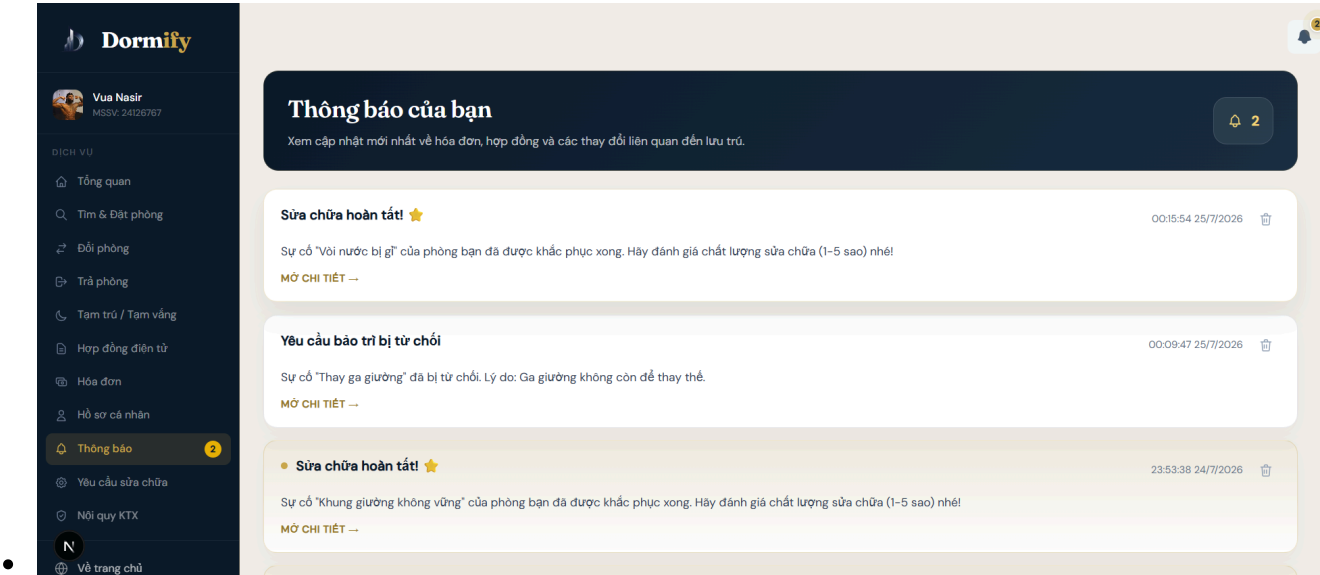
Postconditions:

- Read status and any sent reply are stored.

Special Requirements:

- None.

Prototype Screens:



- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-NOT-02.

Relationships:

- None.

Traceability: ST26

UC-NOT-03 — Send Announcement or Message

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Use-case ID: UC-NOT-03

Use-case name: Send Announcement or Message

Actor(s): Dormitory Manager

Supporting Actor(s): None

Description: The manager sends information to selected students, rooms, floors, buildings, or all residents.

Trigger: The manager opens Announcement / Message Management.

Preconditions:

1. The actor is authenticated as Dormitory Manager.

Basic Flow (Main Success Scenario):

1. The manager enters title, message, category, priority, and optional attachment.
2. The manager selects target audience: individual students, rooms, floors, buildings, or all students.
3. The system previews recipient count and validates the target.
4. The manager confirms sending.

- The system creates recipient notifications/messages, delivers real-time updates where possible, and records broadcast history.

Alternative Flows:

- A1. No eligible recipients: sending is blocked.
- A2. Partial delivery failure: successful deliveries remain and failures are reported/retried.
- A3. Scheduled announcement, if enabled: the system stores it and sends at the configured time.

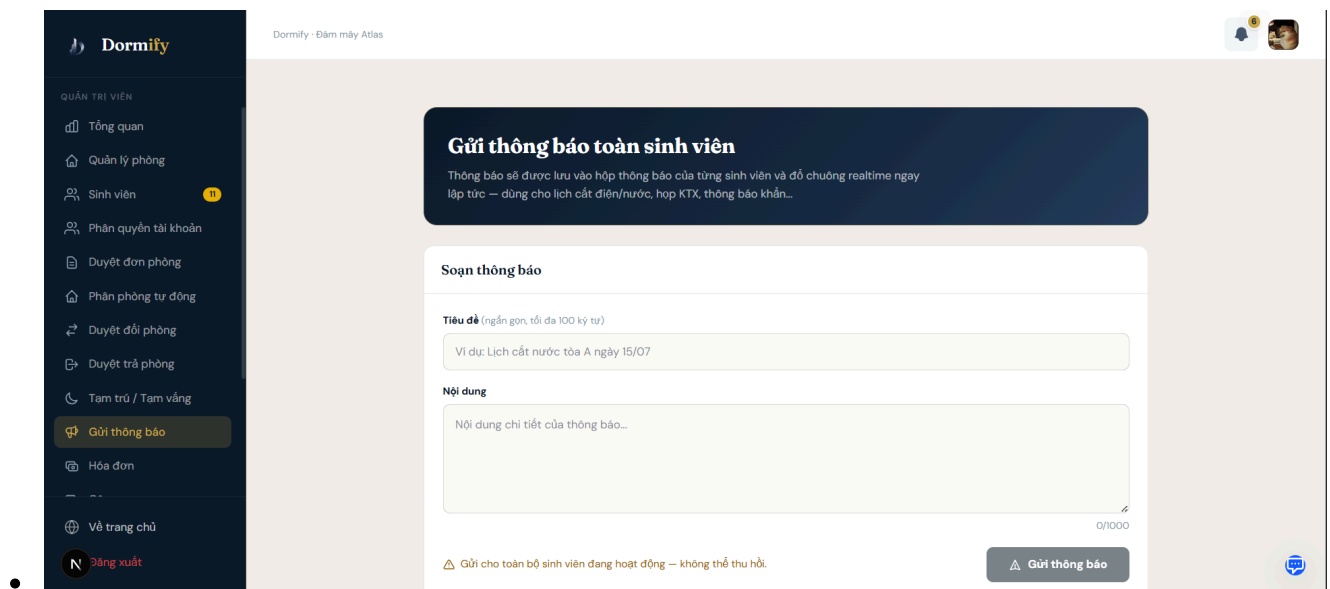
Postconditions:

- Messages/notifications are created for targeted recipients and a send history exists.

Special Requirements:

- Access to broadcast targeting must be restricted and audited.

Prototype Screens:





- PA3 asset screenshot evidence covering the basic flow and alternative flows for UC-NOT-03.

Relationships:

- None.

Traceability: FR11, FR22, FR23, FR25, FR26, FR28, FR29

13. Functional Requirement Traceability

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Functional requirement	Related use cases	Coverage
FR01	UC-AUTH-01	Covered
FR02	UC-AUTH-02, UC-AUTH-03, UC-AUTH-04, UC-PRO-03	Covered
FR03	UC-PRO-01, UC-PRO-02, UC-STAY-01, UC-STAY-02, UC-STAY-03, UC-STU-01	Covered
FR04	UC-ADM-01	Covered
FR05	UC-ADM-02	Covered
FR06	UC-ADM-03	Covered
FR07	UC-ADM-04	Covered
FR08	—	Removed from scope
FR09	UC-ROOM-01	Covered
FR10	UC-STU-01	Covered
FR11	UC-ROOM-04, UC-ROOM-05, UC-ROOM-06	Covered
FR12	UC-ROOM-07	Covered
FR13	UC-ROOM-08, UC-ROOM-09, UC-ROOM-10	Covered
FR14	UC-CON-01, UC-CON-02, UC-CON-03, UC-CON-04	Covered
FR15	UC-CON-02	Covered
FR16	UC-CON-03	Covered
FR17	UC-CON-04, UC-CON-06	Covered
FR18	UC-CHK-01, UC-CHK-02	Covered
FR19	UC-CHK-02, UC-CHK-03	Covered
FR20	UC-CHK-03	Covered
FR21	UC-CHK-04	Covered
FR22	UC-FIN-01, UC-FIN-02, UC-FIN-03, UC-FIN-07, UC-FIN-08, UC-FIN-09, UC-FIN-10, UC-FIN-11	Covered
FR23	UC-FIN-04, UC-FIN-05, UC-FIN-07	Covered
FR24	UC-FIN-06	Covered
FR25	UC-RES-01, UC-RES-05	Covered
FR26	UC-MNT-01, UC-MNT-03	Covered
FR27	UC-MNT-03	Covered

Functional requirement	Related use cases	Coverage
FR28	UC-MNT-02, UC-MNT-04, UC-MNT-05, UC-MNT-06, UC-MNT-07, UC-MNT-08, UC-MNT-09, UC-MNT-10	Covered
FR29	UC-COND-01, UC-COND-05	Covered
FR30	UC-COND-02, UC-COND-03, UC-COND-04	Covered

14. Student Feature Traceability

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Student features	Related use cases
ST01–ST03 Personal profile	UC-PRO-01, UC-PRO-02, UC-PRO-03
ST04–ST06 Residence information	UC-STAY-01, UC-STAY-02, UC-STAY-03
ST07–ST10 Room rental	UC-ROOM-02, UC-ROOM-03, UC-ROOM-04, UC-ROOM-05
ST11–ST12 Contract	UC-CON-05, UC-CON-06
ST13–ST16 Invoice and payment	UC-FIN-08, UC-FIN-09, UC-FIN-10, UC-FIN-11
ST17–ST18 Repair requests	UC-MNT-01, UC-MNT-02
ST19–ST20 Feedback and suggestions	UC-FBK-01, UC-FBK-02
ST21–ST24 Residence declarations	UC-RES-01, UC-RES-02, UC-RES-03, UC-RES-04
ST25–ST26 Notifications	UC-NOT-01, UC-NOT-02
ST27–ST28 Evaluation and violations	UC-COND-04, UC-COND-05

15. Maintenance Staff Feature Traceability

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Đào Duy Anh

Maintenance features	Related use cases
MT01	UC-MNT-05
MT02	UC-MNT-06
MT03–MT04	UC-MNT-07
MT05–MT06	UC-MNT-08
MT07	UC-MNT-09
MT08	UC-MNT-10

16. Floor Manager Function Reassignment

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Đào Duy Anh

Former Floor Manager functions	Responsible role and use cases
FM01–FM02 View students and residence information	Dormitory Manager — UC-STU-01
FM03 Track room conditions	Dormitory Manager — UC-ROOM-01
FM04–FM09 Manage meter readings	Dormitory Manager — UC-FIN-01
FM10–FM12 Track residence declarations	Dormitory Manager — UC-RES-05
FM13 Record violations	Dormitory Manager — UC-COND-01
FM14 Perform periodic evaluations	Dormitory Manager — UC-COND-03
FM15 Receive feedback	Dormitory Manager — UC-FBK-03
FM16 Track repair requests	Dormitory Manager — UC-MNT-04

End of Use-Case Specification — Version 1.0 / PA3